# Linear Algebra and Its Applications

## Chapter 1

Detailed worked solutions

Mathjock

**Text and numbering.** David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

**Source scope.** The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

**How to read the solutions.** Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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## 1.1 Systems of linear equations

A solution is one ordered list of values satisfying *every* equation. The solution set is the set of all such lists. A consistent system has at least one solution; an inconsistent system has none. Two systems are equivalent when their solution sets are equal.

The three permitted operations are swapping two equations, multiplying an equation by a nonzero scalar, and adding a multiple of one equation to another. In matrix notation $R_i \leftarrow R_i + cR_j$ means to replace the *whole* row, including its last entry. We use a vertical bar to separate the coefficient columns from the constant column. A zero row means $0 = 0$; a row $[0 \ \cdots \ 0 \mid d]$ with $d \neq 0$ means an impossible equation.

### Practice problem 1.1.P1: Choose the next elimination step

#### (a) Use the simpler coefficient to make the next pivot.

Swap equations 3 and 4, so the coefficient of $x_3$ in the third equation is 1 instead of 5. Next one would replace equation 4 by equation 4 minus five times equation 3. This eliminates $x_3$ from the last equation. Do not restart elimination on $x_2$: the forward phase first establishes successive pivots below the previous ones.

#### (b) Begin the backward phase at the last pivot.

The fourth equation is $x_4 = 1$. Add twice equation 4 to equation 1 to remove its term $-2x_4$. The other two equations already have no $x_4$ term. After that, divide equation 3 by 2 and use it to clear the $x_3$ terms above it. Each operation preserves all solutions.

### Practice problem 1.1.P2: Consistency of a triangular system

**What we must show.** Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

**Plan.** Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & 5 & 2 & -6 \\ 0 & 4 & -7 & 2 \\ 0 & 0 & 5 & 0 \end{array}\right].$$

Use $R_2 \leftarrow (\frac{1}{4})R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 5 & 2 & -6 \\ 0 & 1 & -\frac{7}{4} & \frac{1}{2} \\ 0 & 0 & 5 & 0 \end{array}\right].$$

Use $R_3 \leftarrow (\frac{1}{5})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 5 & 2 & -6 \\ 0 & 1 & -\frac{7}{4} & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (2)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 5 & 0 & -6 \\ 0 & 1 & -\frac{7}{4} & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-\frac{7}{4})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 5 & 0 & -6 \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (5)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -\frac{17}{2} \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{array}\right].$$

**Read the equations, keeping every free variable.**

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = -\frac{17}{2} \\ x_2 = \frac{1}{2} \\ x_3 = 0 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

**Conclusion.** The displayed values give the unique solution.

### Practice problem 1.1.P3: Test a proposed solution in every equation

Substitute $(x_1, x_2, x_3) = (3, 4, -2)$ into the three left sides:

$$5(3) - 4 + 2(-2) = 7, \qquad -2(3) + 6(4) + 9(-2) = 0,$$

$$-7(3) + 5(4) - 3(-2) = 5.$$

The first two constants match, but the third required constant is $-7$, not 5. One failed equation is enough to reject the proposed solution.

**Conclusion.** $(3, 4, -2)$ is not a solution.

### Practice problem 1.1.P4: A consistency condition on two constants

The second left side is $-3$ times the first. Replace equation 2 by equation 2 plus three times equation 1. The resulting last equation is $0 = k + 3h$. It can hold precisely when $k = -3h$. In that case choose any $x_2 = t$ and put $x_1 = (h + t)/2$; these values satisfy both equations.

**Conclusion.** Consistency holds if and only if $k = -3h$.

### Exercise 1.1.1: A two-equation elimination

**What we must show.** Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

**Plan.** Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cc|c} 1 & 5 & 7 \\ -2 & -7 & -5 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-2)R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & 5 & 7 \\ 0 & 3 & 9 \end{array}\right].$$

Use $R_2 \leftarrow (\frac{1}{3})R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 5 & 7 \\ 0 & 1 & 3 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (5)R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 0 & -8 \\ 0 & 1 & 3 \end{array}\right].$$

**Read the equations, keeping every free variable.**

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = -8 \\ x_2 = 3 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

**Conclusion.** The displayed values give the unique solution.

### Exercise 1.1.2: A second two-equation elimination

**What we must show.** Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

**Plan.** Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cc|c} 3 & 6 & -3 \\ 5 & 7 & 10 \end{array}\right].$$

Use $R_1 \leftarrow (\frac{1}{3})R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & 2 & -1 \\ 5 & 7 & 10 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (5)R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & 2 & -1 \\ 0 & -3 & 15 \end{array}\right].$$

Use $R_2 \leftarrow (-\frac{1}{3})R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 2 & -1 \\ 0 & 1 & -5 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (2)R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 0 & 9 \\ 0 & 1 & -5 \end{array}\right].$$

**Read the equations, keeping every free variable.**

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = 9 \\ x_2 = -5 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

**Conclusion.** The displayed values give the unique solution.

### Exercise 1.1.3: The intersection of two lines

**What we must show.** Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

**Plan.** Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cc|c} 1 & 2 & 4 \\ 1 & -1 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (1)R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & 2 & 4 \\ 0 & -3 & -3 \end{array}\right].$$

Use $R_2 \leftarrow (-\frac{1}{3})R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 2 & 4 \\ 0 & 1 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (2)R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array}\right].$$

**Read the equations, keeping every free variable.**

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = 2 \\ x_2 = 1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

**Conclusion.** The displayed values give the unique solution.

### Exercise 1.1.4: A second line intersection

**What we must show.** Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

**Plan.** Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cc|c} 1 & 2 & -13 \\ 3 & -2 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (3)R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & 2 & -13 \\ 0 & -8 & 40 \end{array}\right].$$

Use $R_2 \leftarrow (-\frac{1}{8})R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 2 & -13 \\ 0 & 1 & -5 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (2)R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 0 & -3 \\ 0 & 1 & -5 \end{array}\right].$$

**Read the equations, keeping every free variable.**

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = -3 \\ x_2 = -5 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

**Conclusion.** The displayed values give the unique solution.

### Exercise 1.1.5: Begin backward elimination

The last two equations already give $x_3 = 2$ and $x_4 = -5$. To begin clearing above the $x_3$ pivot, subtract four times row 3 from row 2, then add three times row 3 to row 1:

$$R_2 \leftarrow R_2 - 4R_3, \qquad R_1 \leftarrow R_1 + 3R_3.$$

The two changed equations become $x_2 = -2$ and $x_1 - 4x_2 = 13$. Both operations act on all entries, including the constants.

**Conclusion.** Subtract four times row 3 from row 2; add three times row 3 to row 1.

### Exercise 1.1.6: Create the last pivot

Use the coefficient 1 in row 3 to remove the coefficient 4 below it:

$$R_4 \leftarrow R_4 - 4R_3 \quad \Longrightarrow \quad [0\ 0\ 0\ -7 \mid 14].$$

Now multiply the last row by $-1/7$, giving $[0\ 0\ 0\ 1 \mid -2]$. These steps create the next pivot without disturbing the earlier zeros.

**Conclusion.** Subtract four times row 3 from row 4, then divide row 4 by $-7$.

### Exercise 1.1.7: A contradiction row cannot be repaired

**What we must show.** Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

**Plan.** Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & 7 & 3 & -4 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & -2 \end{array}\right].$$

Use $R_3 \leftrightarrow R_4$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 7 & 3 & -4 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-4)R_4$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 7 & 3 & 0 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (3)R_4$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 7 & 3 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-2)R_4$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 7 & 3 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (3)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 7 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-1)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 7 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (7)R_2$ to obtain

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right].$$

**Conclusion.** A row with all variable coefficients zero and a nonzero constant asserts $0 = 1$ after scaling. This is impossible, so the system is inconsistent and its solution set is empty.

### Exercise 1.1.8: Four variables fixed at zero

**What we must show.** Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

**Plan.** Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cccc|c} 1 & -5 & 4 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \end{array}\right].$$

Use $R_3 \leftarrow (\frac{1}{3})R_3$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -5 & 4 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \end{array}\right].$$

Use $R_4 \leftarrow (\frac{1}{2})R_4$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -5 & 4 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (1)R_4$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -5 & 4 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (4)R_3$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -5 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-5)R_2$ to obtain

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right].$$

**Read the equations, keeping every free variable.**

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = 0 \\ x_2 = 0 \\ x_3 = 0 \\ x_4 = 0 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

**Conclusion.** The displayed values give the unique solution.

### Exercise 1.1.9: Back substitution through four pivots

**What we must show.** Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

**Plan.** Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cccc|c} 1 & -1 & 0 & 0 & -5 \\ 0 & 1 & -2 & 0 & -7 \\ 0 & 0 & 1 & -3 & 2 \\ 0 & 0 & 0 & 1 & 4 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-3)R_4$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -1 & 0 & 0 & -5 \\ 0 & 1 & -2 & 0 & -7 \\ 0 & 0 & 1 & 0 & 14 \\ 0 & 0 & 0 & 1 & 4 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-2)R_3$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -1 & 0 & 0 & -5 \\ 0 & 1 & 0 & 0 & 21 \\ 0 & 0 & 1 & 0 & 14 \\ 0 & 0 & 0 & 1 & 4 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-1)R_2$ to obtain

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 16 \\ 0 & 1 & 0 & 0 & 21 \\ 0 & 0 & 1 & 0 & 14 \\ 0 & 0 & 0 & 1 & 4 \end{array}\right].$$

**Read the equations, keeping every free variable.**

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = 16 \\ x_2 = 21 \\ x_3 = 14 \\ x_4 = 4 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

**Conclusion.** The displayed values give the unique solution.

### Exercise 1.1.10: A triangular four-variable system

**What we must show.** Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

**Plan.** Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cccc|c} 1 & 3 & 0 & -2 & -7 \\ 0 & 1 & 0 & 3 & 6 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & -2 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-2)R_4$ to obtain

$$\left[\begin{array}{cccc|c} 1 & 3 & 0 & 0 & -11 \\ 0 & 1 & 0 & 3 & 6 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & -2 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (3)R_4$ to obtain

$$\left[\begin{array}{cccc|c} 1 & 3 & 0 & 0 & -11 \\ 0 & 1 & 0 & 0 & 12 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & -2 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (3)R_2$ to obtain

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & -47 \\ 0 & 1 & 0 & 0 & 12 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & -2 \end{array}\right].$$

**Read the equations, keeping every free variable.**

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = -47 \\ x_2 = 12 \\ x_3 = 2 \\ x_4 = -2 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

**Conclusion.** The displayed values give the unique solution.

### Exercise 1.1.11: A system with a zero leading coefficient

**What we must show.** Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

**Plan.** Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 0 & 1 & 5 & -4 \\ 1 & 4 & 3 & -2 \\ 2 & 7 & 1 & -2 \end{array}\right].$$

Use $R_1 \leftrightarrow R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 4 & 3 & -2 \\ 0 & 1 & 5 & -4 \\ 2 & 7 & 1 & -2 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (2)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 4 & 3 & -2 \\ 0 & 1 & 5 & -4 \\ 0 & -1 & -5 & 2 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-1)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 4 & 3 & -2 \\ 0 & 1 & 5 & -4 \\ 0 & 0 & 0 & -2 \end{array}\right].$$

Use $R_3 \leftarrow (-\frac{1}{2})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 4 & 3 & -2 \\ 0 & 1 & 5 & -4 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-2)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 4 & 3 & 0 \\ 0 & 1 & 5 & -4 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-4)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 4 & 3 & 0 \\ 0 & 1 & 5 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (4)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -17 & 0 \\ 0 & 1 & 5 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

**Conclusion.** A row with all variable coefficients zero and a nonzero constant asserts $0 = 1$ after scaling. This is impossible, so the system is inconsistent and its solution set is empty.

### Exercise 1.1.12: A three-variable contradiction

**What we must show.** Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

**Plan.** Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{rrr|r} 1 & -5 & 4 & -3 \\ 2 & -7 & 3 & -2 \\ -2 & 1 & 7 & -1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (2)R_1$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -5 & 4 & -3 \\ 0 & 3 & -5 & 4 \\ -2 & 1 & 7 & -1 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-2)R_1$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -5 & 4 & -3 \\ 0 & 3 & -5 & 4 \\ 0 & -9 & 15 & -7 \end{array}\right].$$

Use $R_2 \leftarrow (\frac{1}{3})R_2$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -5 & 4 & -3 \\ 0 & 1 & -\frac{5}{3} & \frac{4}{3} \\ 0 & -9 & 15 & -7 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-9)R_2$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -5 & 4 & -3 \\ 0 & 1 & -\frac{5}{3} & \frac{4}{3} \\ 0 & 0 & 0 & 5 \end{array}\right].$$

Use $R_3 \leftarrow (\frac{1}{5})R_3$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -5 & 4 & -3 \\ 0 & 1 & -\frac{5}{3} & \frac{4}{3} \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-3)R_3$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -5 & 4 & 0 \\ 0 & 1 & -\frac{5}{3} & \frac{4}{3} \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (\frac{4}{3})R_3$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -5 & 4 & 0 \\ 0 & 1 & -\frac{5}{3} & 0 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-5)R_2$ to obtain

$$\left[\begin{array}{rrr|r} 1 & 0 & -\frac{13}{3} & 0 \\ 0 & 1 & -\frac{5}{3} & 0 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

**Conclusion.** A row with all variable coefficients zero and a nonzero constant asserts $0 = 1$ after scaling. This is impossible, so the system is inconsistent and its solution set is empty.

### Exercise 1.1.13: A unique three-variable solution

**What we must show.** Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

**Plan.** Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & 8 \\ 2 & 2 & 9 & 7 \\ 0 & 1 & 5 & -2 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (2)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & 8 \\ 0 & 2 & 15 & -9 \\ 0 & 1 & 5 & -2 \end{array}\right].$$

Use $R_2 \leftarrow (\frac{1}{2})R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & 8 \\ 0 & 1 & \frac{15}{2} & -\frac{9}{2} \\ 0 & 1 & 5 & -2 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (1)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & 8 \\ 0 & 1 & \frac{15}{2} & -\frac{9}{2} \\ 0 & 0 & -\frac{5}{2} & \frac{5}{2} \end{array}\right].$$

Use $R_3 \leftarrow (-\frac{2}{5})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & 8 \\ 0 & 1 & \frac{15}{2} & -\frac{9}{2} \\ 0 & 0 & 1 & -1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-3)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & \frac{15}{2} & -\frac{9}{2} \\ 0 & 0 & 1 & -1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (\frac{15}{2})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -1 \end{array}\right].$$

**Read the equations, keeping every free variable.**

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = 5 \\ x_2 = 3 \\ x_3 = -1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

**Conclusion.** The displayed values give the unique solution.

### Exercise 1.1.14: Solve after scaling a leading row

**What we must show.** Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

**Plan.** Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 2 & 0 & -6 & -8 \\ 0 & 1 & 2 & 3 \\ 3 & 6 & -2 & -4 \end{array}\right].$$

Use $R_1 \leftarrow (\frac{1}{2})R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & -4 \\ 0 & 1 & 2 & 3 \\ 3 & 6 & -2 & -4 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (3)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & -4 \\ 0 & 1 & 2 & 3 \\ 0 & 6 & 7 & 8 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (6)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & -4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & -5 & -10 \end{array}\right].$$

Use $R_3 \leftarrow (-\frac{1}{5})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & -4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-3)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (2)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{array}\right].$$

**Read the equations, keeping every free variable.**

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = 2 \\ x_2 = -1 \\ x_3 = 2 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

**Conclusion.** The displayed values give the unique solution.

### Exercise 1.1.15: Detect inconsistency without solving

The augmented rows are

$$\begin{bmatrix} 1 & -6 & 0 & 0 & 5 \\ 0 & 1 & -4 & 1 & 0 \\ -1 & 6 & 1 & 5 & 3 \\ 0 & -1 & 5 & 4 & 0 \end{bmatrix}.$$

Use $R_3 \leftarrow R_3 + R_1$ and $R_4 \leftarrow R_4 + R_2$. Rows 3 and 4 become $[0\ 0\ 1\ 5 \mid 8]$ and $[0\ 0\ 1\ 5 \mid 0]$. Then $R_4 \leftarrow R_4 - R_3$ gives $[0\ 0\ 0\ 0 \mid -8]$. Its equation $0 = -8$ is impossible.

**Conclusion.** The system is inconsistent. No further solving is needed.

### Exercise 1.1.16: Detect consistency without solving

Divide row 1 by 2, then use $R_4 \leftarrow R_4 + 3R_1$:

$$\begin{bmatrix} 1 & 0 & 0 & -2 & -5 \\ 0 & 3 & 3 & 0 & 0 \\ 0 & 0 & 1 & 4 & -1 \\ 0 & 2 & 3 & -5 & -10 \end{bmatrix}.$$

Now $R_4 \leftarrow R_4 - \frac{2}{3}R_2$ gives $[0\ 0\ 1\ -5 \mid -10]$, and $R_4 \leftarrow R_4 - R_3$ gives $[0\ 0\ 0\ -9 \mid -9]$. There is a nonzero pivot in each variable column and no contradiction row. Back substitution is therefore possible from the last equation upward.

**Conclusion.** The system is consistent. The exercise does not require its values.

### Exercise 1.1.17: Do three lines meet at one point?

**What we must show.** Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

**Plan.** Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cc|c} 2 & 3 & -1 \\ 6 & 5 & 0 \\ 2 & -5 & 7 \end{array}\right].$$

Use $R_1 \leftarrow (\frac{1}{2})R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{3}{2} & -\frac{1}{2} \\ 6 & 5 & 0 \\ 2 & -5 & 7 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (6)R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{3}{2} & -\frac{1}{2} \\ 0 & -4 & 3 \\ 2 & -5 & 7 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (2)R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{3}{2} & -\frac{1}{2} \\ 0 & -4 & 3 \\ 0 & -8 & 8 \end{array}\right].$$

Use $R_2 \leftarrow (-\frac{1}{4})R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{3}{2} & -\frac{1}{2} \\ 0 & 1 & -\frac{3}{4} \\ 0 & -8 & 8 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-8)R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{3}{2} & -\frac{1}{2} \\ 0 & 1 & -\frac{3}{4} \\ 0 & 0 & 2 \end{array}\right].$$

Use $R_3 \leftarrow (\frac{1}{2})R_3$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{3}{2} & -\frac{1}{2} \\ 0 & 1 & -\frac{3}{4} \\ 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-\frac{1}{2})R_3$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{3}{2} & 0 \\ 0 & 1 & -\frac{3}{4} \\ 0 & 0 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-\frac{3}{4})R_3$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{3}{2} & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (\frac{3}{2})R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right].$$

**Conclusion.** A row with all variable coefficients zero and a nonzero constant asserts $0 = 1$ after scaling. This is impossible, so the system is inconsistent and its solution set is empty.

### Exercise 1.1.18: Do three planes have a common point?

**What we must show.** Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

**Plan.** Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 2 & 4 & 4 & 4 \\ 0 & 1 & -2 & -2 \\ 2 & 3 & 0 & 0 \end{array}\right].$$

Use $R_1 \leftarrow (\frac{1}{2})R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & 1 & -2 & -2 \\ 2 & 3 & 0 & 0 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (2)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & 1 & -2 & -2 \\ 0 & -1 & -4 & -4 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-1)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & 1 & -2 & -2 \\ 0 & 0 & -6 & -6 \end{array}\right].$$

Use $R_3 \leftarrow (-\frac{1}{6})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & 1 & -2 & -2 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (2)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 1 & -2 & -2 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-2)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (2)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

**Read the equations, keeping every free variable.**

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = 0 \\ x_2 = 0 \\ x_3 = 1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

**Conclusion.** The displayed values give the unique solution.

### Exercise 1.1.19: A forbidden value of the parameter

Replace row 2 by row 2 minus three times row 1:

$$\begin{bmatrix} 1 & h & 4 \\ 0 & 6-3h & -4 \end{bmatrix}.$$

If $h = 2$, the last equation is $0 = -4$, so there is no solution. If $h \neq 2$, the second variable is $-4/(6-3h)$ and the first is then determined by row 1. Thus a solution does exist in every other case.

**Conclusion.** $h \neq 2$.

### Exercise 1.1.20: A second forbidden parameter value

Use $R_2 \leftarrow R_2 - 2R_1$ to obtain $[0\ -8-2h \mid 16]$. If $h = -4$, this asserts $0 = 16$. For $h \neq -4$ the nonzero coefficient determines $x_2$, then row 1 determines $x_1$.

**Conclusion.** Consistency holds precisely for $h \neq -4$.

### Exercise 1.1.21: A parameter that never destroys consistency

After $R_2 \leftarrow R_2 - 3R_1$, the last row is $[0\ h-12 \mid 0]$. If $h \neq 12$ it says $x_2 = 0$ and then $x_1 = -2$. If $h = 12$ it says $0 = 0$, leaving $x_2$ free and $x_1 = -2 - 4x_2$. Neither case produces a contradiction.

**Conclusion.** Every real $h$ gives a consistent system.

### Exercise 1.1.22: Match a multiple equation

Use $R_1 \leftarrow R_1 + 2R_2$. Its result is $[0\ 0 \mid h-6]$. Thus consistency requires $h = 6$. Conversely, if $h = 6$, the first equation is $-2$ times the second, so solving $2x_1 - 6x_2 = -3$ supplies solutions of both.

**Conclusion.** $h = 6$.

### Exercise 1.1.23: Definitions and reversible operations

(a) True. Swapping twice restores the original rows. Scaling by $c \neq 0$ is reversed by scaling by $1/c$. Adding $cR_j$ to $R_i$ is reversed by adding $-cR_j$ to $R_i$.

(b) False. In an $m \times n$ matrix, $m$ counts rows and $n$ counts columns. A $5 \times 6$ matrix has five rows and six columns.

(c) False. The described list is *one solution*. The solution *set* contains all lists satisfying the system, and may be empty, a singleton, or infinite.

(d) True. Existence asks whether at least one solution exists; uniqueness asks whether exactly one exists. These are separate questions.

### Exercise 1.1.24: Row equivalence and consistency

(a) False. Equal row counts do not imply row equivalence. For example, [0 0] cannot be row reduced to [1 0]: all allowed operations keep a zero row zero.

(b) True. Each row operation replaces equations by reversible consequences, so it preserves the common solution set.

(c) False. Having identical solution sets is the definition of equivalent systems; different sets would contradict that definition.

(d) True. "Consistent" means that at least one solution exists.

### Exercise 1.1.25: Compatibility of dependent rows

The coefficient rows satisfy $r_3 = -2r_1 - r_2$. Make this relation explicit by $R_3 \leftarrow R_3 + 2R_1$ and then $R_3 \leftarrow R_3 + R_2$. The augmented last row becomes $[0\ 0\ 0 \mid k + 2g + h]$. Therefore consistency requires $k + 2g + h = 0$. If it holds, the remaining first two rows have nonzero pivots in columns 1 and 2, so they can be solved with $x_3$ free.

**Conclusion.** $2g + h + k = 0$ is necessary and sufficient.

### Exercise 1.1.26: Consistency for every right side

Replace row 2 by row 2 minus $c/2$ times row 1. This gives

$$[0\ d - 2c \mid g - cf/2].$$

If $d = 2c$, choose $f = 0, g = 1$. The resulting row asserts $0 = 1$, contradicting the assumption of consistency for *every* right side. Thus $d \neq 2c$. Conversely, that inequality supplies a second pivot for any $f, g$, so it also suffices.

**Conclusion.** $d \neq 2c$.

### Exercise 1.1.27: The condition on a general two-by-two system

Because $a \neq 0$, use $R_2 \leftarrow R_2 - (c/a)R_1$. The new row is

$$[0\ d - cb/a \mid g - cf/a].$$

Its coefficient cannot be zero: otherwise $f = 0, g = 1$ would make the system inconsistent. Multiplying $d - cb/a \neq 0$ by the nonzero $a$ gives $ad - bc \neq 0$. Conversely, this condition makes the second pivot nonzero for every right side.

**Conclusion.** $ad - bc \neq 0$. The condition states exactly that the two equations have independent left sides.

### Exercise 1.1.28: Construct three equivalent augmented matrices

Start with the equations $x_1 = 3$, $x_2 = -2$, $x_3 = -1$. Three possible augmented matrices are

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & -1 \end{array}\right], \qquad \left[\begin{array}{ccc|c} 2 & 0 & 0 & 6 \\ 0 & 3 & 0 & -6 \\ 0 & 0 & 4 & -4 \end{array}\right],$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & -3 \\ 0 & 0 & 1 & -1 \end{array}\right].$$

The second merely rescales equations by nonzero numbers. The third comes from adding row 2 to row 1 and then the original row 3 to row 2. Alternatively back substitution in the third gives $x_3 = -1$, $x_2 = -2$, $x_1 = 3$. Thus all three have exactly the requested single solution.

### Exercise 1.1.29: Reverse a row interchange

The first and third rows exchange places; the middle row stays fixed.

$$\text{Forward: } R_1 \leftrightarrow R_3, \qquad \text{reverse: } R_1 \leftrightarrow R_3.$$

The reverse operation cancels exactly the change made by the forward one; unchanged rows supply the same entries in both directions.

### Exercise 1.1.30: Reverse a row scaling

Each entry in $[0, -5, 10]$ is multiplied by $-1/5$, giving $[0, 1, -2]$.

$$\text{Forward: } R_3 \leftarrow -\tfrac{1}{5}R_3, \qquad \text{reverse: } R_3 \leftarrow -5R_3.$$

The reverse operation cancels exactly the change made by the forward one; unchanged rows supply the same entries in both directions.

### Exercise 1.1.31: Reverse a multiple-row subtraction

The subtraction is $[4, -1, 3, -6] - 4[1, -2, 1, 0] = [0, 7, -1, -6]$.

$$\text{Forward: } R_3 \leftarrow R_3 - 4R_1, \qquad \text{reverse: } R_3 \leftarrow R_3 + 4R_1.$$

The reverse operation cancels exactly the change made by the forward one; unchanged rows supply the same entries in both directions.

### Exercise 1.1.32: Reverse a second multiple-row subtraction

The subtraction is $[0, 4, -12, 7] - 4[0, 1, -3, -2] = [0, 0, 0, 15]$.

$$\text{Forward: } R_3 \leftarrow R_3 - 4R_2, \qquad \text{reverse: } R_3 \leftarrow R_3 + 4R_2.$$

The reverse operation cancels exactly the change made by the forward one; unchanged rows supply the same entries in both directions.

### Exercise 1.1.33: Set up the four-node temperature model

At each node, multiply "temperature equals the average of its four neighbors" by 4. The two fixed neighboring boundary values contribute to the constant; the two interior neighbors stay as unknowns. Reading the labels in the figure gives

$$\begin{aligned} 4T_1 - T_2 - T_4 &= 10 + 20 = 30, \\ -T_1 + 4T_2 - T_3 &= 20 + 40 = 60, \\ -T_2 + 4T_3 - T_4 &= 40 + 30 = 70, \\ -T_1 - T_3 + 4T_4 &= 10 + 30 = 40. \end{aligned}$$

**Conclusion.** These four equations are the required equilibrium model. The order of nodes around the square is $1, 2, 3, 4$, not a row-wise ordering.

### Exercise 1.1.34: Solve the temperature model

Use the equations from Exercise 33. Adding the first and third gives $4(T_1 + T_3) - 2(T_2 + T_4) = 100$. Adding the second and fourth gives $-2(T_1 + T_3) + 4(T_2 + T_4) = 100$. Let $S = T_1 + T_3$ and $U = T_2 + T_4$. Dividing both equations by 2 gives $2S - U = 50$ and $-S + 2U = 50$. Thus $S = U = 50$. Now substitute these pair sums into the original equations:

$$T_1 = (30 + U)/4 = 20, \quad T_2 = (60 + S)/4 = 55/2,$$

$$T_3 = (70 + U)/4 = 30, \quad T_4 = (40 + S)/4 = 45/2.$$

For example, the first average is $(10 + 20 + 55/2 + 45/2)/4 = 20$; the other three check in the same way by the four displayed equations. Every solution must have these pair sums and hence these values, so uniqueness is also established.

**Conclusion.** $T_1 = 20$, $T_2 = 27.5$, $T_3 = 30$, $T_4 = 22.5$ degrees.

## 1.2 Row reduction and echelon forms

A pivot is a leading nonzero entry of an echelon row. In echelon form, zero rows are at the bottom, pivots move strictly to the right as one moves downward, and all entries below a pivot are zero. In reduced echelon form (RREF), every pivot is 1 and is the only nonzero entry of its column. RREF is unique even though the sequence of operations leading to it is not.

For a system, pivot variable columns correspond to basic variables, and nonpivot variable columns correspond to free variables. *First test consistency*: a pivot in the augmented column means no solution. Only a consistent system with a free variable has infinitely many solutions.

### Practice problem 1.2.P1: Describe every solution with one parameter

**What we must show.** Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

**Plan.** Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & -3 & -5 & 0 \\ 0 & 1 & 1 & 3 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-3)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -2 & 9 \\ 0 & 1 & 1 & 3 \end{array}\right].$$

**Read the equations, keeping every free variable.**

The nonpivot variable is $x_3$. Set $x_3 = t_1$, where the parameters range independently over $\mathbb{R}$.

$$\begin{cases} x_1 = 2t_1 + 9 \\ x_2 = 3 - t_1 \\ x_3 = t_1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

**Conclusion.** Infinitely many solutions, exactly as parametrized above.

### Practice problem 1.2.P2: Elimination with two free variables

**What we must show.** Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

**Plan.** Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 0 \\ -2 & 4 & 5 & -5 & 3 \\ 3 & -6 & -6 & 8 & 2 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-2)R_1$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 0 \\ 0 & 0 & 3 & 1 & 3 \\ 3 & -6 & -6 & 8 & 2 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (3)R_1$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 0 \\ 0 & 0 & 3 & 1 & 3 \\ 0 & 0 & -3 & -1 & 2 \end{array}\right].$$

Use $R_2 \leftarrow (\frac{1}{3})R_2$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 0 \\ 0 & 0 & 1 & \frac{1}{3} & 1 \\ 0 & 0 & -3 & -1 & 2 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-3)R_2$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 0 \\ 0 & 0 & 1 & \frac{1}{3} & 1 \\ 0 & 0 & 0 & 0 & 5 \end{array}\right].$$

Use $R_3 \leftarrow (\frac{1}{5})R_3$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 0 \\ 0 & 0 & 1 & \frac{1}{3} & 1 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (1)R_3$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 0 \\ 0 & 0 & 1 & \frac{1}{3} & 0 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-1)R_2$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -2 & 0 & \frac{10}{3} & 0 \\ 0 & 0 & 1 & \frac{1}{3} & 0 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right].$$

**Conclusion.** A row with all variable coefficients zero and a nonzero constant asserts $0 = 1$ after scaling. This is impossible, so the system is inconsistent and its solution set is empty.

### Exercise 1.2.1: Distinguish echelon form from reduced form

(a) Echelon only. The first pivot is 10, not 1, so reduced form fails, although pivots move right and their columns are zero below them.

(b) Reduced echelon form. The pivots are 1 in columns 1, 2, 4, and these pivot columns have no other nonzero entries. The 1 in column 3 of the first row does not matter because column 3 is not a pivot column.

(c) Neither. Row 3 is zero but row 4 is nonzero; zero rows must be last.

(d) Echelon only. The successive pivots $1, 2, 3, 4$ move right, but three pivots are not 1, and pivot columns may have entries above them.

### Exercise 1.2.2: Check all echelon requirements

(a) Reduced echelon form: pivots in columns 1 and 2 are 1 and are the only nonzero entries in their columns; the zero row is last.

(b) Echelon only: the middle pivot is 2, so it is not reduced.

(c) Neither: the zero row is above nonzero rows.

(d) Echelon only: pivots in columns 2, 3, 5 move right and are 1, but there are nonzero entries above the latter pivots. A pivot being 1 by itself does not make a matrix reduced.

### Exercise 1.2.3: Locate two pivot positions

**What we must show.** Reduce the matrix itself; it is not being declared an augmented matrix.

$$\begin{bmatrix} 1 & 2 & 4 & 8 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 9 & 12 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - (2)R_1$ to obtain

$$\begin{bmatrix} 1 & 2 & 4 & 8 \\ 0 & 0 & -2 & -8 \\ 3 & 6 & 9 & 12 \end{bmatrix}.$$

Use $R_3 \leftarrow R_3 - (3)R_1$ to obtain

$$\begin{bmatrix} 1 & 2 & 4 & 8 \\ 0 & 0 & -2 & -8 \\ 0 & 0 & -3 & -12 \end{bmatrix}.$$

Use $R_2 \leftarrow (-\frac{1}{2})R_2$ to obtain

$$\begin{bmatrix} 1 & 2 & 4 & 8 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & -3 & -12 \end{bmatrix}.$$

Use $R_3 \leftarrow R_3 - (-3)R_2$ to obtain

$$\begin{bmatrix} 1 & 2 & 4 & 8 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Use $R_1 \leftarrow R_1 - (4)R_2$ to obtain

$$\begin{bmatrix} 1 & 2 & 0 & -8 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

**Conclusion.** The pivot positions are $(1,1)$ and $(2,3)$ in both the final matrix and the original matrix; the pivot columns are columns 1 and 3.

The circled entries mark those positions explicitly:

$$\begin{bmatrix} \textcircled{1} & 2 & 4 & 8 \\ 2 & 4 & \textcircled{6} & 8 \\ 3 & 6 & 9 & 12 \end{bmatrix} \longrightarrow \begin{bmatrix} \textcircled{1} & 2 & 0 & -8 \\ 0 & 0 & \textcircled{1} & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

A pivot position refers to a location, not necessarily the same numerical entry before and after reduction.

### Exercise 1.2.4: Locate three pivot positions

**What we must show.** Reduce the matrix itself; it is not being declared an augmented matrix.

$$\begin{bmatrix} 1 & 2 & 4 & 5 \\ 2 & 4 & 5 & 4 \\ 4 & 5 & 4 & 2 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - (2)R_1$ to obtain

$$\begin{bmatrix} 1 & 2 & 4 & 5 \\ 0 & 0 & -3 & -6 \\ 4 & 5 & 4 & 2 \end{bmatrix}.$$

Use $R_3 \leftarrow R_3 - (4)R_1$ to obtain

$$\begin{bmatrix} 1 & 2 & 4 & 5 \\ 0 & 0 & -3 & -6 \\ 0 & -3 & -12 & -18 \end{bmatrix}.$$

Use $R_2 \leftrightarrow R_3$ to obtain

$$\begin{bmatrix} 1 & 2 & 4 & 5 \\ 0 & -3 & -12 & -18 \\ 0 & 0 & -3 & -6 \end{bmatrix}.$$

Use $R_2 \leftarrow (-\frac{1}{3})R_2$ to obtain

$$\begin{bmatrix} 1 & 2 & 4 & 5 \\ 0 & 1 & 4 & 6 \\ 0 & 0 & -3 & -6 \end{bmatrix}.$$

Use $R_3 \leftarrow (-\frac{1}{3})R_3$ to obtain

$$\begin{bmatrix} 1 & 2 & 4 & 5 \\ 0 & 1 & 4 & 6 \\ 0 & 0 & 1 & 2 \end{bmatrix}.$$

Use $R_1 \leftarrow R_1 - (4)R_3$ to obtain

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ 0 & 1 & 4 & 6 \\ 0 & 0 & 1 & 2 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - (4)R_3$ to obtain

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 2 \end{bmatrix}.$$

Use $R_1 \leftarrow R_1 - (2)R_2$ to obtain

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 2 \end{bmatrix}.$$

**Conclusion.** The pivot positions are $(1,1)$, $(2,2)$, and $(3,3)$ in both the final matrix and the original matrix; the pivot columns are columns 1, 2, and 3.

The original and reduced pivot positions can be circled as follows:

$$\begin{bmatrix} \textcircled{1} & 2 & 4 & 5 \\ 2 & \textcircled{4} & 5 & 4 \\ 4 & 5 & \textcircled{4} & 2 \end{bmatrix} \longrightarrow \begin{bmatrix} \textcircled{1} & 0 & 0 & 1 \\ 0 & \textcircled{1} & 0 & -2 \\ 0 & 0 & \textcircled{1} & 2 \end{bmatrix}.$$

### Exercise 1.2.5: All nonzero two-by-two echelon patterns

Let ■ mean a nonzero pivot and $*$ an arbitrary entry. A nonzero $2 \times 2$ matrix has either one or two nonzero echelon rows. With one nonzero row, its pivot is in column 1 or column 2. With two, pivots must be in columns 1 then 2. Thus all possibilities are

$$\begin{bmatrix} ■ & * \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & ■ \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} ■ & * \\ 0 & ■ \end{bmatrix}.$$

There are no other choices: a pivot cannot occur to the left of the pivot above it, and zero rows cannot precede nonzero rows.

### Exercise 1.2.6: All nonzero three-by-two echelon patterns

There can be at most two pivots because there are only two columns. As in Exercise 5, either the sole pivot is in column 1, the sole pivot is in column 2, or the two pivots occupy columns 1 and 2 in succession:

$$\begin{bmatrix} ■ & * \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & ■ \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} ■ & * \\ 0 & ■ \\ 0 & 0 \end{bmatrix}.$$

The third row must be zero in every case, and nonzeroness excludes the all-zero matrix. These exhaust the echelon possibilities.

### Exercise 1.2.7: Two pivots and one free variable

**What we must show.** Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

**Plan.** Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & 3 & 4 & 7 \\ 3 & 9 & 7 & 6 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (3)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 3 & 4 & 7 \\ 0 & 0 & -5 & -15 \end{array}\right].$$

Use $R_2 \leftarrow (-\frac{1}{5})R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 3 & 4 & 7 \\ 0 & 0 & 1 & 3 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (4)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 3 & 0 & -5 \\ 0 & 0 & 1 & 3 \end{array}\right].$$

**Read the equations, keeping every free variable.**

The nonpivot variable is $x_2$. Set $x_2 = t_1$, where the parameters range independently over $\mathbb{R}$.

$$\begin{cases} x_1 = -3t_1 - 5 \\ x_2 = t_1 \\ x_3 = 3 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

**Conclusion.** Infinitely many solutions, exactly as parametrized above.

### Exercise 1.2.8: A variable absent from both equations

**What we must show.** Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

**Plan.** Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & -3 & 0 & -5 \\ -3 & 7 & 0 & 9 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-3)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -3 & 0 & -5 \\ 0 & -2 & 0 & -6 \end{array}\right].$$

Use $R_2 \leftarrow (-\frac{1}{2})R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -3 & 0 & -5 \\ 0 & 1 & 0 & 3 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-3)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 3 \end{array}\right].$$

**Read the equations, keeping every free variable.**

The nonpivot variable is $x_3$. Set $x_3 = t_1$, where the parameters range independently over $\mathbb{R}$.

$$\begin{cases} x_1 = 4 \\ x_2 = 3 \\ x_3 = t_1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

**Conclusion.** Infinitely many solutions, exactly as parametrized above.

### Exercise 1.2.9: Interchange before eliminating

**What we must show.** Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

**Plan.** Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 0 & 1 & -2 & 3 \\ 1 & -3 & 4 & -6 \end{array}\right].$$

Use $R_1 \leftrightarrow R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -3 & 4 & -6 \\ 0 & 1 & -2 & 3 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-3)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -2 & 3 \\ 0 & 1 & -2 & 3 \end{array}\right].$$

**Read the equations, keeping every free variable.**

The nonpivot variable is $x_3$. Set $x_3 = t_1$, where the parameters range independently over $\mathbb{R}$.

$$\begin{cases} x_1 = 2t_1 + 3 \\ x_2 = 2t_1 + 3 \\ x_3 = t_1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

**Conclusion.** Infinitely many solutions, exactly as parametrized above.

### Exercise 1.2.10: A nonzero second pivot after elimination

**What we must show.** Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

**Plan.** Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & -2 & -1 & 4 \\ -2 & 4 & -5 & 6 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-2)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -2 & -1 & 4 \\ 0 & 0 & -7 & 14 \end{array}\right].$$

Use $R_2 \leftarrow (-\frac{1}{7})R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -2 & -1 & 4 \\ 0 & 0 & 1 & -2 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-1)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -2 & 0 & 2 \\ 0 & 0 & 1 & -2 \end{array}\right].$$

**Read the equations, keeping every free variable.**

The nonpivot variable is $x_2$. Set $x_2 = t_1$, where the parameters range independently over $\mathbb{R}$.

$$\begin{cases} x_1 = 2t_1 + 2 \\ x_2 = t_1 \\ x_3 = -2 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

**Conclusion.** Infinitely many solutions, exactly as parametrized above.

### Exercise 1.2.11: Repeated equations leave two free variables

**What we must show.** Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

**Plan.** Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 3 & -2 & 4 & 0 \\ 9 & -6 & 12 & 0 \\ 6 & -4 & 8 & 0 \end{array}\right].$$

Use $R_1 \leftarrow (\frac{1}{3})R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -\frac{2}{3} & \frac{4}{3} & 0 \\ 9 & -6 & 12 & 0 \\ 6 & -4 & 8 & 0 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (9)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -\frac{2}{3} & \frac{4}{3} & 0 \\ 0 & 0 & 0 & 0 \\ 6 & -4 & 8 & 0 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (6)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -\frac{2}{3} & \frac{4}{3} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

**Read the equations, keeping every free variable.**

The nonpivot variables are $x_2$, $x_3$. Set $x_2 = t_1$, $x_3 = t_2$, where the parameters range independently over $\mathbb{R}$.

$$\begin{cases} x_1 = \frac{2t_1}{3} - \frac{4t_2}{3} \\ x_2 = t_1 \\ x_3 = t_2 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

**Conclusion.** Infinitely many solutions, exactly as parametrized above.

### Exercise 1.2.12: An augmented-column pivot rules out solutions

**What we must show.** Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

**Plan.** Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cccc|c} 1 & 0 & -9 & 0 & 4 \\ 0 & 1 & 3 & 0 & -1 \\ 0 & 0 & 0 & 1 & -7 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (4)R_4$ to obtain

$$\left[\begin{array}{cccc|c} 1 & 0 & -9 & 0 & 0 \\ 0 & 1 & 3 & 0 & -1 \\ 0 & 0 & 0 & 1 & -7 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-1)R_4$ to obtain

$$\left[\begin{array}{cccc|c} 1 & 0 & -9 & 0 & 0 \\ 0 & 1 & 3 & 0 & 0 \\ 0 & 0 & 0 & 1 & -7 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-7)R_4$ to obtain

$$\left[\begin{array}{cccc|c} 1 & 0 & -9 & 0 & 0 \\ 0 & 1 & 3 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right].$$

**Conclusion.** A row with all variable coefficients zero and a nonzero constant asserts $0 = 1$ after scaling. This is impossible, so the system is inconsistent and its solution set is empty.

### Exercise 1.2.13: Five variables and two free choices

**What we must show.** Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

**Plan.** Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccccc|c} 1 & -3 & 0 & -1 & 0 & -2 \\ 0 & 1 & 0 & 0 & -4 & 1 \\ 0 & 0 & 0 & 1 & 9 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-1)R_3$ to obtain

$$\left[\begin{array}{rrrrr|r} 1 & -3 & 0 & 0 & 9 & 2 \\ 0 & 1 & 0 & 0 & -4 & 1 \\ 0 & 0 & 0 & 1 & 9 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-3)R_2$ to obtain

$$\left[\begin{array}{rrrrr|r} 1 & 0 & 0 & 0 & -3 & 5 \\ 0 & 1 & 0 & 0 & -4 & 1 \\ 0 & 0 & 0 & 1 & 9 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

**Read the equations, keeping every free variable.**

The nonpivot variables are $x_3$, $x_5$. Set $x_3 = t_1$, $x_5 = t_2$, where the parameters range independently over $\mathbb{R}$.

$$\begin{cases} x_1 = 3t_2 + 5 \\ x_2 = 4t_2 + 1 \\ x_3 = t_1 \\ x_4 = 4 - 9t_2 \\ x_5 = t_2 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

**Conclusion.** Infinitely many solutions, exactly as parametrized above.

### Exercise 1.2.14: A zero basic variable simplifies the parameters

**What we must show.** Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

**Plan.** Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{rrrrr|r} 1 & 0 & -5 & 0 & -8 & 3 \\ 0 & 1 & 4 & -1 & 0 & 6 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-8)R_3$ to obtain

$$\left[\begin{array}{rrrrr|r} 1 & 0 & -5 & 0 & 0 & 3 \\ 0 & 1 & 4 & -1 & 0 & 6 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

**Read the equations, keeping every free variable.**

The nonpivot variables are $x_3$, $x_4$. Set $x_3 = t_1$, $x_4 = t_2$, where the parameters range independently over $\mathbb{R}$.

$$\begin{cases} x_1 = 5t_1 + 3 \\ x_2 = -4t_1 + t_2 + 6 \\ x_3 = t_1 \\ x_4 = t_2 \\ x_5 = 0 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

**Conclusion.** Infinitely many solutions, exactly as parametrized above.

### Exercise 1.2.15: Read two symbolic echelon patterns

#### (a) Two pivots among three variable columns.

The last row is zero, and the augmented column is not a pivot column. Thus the system is consistent. Columns 1 and 2 are pivot columns; column 3 supplies one free variable. Every choice of that variable allows back substitution, so there are infinitely many solutions.

#### (b) Three pivots among four variable columns.

The pivot columns are 2, 3, 4; column 5 is the augmented column and has no pivot. Thus the system is consistent. The first variable is free, so the solution is not unique: there are infinitely many solutions.

**Conclusion.** Both systems are consistent and neither has a unique solution.

### Exercise 1.2.16: Contrast variable pivots with a constant pivot

#### (a) A pivot in the constant column.

There are three columns total, so column 3 is augmented. The last row is $[0\ 0 \mid \blacksquare]$, where $\blacksquare \neq 0$. It asserts $0 = \blacksquare$, a contradiction. The system is inconsistent.

#### (b) No pivot in the constant column.

Column 5 is augmented and has no pivot; the pivots are in columns 1, 3, 4. The system is consistent, and variable 2 is free. It consequently has infinitely many solutions.

**Conclusion.** Part (a) has no solution; part (b) is consistent but not unique.

### Exercise 1.2.17: A parameter in the constant column

Eliminate the first coefficient in row 2 by $R_2 \leftarrow R_2 + 2R_1$. The last row becomes $[0\ 1 \mid h + 8]$. Its variable coefficient is 1 for every $h$, so it never produces a contradiction. Set $x_2 = h + 8$ and then $x_1 = 4 + x_2 = h + 12$.

**Conclusion.** The system is consistent for every real $h$.

### Exercise 1.2.18: A parameter that can remove one equation

Use $R_2 \leftarrow R_2 - hR_1$. Its result is $[0\ 6+3h \mid -2-h]$. If $h \neq -2$, the variable coefficient is nonzero and gives a solution by back substitution. If $h = -2$, both entries are zero, so the second row is merely $0 = 0$; solutions still exist.

**Conclusion.** All real $h$ work. At $h = -2$ the solutions are not unique; at every other $h$ there is a unique solution.

### Exercise 1.2.19: Choose parameters for all three outcomes

Elimination gives

$$\begin{bmatrix} 1 & h & 2 \\ 0 & 8-4h & k-8 \end{bmatrix}.$$

If $h \neq 2$, the second coefficient is nonzero, giving a unique solution for every $k$. If $h = 2$, the last equation is $0 = k - 8$. It is impossible for $k \neq 8$ and imposes no constraint for $k = 8$, leaving $x_2$ free.

**Conclusion.** (a) No solution: $h = 2, k \neq 8$, for example $(2, 0)$. (b) Unique: $h \neq 2$, any $k$, for example $(0, 0)$. (c) Infinitely many: $h = 2, k = 8$.

### Exercise 1.2.20: A second three-outcome parameter family

Replace row 2 by row 2 minus twice row 1. The last row is $[0\ h+6 \mid k-2]$. If $h \neq -6$ it determines $x_2$ uniquely. If $h = -6$, it is compatible precisely when $k = 2$, and then $x_2$ is unrestricted.

**Conclusion.** (a) No solution: $h = -6, k \neq 2$, for example $(-6, 0)$. (b) Unique: $h \neq -6$, any $k$, for example $(0, 0)$. (c) Infinitely many: $h = -6, k = 2$.

### Exercise 1.2.21: Definitions of basic variables and solutions

(a) False. The uniqueness-of-RREF theorem says every matrix has exactly one reduced echelon form; different operation sequences give that same matrix.

(b) False. Row reduction applies to any rectangular matrix, whether or not its last column represents constants in a system.

(c) True. A basic variable corresponds by definition to a pivot column of the coefficient matrix.

(d) True. Solving a system means describing all its solutions. A parametric description does this while recording the free choices.

(e) False. In $[0\ 0\ 0\ 5 \mid 0]$ the nonzero entry is in a variable column: the equation is $5x_4 = 0$, which is satisfiable.

### Exercise 1.2.22: Uniqueness of RREF and consistency cautions

(a) True. This is precisely the uniqueness theorem for RREF.

(b) False. A pivot in *every* augmented-matrix column includes a pivot in the final column, implying an impossible equation. For example [0 | 1] already exhibits the obstruction (and an identity augmented matrix gives examples with every column a pivot column).

(c) False. Pivot positions are determined from the unique RREF, not by the particular path of row operations.

(d) True. A general solution explicitly describes all solutions, including the parameters and their allowed values.

(e) False. Consistency is also required. The system $0x_1 + 0x_2 = 1$ has nonpivot variable columns but no solutions.

### Exercise 1.2.23: Four coefficient pivots force uniqueness

There are four variable columns and all four are pivot columns. In echelon form, the four pivots occupy all four rows, so the final augmented column cannot contain an additional pivot. Hence there is no contradiction row and the system is consistent. Since all variables are basic, no free choice remains; back substitution determines exactly one value for each variable.

**Conclusion.** The system has a unique solution.

### Exercise 1.2.24: No pivot in the last augmented column

The fifth column is the constant column. By hypothesis it is not a pivot column. Thus RREF has no row $[0\ 0\ 0\ 0 \mid d]$ with $d \neq 0$, which is exactly the consistency criterion.

**Conclusion.** Yes, the system is consistent. It cannot have a unique solution: four variable columns but only three rows leave at least one free variable.

### Exercise 1.2.25: A pivot in every coefficient row

A pivot in every coefficient-matrix row uses all available rows. Appending a right side cannot change the coefficient entries or create an additional row in which all coefficients vanish but the constant is nonzero. More concretely, apply to the right side the same row operations that put the coefficient matrix in echelon form. Every resulting row has a variable pivot, so no resulting row can assert $0 = d \neq 0$.

**Conclusion.** The system is consistent for every possible right side.

### Exercise 1.2.26: Three pivots in a three-by-five coefficient matrix

There are three rows and three pivot columns. Different pivots occupy different rows, so there is a pivot in every coefficient row. By the argument in Exercise 25, there can be no contradiction row for any right side. There are five variables but only three pivots, so two variables are free.

**Conclusion.** The system is consistent and has infinitely many solutions.

### Exercise 1.2.27: Restate uniqueness in pivot language

The exact completion is: "every column of the coefficient matrix is a pivot column." A nonpivot variable column is equivalent to a free variable. In a consistent system a free variable gives infinitely many solutions; without any free variables, all variables are fixed.

**Conclusion.** For a consistent system, uniqueness is equivalent to a pivot in every variable column.

### Exercise 1.2.28: Combine existence and uniqueness conditions

Two facts are needed. The last (augmented) column must *not* be a pivot column, ensuring consistency. Every preceding (coefficient) column must be a pivot column, ensuring that there are no free variables. Conversely those two facts give existence and uniqueness by Theorem 2.

**Conclusion.** Pivots in every coefficient column, and no pivot in the augmented column. Neither requirement alone is sufficient.

### Exercise 1.2.29: Why underdetermined systems cannot be unique

If there are $m$ equations and $n$ unknowns with $m < n$, there can be at most $m$ pivots. Thus at least $n - m > 0$ variable columns are nonpivot. If the system is consistent, these give free variables and infinitely many solutions. If it is inconsistent, it has no solutions. Neither case has exactly one solution.

**Conclusion.** No underdetermined linear system can have a unique solution.

### Exercise 1.2.30: An inconsistent underdetermined example

For example,

$$x_1 + x_2 + x_3 = 0, \qquad x_1 + x_2 + x_3 = 1.$$

There are only two equations for three unknowns. Subtracting the first from the second gives $0 = 1$, so the system is inconsistent.

**Conclusion.** Underdetermined does not mean consistent.

### Exercise 1.2.31: A consistent overdetermined example

Yes. Consider

$$x_1 = 1, \qquad x_2 = 2, \qquad x_1 + x_2 = 3.$$

The pair $(1, 2)$ satisfies every equation, so this three-equation, two-variable system is consistent. The third equation is redundant: it is the sum of the first two, rather than an incompatible additional constraint.

**Conclusion.** An overdetermined system can be consistent; equation counts alone do not decide consistency.

### Exercise 1.2.32: Estimate the backward share of reduction cost

The numerical note estimates the forward cost by $F_n = 2n^3/3 + n^2/2 - 7n/6$ and bounds the backward cost by $B_n = n^2$. Using the leading forward term, the approximate backward fraction of *total* cost is

$$\frac{n^2}{(2/3)n^3 + n^2} = \frac{3}{2n+3}.$$

At $n = 20$ this is $3/43 \approx 0.06977$, about 7%; at $n = 200$ it is $3/403 \approx 0.007444$, about 0.74%. Using the full stated forward count instead gives $400/(5510 + 400) \approx 6.77\%$ and $40000/(5353100 + 40000) \approx 0.742\%$. The cost of the backward phase is stated as an upper bound, so these are cost-model estimates, not an exact count for every input matrix.

**Conclusion.** About 7% for $n = 20$ and 0.74% for $n = 200$.

### Exercise 1.2.33: Interpolate three points with a quadratic

The coefficient vector is $(a_0, a_1, a_2)$. Its augmented matrix is

$$\begin{bmatrix} 1 & 1 & 1 & 6 \\ 1 & 2 & 4 & 15 \\ 1 & 3 & 9 & 28 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - R_1$ and $R_3 \leftarrow R_3 - R_1$ to get rows [0 1 3 | 9] and [0 2 8 | 22] below the first. Then $R_3 \leftarrow R_3 - 2R_2$ gives [0 0 2 | 4]. Thus $a_2 = 2$, $a_1 = 9 - 3(2) = 3$, and $a_0 = 6 - 3 - 2 = 1$.

$$p(t) = 1 + 3t + 2t^2.$$

Check: $p(1) = 6$, $p(2) = 15$, and $p(3) = 28$.

**Conclusion.** The unique interpolating quadratic is $1 + 3t + 2t^2$.

### Exercise 1.2.34: Interpolate wind-tunnel measurements

**What we must show.** Interpolate all six measured data points and convert the requested velocity into the table's units.

**Set up six equations for six coefficients.**

For each listed pair $(t, F)$, use $a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5 = F$. The coefficient matrix has rows $[1, t, t^2, t^3, t^4, t^5]$ for $t = 0, 2, 4, 6, 8, 10$; the right side is $[0, 29/10, 74/5, 198/5, 743/10, 119]^T$. Treating the printed decimals as the given exact data, elimination gives

$$\begin{aligned} a_0 &= 0, & a_1 &= 137/80, & a_2 &= -1147/960, \\ a_3 &= 127/192, & a_4 &= -269/3840, & a_5 &= 1/384. \end{aligned}$$

Thus

$$p(t) = \frac{137t}{80} - \frac{1147t^2}{960} + \frac{127t^3}{192} - \frac{269t^4}{3840} + \frac{t^5}{384}.$$

Substitution at the six inputs gives exactly the six measured outputs, so the computed coefficients satisfy the full system. They are unique: a difference of two degree-at-most-five interpolants would have six distinct zeros and hence would be the zero polynomial.

**Use the measurement units.**

The velocity 750 ft/sec corresponds to $t = 7.5$, not $t = 750$. The force is tabulated in hundreds of pounds. Consequently

$$p(7.5) = \frac{132789}{2048} \approx 64.83838, \qquad \text{force} \approx 6483.84 \text{ lb}.$$

**Explain why lowering the degree fails.**

The unique degree-at-most-five interpolant has nonzero $t^5$ coefficient $1/384$. Thus no polynomial of degree less than five can pass through all six data points. For a direct cubic check, fitting the first four points gives

$$q(t) = \frac{13t^3}{160} + \frac{51t^2}{80} - \frac{3t}{20}.$$

But $q(8) = 81.2 \neq 74.3$ and $q(10) = 143.5 \neq 119$. The six-equation cubic system is inconsistent, rather than merely less accurate.

**Conclusion.** The displayed quintic interpolates the data; its prediction at 750 ft/sec is approximately 6484 lb.

## 1.3 Vector equations

A vector is an ordered list; its coordinates are added and multiplied by scalars coordinate by coordinate. We display vectors as columns unless explicitly discussing points. A linear combination $c_1v_1+\cdots+c_kv_k$ uses scalar weights $c_i$. The span is the *set of all* combinations, not merely the finite list of generating vectors.

To test whether $b$ is in a span, put the generators into the columns of a matrix and put $b$ in the augmented column. Consistency of that system means the needed weights exist. An inconsistent system means $b$ is outside the span. Neither the weights nor a solution should be guessed from a picture when exact coordinates are supplied.

### Practice problem 1.3.P1: Commutativity from the coordinate definition

Take arbitrary $u = (u_1, \ldots, u_n)$ and $v = (v_1, \ldots, v_n)$. Their $i$th sum coordinate is $u_i+v_i$. Real-number commutativity gives $u_i + v_i = v_i + u_i$ for each $i$. Thus

$$u + v = (u_1 + v_1, \ldots, u_n + v_n) = (v_1 + u_1, \ldots, v_n + u_n) = v + u.$$

**Conclusion.** The vector identity holds because all corresponding coordinates are equal, not because the vector symbols can be rearranged without reason.

### Practice problem 1.3.P2: Find the height that lies in a span

Seek weights $x_1, x_2, x_3$ and reduce

$$\left[\begin{array}{rrr|r} 1 & 5 & -3 & -4 \\ -1 & -4 & 1 & 3 \\ -2 & -7 & 0 & h \end{array}\right].$$

Use $R_2 \leftarrow R_2 + R_1$ and $R_3 \leftarrow R_3 + 2R_1$:

$$\left[\begin{array}{rrr|r} 1 & 5 & -3 & -4 \\ 0 & 1 & -2 & -1 \\ 0 & 3 & -6 & h-8 \end{array}\right].$$

Now $R_3 \leftarrow R_3 - 3R_2$ gives $[0\ 0\ 0 \mid h - 5]$. Consequently consistency requires $h = 5$. If $h = 5$, choose $x_3 = 0$; row 2 gives $x_2 = -1$, and row 1 gives $x_1 = 1$. Indeed $v_1 - v_2 = (-4, 3, 5)^T$.

**Conclusion.** Exactly $h = 5$ works.

### Exercise 1.3.1: Add vectors and subtract twice a vector

Add coordinates in the same positions, and first double $v$ for the second expression:

$$u + v = \begin{bmatrix} -1 + (-3) \\ 2 + (-1) \end{bmatrix} = \begin{bmatrix} -4 \\ 1 \end{bmatrix}, \qquad u - 2v = \begin{bmatrix} -1 - 2(-3) \\ 2 - 2(-1) \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}.$$

**Conclusion.** $u + v = (-4, 1)^T$ and $u - 2v = (5, 4)^T$.

### Exercise 1.3.2: A second coordinate calculation

Coordinatewise addition and scalar multiplication give

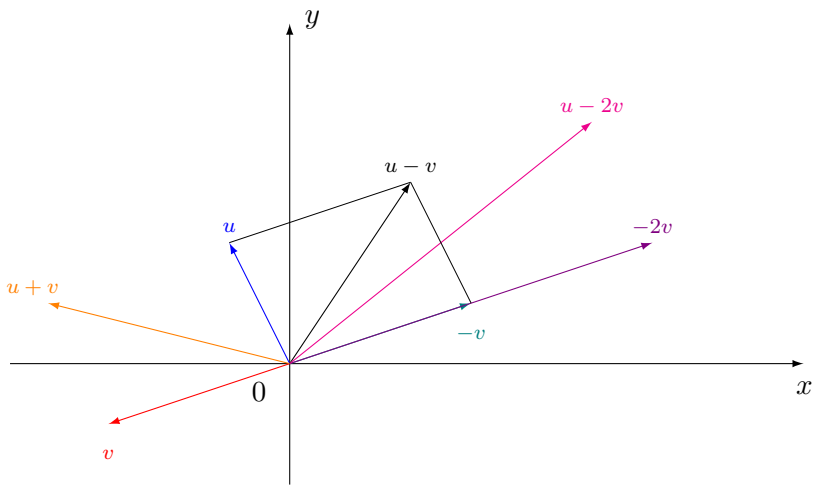
$$u+v=\begin{bmatrix}3+2\\2-1\end{bmatrix}=\begin{bmatrix}5\\1\end{bmatrix},\qquad u-2v=\begin{bmatrix}3-4\\2-2(-1)\end{bmatrix}=\begin{bmatrix}-1\\4\end{bmatrix}.$$

**Conclusion.** $u+v=(5,1)^T$ and $u-2v=(-1,4)^T$.

### Exercise 1.3.3: Seven vectors and their parallelogram

Compute the coordinates before drawing. Every arrow below begins at the origin and ends at its vector coordinates.

| vector | $x$ | $y$ |
|---|---|---|
| $u$ | $-1$ | $2$ |
| $v$ | $-3$ | $-1$ |
| $-v$ | $3$ | $1$ |
| $-2v$ | $6$ | $2$ |
| $u+v$ | $-4$ | $1$ |
| $u-v$ | $2$ | $3$ |
| $u-2v$ | $5$ | $4$ |

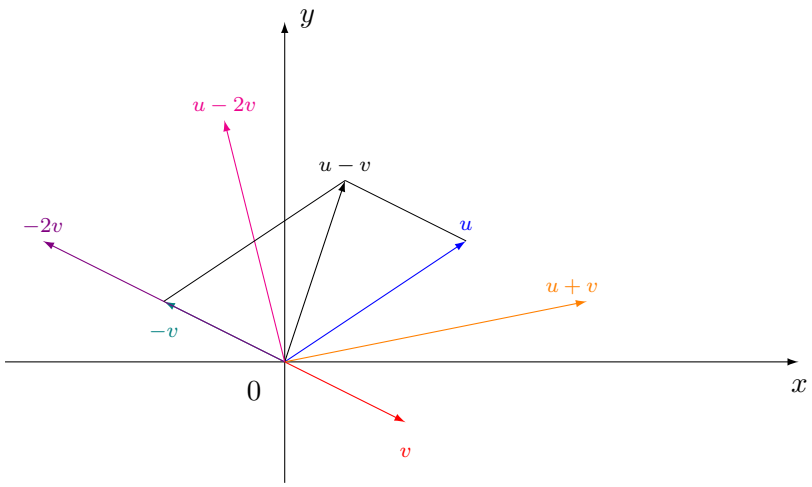



The thin extra sides join $u$ to $u-v$ and $-v$ to $u-v$. They complete the parallelogram with vertices $0, u, u-v, -v$. The coordinate table fixes the exact endpoints even where arrows cross.

### Exercise 1.3.4: A second arrow diagram

Compute the coordinates before drawing. Every arrow below begins at the origin and ends at its vector coordinates.

| vector | $x$ | $y$ |
|---|---|---|
| $u$ | 3 | 2 |
| $v$ | 2 | $-1$ |
| $-v$ | $-2$ | 1 |
| $-2v$ | $-4$ | 2 |
| $u+v$ | 5 | 1 |
| $u-v$ | 1 | 3 |
| $u-2v$ | $-1$ | 4 |




The thin extra sides join $u$ to $u-v$ and $-v$ to $u-v$. They complete the parallelogram with vertices $0, u, u-v, -v$. The coordinate table fixes the exact endpoints even where arrows cross.

### Exercise 1.3.5: Read a vector equation coordinate by coordinate

The left side has coordinates $(3x_1+5x_2, -2x_1, 8x_1-9x_2)$. Two vectors are equal exactly when each corresponding coordinate is equal. Thus the equivalent system is

$$3x_1+5x_2=2, \qquad -2x_1=-3, \qquad 8x_1-9x_2=8.$$

**Conclusion.** These three scalar equations express precisely the given vector equation.

### Exercise 1.3.6: Three weights but two coordinate equations

Multiply each column by its weight, add the two coordinates separately, and compare with the zero vector:

$$3x_1+7x_2-2x_3=0, \qquad -2x_1+3x_2+x_3=0.$$

**Conclusion.** The two scalar equations are equivalent to the vector equation; there are two equations even though there are three unknown weights.

### Exercise 1.3.7: Read four vectors on an oblique grid

The grid consists of lines parallel to $u$ and lines parallel to $v$. Following a line parallel to $u$ changes only the $u$ coefficient; following a line parallel to $v$ changes only the $v$ coefficient. Use the marked $u, v, -u, -v, -2v, 2v$ to fix unit steps. Counting those steps, including the halfway position of $c$, gives

$$a = u - 2v, \qquad b = 2u - 2v, \qquad c = 2u - \frac{7}{2}v, \qquad d = 3u - 4v.$$

For example, the line through $a$ parallel to $u$ meets the $v$-axis at $-2v$, and moving one $u$ step from there reaches $a$. The vectors $u, v$ point in distinct directions, so their two axes are not parallel. Through any plane point draw the corresponding two grid lines; they give one coefficient along each axis.

**Conclusion.** The displayed combinations describe $a, b, c, d$. Yes, every vector in $\mathbb{R}^2$ is a linear combination of these two nonparallel vectors.

### Exercise 1.3.8: Read four more vectors on the grid

Count displacements on the same two oblique axes as in Exercise 7. The marked points give

$$w = -u + 2v, \qquad x = -2u + 2v, \qquad y = -2u + \frac{7}{2}v, \qquad z = -3u + 4v.$$

For instance, starting at $2v$ and moving one step in the $-u$ direction reaches $w$; moving one more step reaches $x$. The halfway placement of $y$ in the $v$ direction gives coefficient $7/2$, not 3 or 4. As the two generating directions are nonparallel, their grid extends across the whole plane.

**Conclusion.** The four combinations are as displayed, and the span is all of $\mathbb{R}^2$.

### Exercise 1.3.9: Collect coefficients into three columns

Read down the coefficients of each variable, including its zero entry:

$$x_1 \begin{bmatrix} 0 \\ 4 \\ -1 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 6 \\ 3 \end{bmatrix} + x_3 \begin{bmatrix} 5 \\ -1 \\ -8 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The first coordinate reproduces $x_2 + 5x_3 = 0$, the second reproduces $4x_1 + 6x_2 - x_3 = 0$, and the third reproduces the last equation. Thus no equation or coefficient has been omitted.

### Exercise 1.3.10: Encode a nonhomogeneous system as a vector equation

Collect the coefficients down the three equations:

$$x_1 \begin{bmatrix} 3 \\ -2 \\ 5 \end{bmatrix} + x_2 \begin{bmatrix} -2 \\ -7 \\ 4 \end{bmatrix} + x_3 \begin{bmatrix} 4 \\ 5 \\ -3 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}.$$

Equality of each coordinate is exactly the corresponding original scalar equation, so the vector equation is equivalent to the system.

### Exercise 1.3.11: A span test with a selected-answer discrepancy

**What we must show.** Decide whether weights exist, rather than whether $b$ equals one of the columns. Form $[a_1\ a_2\ a_3 \mid b]$.

**What we must show.** Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

**Plan.** Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ -2 & 1 & -6 & -1 \\ 0 & 2 & 8 & 6 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-2)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ 0 & 1 & 4 & 3 \\ 0 & 2 & 8 & 6 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (2)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ 0 & 1 & 4 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

**Read the equations, keeping every free variable.**

The nonpivot variable is $x_3$. Set $x_3 = t_1$, where the parameters range independently over $\mathbb{R}$.

$$\begin{cases} x_1 = 2 - 5t_1 \\ x_2 = 3 - 4t_1 \\ x_3 = t_1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

**Conclusion.** Infinitely many solutions, exactly as parametrized above.

For a concrete combination take $x_3 = 0$, giving $b = 2a_1 + 3a_2$. Therefore $b$ belongs to the span. The selected-answer entry for Exercise 11 in the supplied PDF says "No, but the explicit coordinate check $2(1, -2, 0)^T + 3(0, 1, 2)^T = (2, -1, 6)^T$ shows that entry is inconsistent with the printed data.

### Exercise 1.3.12: A span test with unique weights

**What we must show.** Decide whether weights exist, rather than whether $b$ equals one of the columns. Form $[a_1\ a_2\ a_3 \mid b]$.

**What we must show.** Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

**Plan.** Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & -2 & -6 & 11 \\ 0 & 3 & 7 & -5 \\ 1 & -2 & 5 & 9 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (1)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -2 & -6 & 11 \\ 0 & 3 & 7 & -5 \\ 0 & 0 & 11 & -2 \end{array}\right].$$

Use $R_2 \leftarrow (\frac{1}{3})R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -2 & -6 & 11 \\ 0 & 1 & \frac{7}{3} & -\frac{5}{3} \\ 0 & 0 & 11 & -2 \end{array}\right].$$

Use $R_3 \leftarrow (\frac{1}{11})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -2 & -6 & 11 \\ 0 & 1 & \frac{7}{3} & -\frac{5}{3} \\ 0 & 0 & 1 & -\frac{2}{11} \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-6)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -2 & 0 & \frac{109}{11} \\ 0 & 1 & \frac{7}{3} & -\frac{5}{3} \\ 0 & 0 & 1 & -\frac{2}{11} \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (\frac{7}{3})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -2 & 0 & \frac{109}{11} \\ 0 & 1 & 0 & -\frac{41}{33} \\ 0 & 0 & 1 & -\frac{2}{11} \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-2)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & \frac{245}{33} \\ 0 & 1 & 0 & -\frac{41}{33} \\ 0 & 0 & 1 & -\frac{2}{11} \end{array}\right].$$

**Read the equations, keeping every free variable.**

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = \frac{245}{33} \\ x_2 = -\frac{41}{33} \\ x_3 = -\frac{2}{11} \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

**Conclusion.** The displayed values give the unique solution.

Thus $b = (245/33)a_1 - (41/33)a_2 - (2/11)a_3$, so it is a linear combination.

### Exercise 1.3.13: An inconsistent column-span test

**What we must show.** Decide whether weights exist, rather than whether $b$ equals one of the columns. Form $[a_1\ a_2\ a_3 \mid b]$.

**What we must show.** Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

**Plan.** Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & -4 & 2 & 3 \\ 0 & 3 & 5 & -7 \\ -2 & 8 & -4 & -3 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-2)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -4 & 2 & 3 \\ 0 & 3 & 5 & -7 \\ 0 & 0 & 0 & 3 \end{array}\right].$$

Use $R_2 \leftarrow (\frac{1}{3})R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -4 & 2 & 3 \\ 0 & 1 & \frac{5}{3} & -\frac{7}{3} \\ 0 & 0 & 0 & 3 \end{array}\right].$$

Use $R_3 \leftarrow (\frac{1}{3})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -4 & 2 & 3 \\ 0 & 1 & \frac{5}{3} & -\frac{7}{3} \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (3)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -4 & 2 & 0 \\ 0 & 1 & \frac{5}{3} & -\frac{7}{3} \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-\frac{7}{3})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -4 & 2 & 0 \\ 0 & 1 & \frac{5}{3} & 0 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-4)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & \frac{26}{3} & 0 \\ 0 & 1 & \frac{5}{3} & 0 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

**Conclusion.** A row with all variable coefficients zero and a nonzero constant asserts $0 = 1$ after scaling. This is impossible, so the system is inconsistent and its solution set is empty.

There are no weights producing $b$, so it is not a linear combination of the columns.

### Exercise 1.3.14: An explicit combination with one free weight

**What we must show.** Decide whether weights exist, rather than whether $b$ equals one of the columns. Form $[a_1\ a_2\ a_3 \mid b]$.

**What we must show.** Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

**Plan.** Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{rrr|r} 1 & 0 & 5 & 2 \\ -2 & 1 & -6 & -1 \\ 0 & 2 & 8 & 6 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-2)R_1$ to obtain

$$\left[\begin{array}{rrr|r} 1 & 0 & 5 & 2 \\ 0 & 1 & 4 & 3 \\ 0 & 2 & 8 & 6 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (2)R_2$ to obtain

$$\left[\begin{array}{rrr|r} 1 & 0 & 5 & 2 \\ 0 & 1 & 4 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

**Read the equations, keeping every free variable.**

The nonpivot variable is $x_3$. Set $x_3 = t_1$, where the parameters range independently over $\mathbb{R}$.

$$\begin{cases} x_1 = 2 - 5t_1 \\ x_2 = 3 - 4t_1 \\ x_3 = t_1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

**Conclusion.** Infinitely many solutions, exactly as parametrized above.

For a concrete combination take $x_3 = 0$, giving $b = 2a_1 + 3a_2$. Therefore $b$ belongs to the span.

### Exercise 1.3.15: Find the height in a generated plane

Write $x_1a_1 + x_2a_2 = b$ and reduce the augmented matrix:

$$\left[\begin{array}{rr|r} 1 & -5 & 3 \\ 3 & -8 & -5 \\ -1 & 2 & h \end{array}\right].$$

Use $R_2 \leftarrow R_2 - 3R_1$ and $R_3 \leftarrow R_3 + R_1$. The last two rows become $[0\ 7 \mid -14]$ and $[0\ -3 \mid h + 3]$. The former gives $x_2 = -2$, then the first row gives $x_1 = -7$. The last row requires $6 = h + 3$, or $h = 3$. Conversely these weights then give $-7a_1 - 2a_2 = (3, -5, 3)^T$ exactly.

**Conclusion.** $h = 3$, and only that height lies in the plane.

### Exercise 1.3.16: Find a first coordinate in a generated plane

From the second coordinate of $x_1v_1 + x_2v_2 = y$, we immediately get $x_2 = -3$. The third coordinate is $-2x_1 + 7x_2 = -5$, so $-2x_1 - 21 = -5$ and $x_1 = -8$. The first coordinate then requires $h = x_1 - 2x_2 = -8 + 6 = -2$. Substitution verifies all three coordinates.

**Conclusion.** Exactly $h = -2$ works, with $y = -8v_1 - 3v_2$.

### Exercise 1.3.17: Five combinations and their weights

Choose five different pairs of weights and compute their coordinates. The choice of weights itself proves membership in the span.

| $c_1$ | $c_2$ | $c_1v_1 + c_2v_2$ | | |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 3 | 1 | 2 |
| 0 | 1 | −4 | 0 | 1 |
| 1 | 1 | −1 | 1 | 3 |
| 2 | −1 | 10 | 2 | 3 |

**Conclusion.** These are five explicit vectors with their generating weights. A sketch is not needed or requested.

### Exercise 1.3.18: Five combinations of a second generating pair

Choose five different pairs of weights and compute their coordinates. The choice of weights itself proves membership in the span.

| $c_1$ | $c_2$ | $c_1v_1 + c_2v_2$ | | |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 1 | −2 |
| 0 | 1 | −2 | 3 | 0 |
| 1 | 1 | −1 | 4 | −2 |
| 2 | −1 | 4 | −1 | −4 |

**Conclusion.** These are five explicit vectors with their generating weights. A sketch is not needed or requested.

### Exercise 1.3.19: Dependent directions generate a line

We have $v_2 = (3/2)v_1$, since multiplying each coordinate of $(8, 2, -6)^T$ by 3/2 gives $(12, 3, -9)^T$. Therefore $c_1v_1 + c_2v_2 = (c_1 + (3/2)c_2)v_1$. Every combination lies on the line through the origin in direction $v_1$, and every point on that line occurs by taking $c_2 = 0$.

**Conclusion.** The span is the line $\{t(8, 2, -6)^T : t \in \mathbb{R}\}$, not a plane.

### Exercise 1.3.20: Two nonparallel vectors generate a plane

The vectors $(1, 1, -2)^T$ and $(-2, 3, 0)^T$ are not scalar multiples: a multiple of the first with third coordinate zero would have multiplier zero and hence be the zero vector. Two nonparallel directions

through the origin generate a plane. To identify it algebraically, observe that $n = (6, 4, 5)^T$ satisfies $n \cdot v_1 = 6 + 4 - 10 = 0$ and $n \cdot v_2 = -12 + 12 = 0$. Thus every combination obeys $6x + 4y + 5z = 0$. Conversely, given a point on this plane, choose $c_1 = -z/2$ and $c_2 = (y - c_1)/3$. Its second and third coordinates match, and the plane equation forces the first coordinate to match as well.

**Conclusion.** The span is the plane through the origin with equation $6x + 4y + 5z = 0$.

### Exercise 1.3.21: Express every plane vector using two generators

Seek scalars $a, b$ with $a(2, -1)^T + b(2, 1)^T = (h, k)^T$. The two equations are $2a + 2b = h$ and $-a + b = k$. The second gives $b = a + k$; substitute this into the first to get $4a + 2k = h$. Thus

$$a = \frac{h - 2k}{4}, \qquad b = \frac{h + 2k}{4}.$$

These real numbers are defined for every real $h, k$. Their substitution gives the desired coordinates exactly.

**Conclusion.** Every $(h, k)^T$ lies in the span, with the displayed weights.

### Exercise 1.3.22: A nonzero-entry matrix with a proper span

Take

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

All entries of $A$ are nonzero. Each linear combination of its columns has form $(t, t, t)^T$, where $t$ is the sum of the three weights. But $b$ does not have three equal coordinates. Therefore no combination can produce $b$.

**Conclusion.** The displayed $A, b$ meet both requirements.

### Exercise 1.3.23: Vector notation and span definitions

(a) False. A column vector has shape $2 \times 1$, whereas the displayed row has shape $1 \times 2$. Changing the orientation requires a transpose; the two matrix shapes are not interchangeable.

(b) False. If $(-5, 2)$ were a scalar multiple of $(-2, 5)$, the first coordinate would require multiplier $5/2$, but the second would then be $25/2$, not 2. Thus the two points do not lie on one line through zero.

(c) True. The expression $(1/2)v_1$ is $(1/2)v_1 + 0v_2$; a coefficient may be zero.

(d) True. Coordinatewise expansion of the vector equation gives exactly the equations encoded by $[a_1\ a_2\ a_3 \mid b]$.

(e) False. If $u = v \neq 0$ the span is a line, and if $u = v = 0$ it is just the origin. A plane requires two independent directions.

### Exercise 1.3.24: What linear combinations do and do not require

(a) False. With $u = (1, 0)^T, v = (0, 1)^T$, the combination $u + v = (1, 1)^T$ lies on neither generating axis. The span contains combinations involving both directions.

(b) True. $\mathbb{R}^5$ is the set of ordered lists of five real coordinates.

(c) True. By definition, $b$ is in the span exactly when suitable weights exist, which is the same as consistency of the augmented system.

(d) False. Adding $u - v$ to $v$ yields $u - v + v = u$, not generally $v$. For example $u = 0$, $v \neq 0$ gives zero as the result.

(e) False. All zero weights are allowed and produce the zero vector. Consequently every span contains zero.

### Exercise 1.3.25: A generating set is different from its span

#### (a) Distinguish the generating set from its span.

The three columns are $(1, 0, -2)^T$, $(0, 3, 6)^T$, and $(-4, -2, 3)^T$. None equals $b = (4, 1, -4)^T$. Thus $b$ is not in the three-element set $\{a_1, a_2, a_3\}$, which contains exactly three distinct vectors.

#### (b) Produce actual weights.

Direct calculation gives

$$-4a_1 - a_2 - 2a_3 = (-4, 0, 8)^T + (0, -3, -6)^T + (8, 4, -6)^T = (4, 1, -4)^T = b.$$

So $b \in W$. Also $W$ contains all multiples of the nonzero vector $a_1$; different real multiples give different vectors, so $W$ is infinite.

#### (c) Use the definition without reducing a matrix.

We have $a_1 = 1a_1 + 0a_2 + 0a_3$, which explicitly represents $a_1$ as a combination of the generators.

**Conclusion.** The finite generating set does not contain $b$, but its infinite span does contain $b$ and all three generating columns.

### Exercise 1.3.26: A target outside the span but a column inside

#### (a) Check the three coordinate equations for compatible weights.

Writing the weights as $c_1, c_2, c_3$, the first coordinate gives $2c_1 + 6c_3 = 10$, or $c_1 = 5 - 3c_3$. The second is $-c_1 + 8c_2 + 5c_3 = 3$, so after substitution $c_2 + c_3 = 1$. The third would require

$$c_1 - 2c_2 + c_3 = 7.$$

Its left side, using the first two equations, is instead $(5 - 3c_3) - 2(1 - c_3) + c_3 = 3$, independent of $c_3$. Thus it cannot equal 7; no weights exist, and $b \notin W$.

#### (b) Exhibit the trivial combination for a generator.

The second column is $0a_1 + 1a_2 + 0a_3$, so it is certainly in $W$.

**Conclusion.** $b$ is outside the column span, although each generating column itself is automatically inside it.

### Exercise 1.3.27: Translate mine outputs into a vector equation

**(a) Interpret the scale factor together with the units.**

Five days of mine 1 produce $5v_1 = (150, 3000)^T$: 150 metric tons of copper and 3000 kilograms of silver.

**(b) Express the target as the sum of the two outputs.**

The vector equation is

$$x_1 \begin{bmatrix} 30 \\ 600 \end{bmatrix} + x_2 \begin{bmatrix} 40 \\ 380 \end{bmatrix} = \begin{bmatrix} 240 \\ 2824 \end{bmatrix}.$$

The unknowns count days, so physically they must be nonnegative.

**(c) Solve and check.**

Divide the copper equation by 10: $3x_1 + 4x_2 = 24$. Multiply it by 200 and subtract the silver equation to get $420x_2 = 4800 - 2824 = 1976$. Therefore

$$x_2 = \frac{494}{105}, \qquad x_1 = \frac{24 - 4(494/105)}{3} = \frac{544}{315}.$$

Both are positive. In the copper equation their contributions sum to $30(544/315) + 40(494/105) = 240$; in the silver equation they sum to $600(544/315) + 380(494/105) = 2824$.

**Conclusion.** Mine 1 runs $544/315 \approx 1.727$ days and mine 2 runs $494/105 \approx 4.705$ days.

### Exercise 1.3.28: Check three coal-plant output coordinates

**(a) Track just the heat coordinate.**

Burning $x_1$ tons of A and $x_2$ tons of B gives $27.6x_1 + 30.2x_2$ million Btu of heat.

**(b) Keep the three output types in a fixed order.**

With coordinates heat (million Btu), sulfur dioxide (g), and particulates (g), the output vector is

$$x_1 \begin{bmatrix} 27.6 \\ 3100 \\ 250 \end{bmatrix} + x_2 \begin{bmatrix} 30.2 \\ 6400 \\ 360 \end{bmatrix}.$$

**(c) Set this vector equal to the observed output.**

Use right side $(162, 23610, 1623)^T$. The heat and particulate equations, clearing the heat decimal, are

$$276x_1 + 302x_2 = 1620, \qquad 250x_1 + 360x_2 = 1623.$$

Subtracting 250 times the first from 276 times the second gives $23860x_2 = 42948$, so $x_2 = 9/5 = 1.8$. Substitution in either equation gives $x_1 = 39/10 = 3.9$. Now check the remaining equation rather than ignoring the extra datum:

$$3100(3.9) + 6400(1.8) = 12090 + 11520 = 23610.$$

The other outputs are $107.64 + 54.36 = 162$ and $975 + 648 = 1623$.

**Conclusion.** The plant burned 3.9 tons of anthracite and 1.8 tons of bituminous coal, satisfying all three measurements.

### Exercise 1.3.29: Compute a mass-weighted average

The total mass is $4 + 2 + 3 + 5 = 14$ grams. Form the weighted sum one coordinate at a time:

$$\begin{aligned} 4v_1 + 2v_2 + 3v_3 + 5v_4 &= (8, -8, 16) + (-8, 4, 6) \\ &\quad + (12, 0, -6) + (5, -30, 0) \\ &= (17, -34, 16). \end{aligned}$$

Dividing by the total mass gives

$$\overline{v} = \left(\frac{17}{14}, -\frac{17}{7}, \frac{8}{7}\right).$$

The division is by total *mass*, not by the number of points.

**Conclusion.** The center of mass is $(17/14, -17/7, 8/7)$.

### Exercise 1.3.30: A center of mass is a linear combination

Assume the total mass $m$ is nonzero, as required to define the center of mass. Rewrite its formula as

$$\overline{v} = \frac{m_1}{m}v_1 + \cdots + \frac{m_k}{m}v_k.$$

The coefficients $m_j/m$ are real scalars. This is precisely a linear combination of the given points viewed as vectors, so it belongs to their span. For positive physical masses the coefficients also sum to one and are nonnegative, although those extra facts are not needed for span membership.

**Conclusion.** Yes; the displayed normalized masses are the required weights.

### Exercise 1.3.31: Move the center of a triangular plate

#### (a) Average the three equal point masses.

The uniform triangular plate has the same center as three 1-gram masses at the vertices, so

$$\overline{v} = \frac{(0, 1) + (8, 1) + (2, 4)}{3} = \left(\frac{10}{3}, 2\right).$$

#### (b) Include the plate's existing mass in the new balance.

The total mass after adding 6 grams will be 9 grams. The plate already has first moment $3\overline{v} = (10, 6)$. Let $w_i$ be the added grams at vertex $v_i$. The new center $(2, 2)$ requires total first moment $(18, 18)$, so

$$w_1 + w_2 + w_3 = 6, \qquad 8w_2 + 2w_3 = 18 - 10 = 8,$$

$$w_1 + w_2 + 4w_3 = 18 - 6 = 12.$$

Subtract the mass equation from the last equation to obtain $3w_3 = 6$, so $w_3 = 2$. The horizontal equation then gives $8w_2 = 4$ and $w_2 = 1/2$. Finally $w_1 = 6 - 2 - 1/2 = 7/2$. All added masses are nonnegative, as the physical request requires.

**Conclusion.** Add 3.5 g at $(0, 1)$, 0.5 g at $(8, 1)$, and 2 g at $(2, 4)$. Their contribution is $(8, 12)$, making the final first moment $(18, 18)$ and the desired center $(2, 2)$.

### Exercise 1.3.32: A plane-spanning diagram with nonunique weights

The figure shows $v_1, v_2$ pointing in nonparallel directions. Their span is therefore the whole plane, so in particular there are weights $a, b$ with $av_1 + bv_2 = b_{\text{target}}$. Taking the third weight to be zero already gives a solution. The same plane contains $v_3$, so write $v_3 = pv_1 + qv_2$. For every $t \in \mathbb{R}$,

$$(a - pt)v_1 + (b - qt)v_2 + tv_3 = av_1 + bv_2 = b_{\text{target}}.$$

These are distinct coefficient triples because their third entries are distinct. Negative weights are allowed; the target need not lie between the positive arrows.

**Conclusion.** A solution exists, and it is not unique: there are infinitely many coefficient triples.

### Exercise 1.3.33: Associativity and distributivity coordinatewise

#### (a) Associativity.

For each coordinate $i$, real-number associativity gives $(u_i + v_i) + w_i = u_i + (v_i + w_i)$. The $i$th coordinates of $(u + v) + w$ and $u + (v + w)$ are therefore equal for every $i$, which proves equality of the vectors.

#### (b) Distributivity over vector addition.

For each $i$, $c(u_i + v_i) = cu_i + cv_i$ by real-number distributivity. The left side is the $i$th coordinate of $c(u + v)$; the right side is the $i$th coordinate of $cu + cv$. Equality coordinate by coordinate gives $c(u + v) = cu + cv$.

**Conclusion.** Both vector laws follow from the corresponding real-number laws.

### Exercise 1.3.34: Additive inverses and scalar associativity

#### (a) The additive inverse.

The $i$th coordinate of $u + (-u)$ is $u_i + (-u_i) = 0$. Similarly the $i$th coordinate of $(-u) + u$ is $-u_i + u_i = 0$. Both sums are therefore the vector $(0, \ldots, 0)$.

#### (b) Two successive scalar multiplications.

The $i$th coordinate of $c(du)$ is $c(du_i)$. Real multiplication is associative, so $c(du_i) = (cd)u_i$, which is the $i$th coordinate of $(cd)u$. This holds for every coordinate and for all real $c, d$, including zero and negative values.

**Conclusion.** $u + (-u) = (-u) + u = 0$ and $c(du) = (cd)u$.

## 1.4 The matrix equation $Ax = b$

A matrix–vector product is a weighted sum of the matrix's columns. If $A = [a_1 \ \cdots \ a_n]$ and $x = (x_1, \ldots, x_n)^T$, then

$$Ax = x_1a_1 + \cdots + x_na_n.$$

Thus $Ax$ is defined only when $x$ has one entry for each column of $A$. The row rule gives the same product: entry $i$ is $a_{i1}x_1 + \cdots + a_{in}x_n$. A vector $b$ belongs to the span of the columns exactly when $Ax = b$ is consistent. Theorem 4 says the columns span the whole target $\mathbb{R}^m$ exactly when $A$ has a pivot in every row. A pivot in an augmented column does not count as a pivot in $A$.

### Practice problem 1.4.P1: Read coefficients from a known solution

The columns are $a_1 = (1, -3, 4)^T$, $a_2 = (5, 1, -8)^T$, $a_3 = (-2, 9, -1)^T$, and $a_4 = (0, -5, 7)^T$. The given solution has entries $(3, -2, 0, -4)^T$, so the definition gives

$$3a_1 - 2a_2 + 0a_3 - 4a_4 = (3, -9, 12)^T + (-10, -2, 16)^T + (0, 20, -28)^T = (-7, 9, 0)^T = b.$$

The coefficients are the entries of the solution, not row entries of $A$.

### Practice problem 1.4.P2: Check additivity by both calculations

Here $A = \begin{bmatrix} 2 & 5 \\ 3 & 1 \end{bmatrix}$, $u = (4, -1)^T$, and $v = (-3, 5)^T$. First $u + v = (1, 4)^T$, so

$$A(u + v) = (2 + 20, 3 + 4)^T = (22, 7)^T.$$

Separately $Au = (8 - 5, 12 - 1)^T = (3, 11)^T$ and $Av = (-6 + 25, -9 + 5)^T = (19, -4)^T$. Their sum is again $(22, 7)^T$. This checks $A(u + v) = Au + Av$ with the actual data.

### Exercise 1.4.1: Check dimensions before multiplying

**What we must show.** Compute the indicated product, if possible, in both requested ways. The matrix $\begin{bmatrix} -4 & 2 \\ 1 & 6 \\ 0 & 1 \end{bmatrix}$ has two columns, but $(3, -2, 7)^T$ has three entries. In the column definition there are only two columns for three proposed weights. In the row rule a row has two entries and cannot be paired with a three-entry vector.

**Conclusion.** The product is undefined; no numerical result can be assigned.

### Exercise 1.4.2: A column matrix is not a row matrix

The matrix $(1, 3, 1)^T$ has size $3 \times 1$, whereas $(5, -1)^T$ has size $2 \times 1$. The inner sizes 1 and 2 differ. The column definition would need one scalar, and each row contains only one entry.

**Conclusion.** The product is undefined by either rule.

### Exercise 1.4.3: Compute one product in two ways

**(a) Weight the two columns.**

$$-2\begin{bmatrix}1\\-3\\1\end{bmatrix}+3\begin{bmatrix}2\\1\\6\end{bmatrix}=\begin{bmatrix}-2+6\\6+3\\-2+18\end{bmatrix}=\begin{bmatrix}4\\9\\16\end{bmatrix}.$$

**(b) Use the rows.**

The three dot products are $1(-2)+2(3)=4$, $(-3)(-2)+1(3)=9$, and $1(-2)+6(3)=16$.

**Conclusion.** Both rules give $(4,9,16)^T$.

### Exercise 1.4.4: Two rows give two output entries

**(a) Use the column definition.**

$$1\begin{bmatrix}1\\3\end{bmatrix}+2\begin{bmatrix}3\\2\end{bmatrix}+1\begin{bmatrix}-4\\1\end{bmatrix}=\begin{bmatrix}1+6-4\\3+4+1\end{bmatrix}=\begin{bmatrix}3\\8\end{bmatrix}.$$

**(b) Use the row rule.**

The first entry is $1(1)+3(2)-4(1)=3$; the second is $3(1)+2(2)+1(1)=8$. Both entries have now been checked independently.

### Exercise 1.4.5: Expand the matrix equation into columns

Read each column vertically and use the corresponding entry of the multiplier vector as its weight:

$$2\begin{bmatrix}1\\-2\end{bmatrix}-\begin{bmatrix}2\\-3\end{bmatrix}+\begin{bmatrix}-3\\1\end{bmatrix}-\begin{bmatrix}1\\-1\end{bmatrix}=\begin{bmatrix}-4\\1\end{bmatrix}.$$

For a direct check, the first coordinate is $2-2-3-1=-4$ and the second is $-4+3+1+1=1$.

### Exercise 1.4.6: A four-entry output from two columns

The equivalent vector equation is

$$-3\begin{bmatrix}2\\3\\8\\-2\end{bmatrix}+5\begin{bmatrix}-3\\2\\-5\\1\end{bmatrix}=\begin{bmatrix}-21\\1\\-49\\11\end{bmatrix}.$$

The coordinates are $-6-15=-21$, $-9+10=1$, $-24-25=-49$, and $6+5=11$.

### Exercise 1.4.7: Assemble the coefficient vectors as columns

Place the coefficient of $x_1$ in column 1, that of $x_2$ in column 2, and that of $x_3$ in column 3. The result is

$$\begin{bmatrix} 4 & -5 & 7 \\ -1 & 3 & -8 \\ 7 & -5 & 0 \\ -4 & 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ -8 \\ 0 \\ -7 \end{bmatrix}.$$

The output has four entries because each original vector lies in $\mathbb{R}^4$.

### Exercise 1.4.8: Four scalar weights and a two-dimensional target

The same column convention gives

$$\begin{bmatrix} 2 & -1 & -4 & 0 \\ -4 & 5 & 3 & 2 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \end{bmatrix} = \begin{bmatrix} 5 \\ 12 \end{bmatrix}.$$

There are four columns, one for each weight, and two rows, one for each coordinate equation.

### Exercise 1.4.9: Keep the missing coefficient equal to zero

The coefficient of $x_1$ in the second equation is zero. Therefore

$$x_1 \begin{bmatrix} 5 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + x_3 \begin{bmatrix} -3 \\ 4 \end{bmatrix} = \begin{bmatrix} 8 \\ 0 \end{bmatrix},$$

and the matrix equation is

$$\begin{bmatrix} 5 & 1 & -3 \\ 0 & 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 0 \end{bmatrix}.$$

### Exercise 1.4.10: Three equations become three coordinates

The vector equation is

$$x_1 \begin{bmatrix} 4 \\ 5 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} -1 \\ 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 8 \\ 2 \\ 1 \end{bmatrix}.$$

Equivalently,

$$\begin{bmatrix} 4 & -1 \\ 5 & 3 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 8 \\ 2 \\ 1 \end{bmatrix}.$$

No equation has been discarded in changing notation.

### Exercise 1.4.11: Solve an augmented system and check its vector

**Form and reduce the augmented matrix.**

$$\left[\begin{array}{ccc|c} 1 & 3 & -4 & -2 \\ 1 & 5 & 2 & 4 \\ -3 & -7 & 6 & 12 \end{array}\right] \xrightarrow[R_3 \leftarrow R_3+3R_1]{R_2 \leftarrow R_2-R_1} \left[\begin{array}{ccc|c} 1 & 3 & -4 & -2 \\ 0 & 2 & 6 & 6 \\ 0 & 2 & -6 & 6 \end{array}\right].$$

Replace $R_3$ by $R_3 - R_2$ to obtain $-12x_3 = 0$. Thus $x_3 = 0$. The second row gives $2x_2 + 6x_3 = 6$, hence $x_2 = 3$. The first gives $x_1 + 9 = -2$, hence $x_1 = -11$.

**Substitute into the original rows.**

They give $-11 + 9 = -2$, $-11 + 15 = 4$, and $33 - 21 = 12$.

**Conclusion.** The solution vector is $x = (-11, 3, 0)^T$.

### Exercise 1.4.12: A second exact augmented reduction

Start with

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ -3 & -4 & 2 & 2 \\ 5 & 2 & 3 & -3 \end{array}\right] \xrightarrow[R_3 \leftarrow R_3-5R_1]{R_2 \leftarrow R_2+3R_1} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 2 & -1 & 5 \\ 0 & -8 & 8 & -8 \end{array}\right].$$

Now $R_3 \leftarrow R_3 + 4R_2$ gives $4x_3 = 12$, so $x_3 = 3$. Then $2x_2 - 3 = 5$ gives $x_2 = 4$, and $x_1 + 8 - 3 = 1$ gives $x_1 = -4$. The original products are $-4 + 8 - 3 = 1$, $12 - 16 + 6 = 2$, and $-20 + 8 + 9 = -3$.

**Conclusion.** $x = (-4, 4, 3)^T$.

### Exercise 1.4.13: Membership in a plane means consistency

**What we must show.** Decide whether $u = (0, 4, 4)^T$ is a linear combination of $(3, -2, 1)^T$ and $(-5, 6, 1)^T$. The equations for the two weights $s, t$ are $3s - 5t = 0$, $-2s + 6t = 4$, and $s + t = 4$. The last equation gives $s = 4 - t$. Substitution into the first gives $12 - 8t = 0$, so $t = 3/2$ and $s = 5/2$. The second equation checks: $-5 + 9 = 4$. Hence

$$u = \tfrac{5}{2}(3, -2, 1)^T + \tfrac{3}{2}(-5, 6, 1)^T.$$

**Conclusion.** Yes: the explicit weights show that $u$ lies in the plane.

### Exercise 1.4.14: A contradictory coordinate rules out membership

The proposed equation $Ax = u$ gives

$$2x_1 + 5x_2 - x_3 = 4, \qquad x_2 - x_3 = -1, \qquad x_1 + 2x_2 = 4.$$

From the last two equations $x_3 = x_2 + 1$ and $x_1 = 4 - 2x_2$. The left side of the first would then be $2(4 - 2x_2) + 5x_2 - (x_2 + 1) = 7$, not 4.

**Conclusion.** No. The system is inconsistent, so $u$ is outside the column span.

### Exercise 1.4.15: Describe every attainable right-hand side

For $A = \begin{bmatrix} 3 & -1 \\ -9 & 3 \end{bmatrix}$, the second row is $-3$ times the first. Eliminating it in $[A \mid b]$ gives the equation $0 = b_2 + 3b_1$. Thus consistency is equivalent to $b_2 = -3b_1$. This is also sufficient: the first column $(3, -9)^T$ alone reaches $(b_1, -3b_1)^T$ with weight $b_1/3$.

**Conclusion.** The attainable vectors are exactly $t(1, -3)^T$, $t \in \mathbb{R}$; for example $(0, 1)^T$ is not attainable, so not every $b$ works.

### Exercise 1.4.16: A condition on three coordinates of the right side

For the given matrix, reduce symbolically:

$$\left[\begin{array}{ccc|c} 1 & -2 & -1 & b_1 \\ -2 & 2 & 0 & b_2 \\ 4 & -1 & 3 & b_3 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} 1 & -2 & -1 & b_1 \\ 0 & -2 & -2 & b_2 + 2b_1 \\ 0 & 7 & 7 & b_3 - 4b_1 \end{array}\right].$$

These steps use $R_2 + 2R_1$ and $R_3 - 4R_1$. Replacing the last row by $2R_3 + 7R_2$ gives $0 = 6b_1 + 7b_2 + 2b_3$. If this condition holds, the remaining two equations have pivots and can be solved, with one free variable. If it fails, the last row is a contradiction.

**Conclusion.** The attainable right sides form the plane $\{b : 6b_1 + 7b_2 + 2b_3 = 0\}$, not all of $\mathbb{R}^3$.

### Exercise 1.4.17: Count pivot rows of the specified matrix

For the matrix $A$ used in Exercises 17 and 19, exact elimination gives

$$A = \begin{bmatrix} 1 & 3 & 0 & 3 \\ -1 & -1 & -1 & 1 \\ 0 & -4 & 2 & -8 \\ 2 & 0 & 3 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & \frac{3}{2} & 0 \\ 0 & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There are three pivot rows, in columns 1, 2, and 4. The fourth row of the coefficient matrix has no pivot. By Theorem 4, $Ax = b$ is solvable for every $b \in \mathbb{R}^4$ if and only if $A$ has a pivot in every row.

**Conclusion.** Three rows contain pivots; some right sides have no solution.

### Exercise 1.4.18: Distinguish the target space from the number of columns

For the matrix $B$ used here and in Exercise 20,

$$B = \begin{bmatrix} 1 & 4 & 1 & 2 \\ 0 & 1 & 3 & -4 \\ 0 & 2 & 6 & 7 \\ 2 & 9 & 5 & -7 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -11 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There are only three pivots and one zero row. Theorem 4 therefore shows that its columns do not span $\mathbb{R}^4$. They do not span $\mathbb{R}^3$ either: the columns themselves have four entries and belong to $\mathbb{R}^4$, so $\mathbb{R}^3$ is not their ambient space.

**Conclusion.** Not every vector of $\mathbb{R}^4$ is a combination of the columns; the proposed $\mathbb{R}^3$ claim also fails.

### Exercise 1.4.19: Translate the pivot calculation into a span statement

The reduction displayed in Exercise 1.4.17 has a zero fourth row and only three pivots. Theorem 4 says this is equivalent to failure of the columns to span $\mathbb{R}^4$. A right side whose transformed fourth entry is nonzero gives an inconsistent augmented system.

**Conclusion.** No to both questions: not every vector is a column combination, and the column span is a proper subset of $\mathbb{R}^4$.

### Exercise 1.4.20: The span and solvability questions are equivalent

Exercise 1.4.18 exhibits three pivot rows for $B$, not four. By Theorem 4, the following are equivalent: the columns span $\mathbb{R}^4$; $Bx = y$ is consistent for every $y \in \mathbb{R}^4$; and $B$ has a pivot in every row. The last condition fails.

**Conclusion.** The columns do not span $\mathbb{R}^4$, and some $y$ give no solution.

### Exercise 1.4.21: Three columns cannot fill four target directions

A $4 \times 3$ matrix formed from these three vectors has at most three pivot positions, because each pivot requires a different column. It therefore cannot have a pivot in every one of its four rows. Theorem 4 excludes spanning $\mathbb{R}^4$. More concretely each vector's coordinate sum is zero; every linear combination also has coordinate sum zero, whereas $(1, 0, 0, 0)^T$ does not.

**Conclusion.** The three vectors do not span $\mathbb{R}^4$.

### Exercise 1.4.22: Three suitable vectors do span three-dimensional space

Put the vectors into columns:

$$V = \begin{bmatrix} 0 & 0 & 4 \\ 0 & -3 & -2 \\ -3 & 9 & -6 \end{bmatrix}.$$

Swap rows 1 and 3. The resulting matrix has pivots $-3, -3, 4$ in successively later columns, so it has three pivot rows. Theorem 4 implies its columns span $\mathbb{R}^3$. One can also solve $Vc = b$ directly: $c_3 = b_1/4$, then $c_2 = -(b_2 + 2c_3)/3$, and finally $c_1 = (9c_2 - 6c_3 - b_3)/3$. These formulas work for every $b$.

**Conclusion.** Yes, the vectors span $\mathbb{R}^3$.

### Exercise 1.4.23: Six true-or-false statements about the matrix equation

(a) **False.** $Ax = b$ is called a matrix equation. Its equivalent column-sum expression is a vector equation.

(b) **True.** By definition, $Ax$ runs through precisely the linear combinations of the columns; existence of a suitable $x$ is the claim.

(c) **False.** A pivot in every row of $[A \mid b]$ can include a pivot in the last column. For instance $[0 \mid 1]$ has a pivot in its only row but represents $0x = 1$, an inconsistent equation.

(d) **True.** The first entry is $a_{11}x_1 + \cdots + a_{1n}x_n$, a sum of scalar products.

(e) **True.** This is the column-span equivalence in Theorem 4.

(f) **True.** If $A$ had a pivot in every row, Theorem 4 would make every right side attainable. The stated inconsistent right side contradicts that possibility.

### Exercise 1.4.24: Six further equivalences and counterexamples

(a) **True.** Expanding $Ax$ into columns changes notation, not the equations or their solutions.

(b) **True.** A solution gives $b = x_1a_1 + \cdots + x_na_n$, which is exactly membership in the column span.

(c) **True.** Place the vectors being combined into the columns of $A$ and their weights into $x$.

(d) **False.** A pivot in every coefficient row guarantees *consistency* for every right side, not inconsistency.

(e) **True.** Both descriptions use the coefficient matrix $A = [a_1\ a_2\ a_3]$ and the same right side $b$.

(f) **False.** Failure to span means that *some* $b$ lies outside the span and gives an inconsistent equation. The right side 0 is always consistent, since $x = 0$ solves $Ax = 0$.

### Exercise 1.4.25: Read a supplied product without repeating elimination

The given multiplier vector is $(-3, -1, 2)^T$. The definition of matrix multiplication therefore already states

$$(-7, -3, 10)^T = -3(4, 5, -6)^T - (-3, -2, 2)^T + 2(1, 5, -3)^T.$$

**Conclusion.** $c_1 = -3$, $c_2 = -1$, and $c_3 = 2$.

### Exercise 1.4.26: Rearrange a known linear relation

The supplied equality $2u - 3v - w = 0$ rearranges to $w = 2u - 3v$. The displayed matrix has columns $u$ and $v$, so its product with $(2, -3)^T$ is $w$. Checking the entries gives $14 - 9 = 5$, $4 - 3 = 1$, and $10 - 9 = 1$.

**Conclusion.** $x_1 = 2$ and $x_2 = -3$; no row operations are needed.

### Exercise 1.4.27: Name every vector and every coefficient

Define

$$v_1 = (-3, 5)^T, \quad v_2 = (5, 8)^T, \quad v_3 = (-4, 1)^T, \quad v_4 = (9, -2)^T, \quad v_5 = (7, -4)^T.$$

Let $(c_1, c_2, c_3, c_4, c_5) = (-3, 1, 2, -1, 2)$ and $b = (11, -11)^T$. The symbolic vector equation is

$$c_1v_1 + c_2v_2 + c_3v_3 + c_4v_4 + c_5v_5 = b.$$

Its coordinate sums are $9 + 5 - 8 - 9 + 14 = 11$ and $-15 + 8 + 2 + 2 - 8 = -11$, checking the interpretation.

### Exercise 1.4.28: Package a symbolic vector equation

Set $Q = [q_1\ q_2\ q_3]$, a $5 \times 3$ matrix, and $x = (x_1, x_2, x_3)^T$. The required matrix equation is $Qx = v$. Its five coordinates reproduce those of $x_1q_1 + x_2q_2 + x_3q_3 = v$, because a product is a weighted column sum.

### Exercise 1.4.29: Construct a spanning matrix outside echelon form

Take

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The leading entry in row 2 lies left of that in row 1, so $A$ is not in echelon form. Swapping rows 1 and 2 produces $I_3$, which has a pivot in every row. Theorem 4 therefore proves its columns span $\mathbb{R}^3$. Explicitly, $(b_1, b_2, b_3)^T$ is obtained with weights $(b_2, b_1, b_3)^T$.

### Exercise 1.4.30: Construct a nonspanning matrix outside echelon form

Take

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

The first two leading entries move left, so it is not echelon. Every product is $(x_2, x_1, x_2)^T$; its first and third entries agree. The vector $(1, 0, 0)^T$ violates that condition and cannot be obtained.

**Conclusion.** This matrix meets both requirements.

### Exercise 1.4.31: More rows than columns leave a missing pivot row

A $3 \times 2$ matrix has at most two pivots, since there are only two columns. It cannot have a pivot in each of three rows. Theorem 4 then gives some $b \in \mathbb{R}^3$ for which $Ax = b$ is inconsistent. More generally, an $m \times n$ matrix with $m > n$ has at most $n < m$ pivots and hence cannot reach every vector of $\mathbb{R}^m$.

**Conclusion.** At least one target direction is missing whenever $m > n$.

### Exercise 1.4.32: Too few spanning vectors

Place the three proposed vectors in the columns of a $4 \times 3$ matrix. At most three of its four rows can contain pivots, so Theorem 4 rules out spanning $\mathbb{R}^4$. In general $n$ vectors in $\mathbb{R}^m$ with $n < m$ give an $m \times n$ matrix with too few columns to pivot in every row.

**Conclusion.** Neither three vectors in $\mathbb{R}^4$ nor $n < m$ vectors in $\mathbb{R}^m$ can span the whole target space.

### Exercise 1.4.33: A unique solution fixes the reduced form

Since the given equation is consistent and has only one solution, none of its three unknowns can be free. Each of the three coefficient columns therefore contains a pivot. In reduced echelon form the pivot columns are unit vectors, with pivots successively lower. Thus the reduced echelon form of $A$ is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The fourth coefficient row must be zero because only three pivots are possible. Consistency ensures the corresponding transformed right-side entry is zero as well.

### Exercise 1.4.34: Add solutions to reach the sum of their images

The additivity identity of Theorem 5 gives

$$A(u_1 + u_2) = Au_1 + Au_2 = v_1 + v_2 = w.$$

The vector $u_1 + u_2$ belongs to $\mathbb{R}^4$, the correct input space. It is an explicit solution to $Ax = w$, so the system is consistent.

### Exercise 1.4.35: Scale a known input

The scalar-multiplication identity of Theorem 5 gives $A(5y) = 5Ay = 5z$. Thus $x = 5y$ is a solution to the new system. The factor multiplies the entire input vector, not just one entry.

### Exercise 1.4.36: A unique square-system solution forces spanning

The unique solution means no unknown is free. All four coefficient columns have pivots, so the $4 \times 4$ matrix has four pivot rows. By Theorem 4 its columns span $\mathbb{R}^4$. Consistency for one special $b$ alone would not be enough; uniqueness supplies the crucial four column pivots.

### Exercise 1.4.37: Use exact row reduction to test four-dimensional spanning

**What we must show.** Decide whether the columns of the given matrix span $\mathbb{R}^4$. These exercises explicitly allow matrix computation. Ordinary Gauss–Jordan elimination (swap a nonzero entry into each pivot position, scale the pivot to 1, then clear its column) gives the exact result

$$A = \begin{bmatrix} 7 & 2 & -5 & 8 \\ -5 & -3 & 4 & -9 \\ 6 & 10 & -2 & 7 \\ -7 & 9 & 2 & 15 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & -\frac{93}{50} \\ 0 & 1 & 0 & \frac{53}{50} \\ 0 & 0 & 1 & -\frac{189}{50} \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2, 3; there are 3 pivot rows. Theorem 4 requires a pivot in every one of the four rows, not merely in every nonzero row. The last row has no pivot, so some right sides are unattainable.

**Conclusion.** The columns do not span $\mathbb{R}^4$.

### Exercise 1.4.38: Use exact row reduction to test four-dimensional spanning

**What we must show.** Decide whether the columns of the given matrix span $\mathbb{R}^4$. These exercises explicitly allow matrix computation. Ordinary Gauss–Jordan elimination (swap a nonzero entry into each pivot position, scale the pivot to 1, then clear its column) gives the exact result

$$A = \begin{bmatrix} 4 & -5 & -1 & 8 \\ 3 & -7 & -4 & 2 \\ 5 & -6 & -1 & 4 \\ 9 & 1 & 10 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2, 4; there are 3 pivot rows. Theorem 4 requires a pivot in every one of the four rows, not merely in every nonzero row. The last row has no pivot, so some right sides are unattainable.

**Conclusion.** The columns do not span $\mathbb{R}^4$.

### Exercise 1.4.39: Use exact row reduction to test four-dimensional spanning

**What we must show.** Decide whether the columns of the given matrix span $\mathbb{R}^4$. These exercises explicitly allow matrix computation. Ordinary Gauss–Jordan elimination (swap a nonzero entry into each pivot position, scale the pivot to 1, then clear its column) gives the exact result

$$A = \begin{bmatrix} 10 & -7 & 1 & 4 & 6 \\ -8 & 4 & -6 & -10 & -3 \\ -7 & 11 & -5 & -1 & -8 \\ 3 & -1 & 10 & 12 & 12 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

The pivot columns are 1, 2, 3, 5; there are 4 pivot rows. Theorem 4 requires a pivot in every one of the four rows, not merely in every nonzero row. That condition holds, so every right side in $\mathbb{R}^4$ is attainable.

**Conclusion.** The columns do span $\mathbb{R}^4$.

### Exercise 1.4.40: Use exact row reduction to test four-dimensional spanning

**What we must show.** Decide whether the columns of the given matrix span $\mathbb{R}^4$. These exercises explicitly allow matrix computation. Ordinary Gauss–Jordan elimination (swap a nonzero entry into each pivot position, scale the pivot to 1, then clear its column) gives the exact result

$$A = \begin{bmatrix} 5 & 11 & -6 & -7 & 12 \\ -7 & -3 & -4 & 6 & -9 \\ 11 & 5 & 6 & -9 & -3 \\ -3 & 4 & -7 & 2 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

The pivot columns are 1, 2, 4, 5; there are 4 pivot rows. Theorem 4 requires a pivot in every one of the four rows, not merely in every nonzero row. That condition holds, so every right side in $\mathbb{R}^4$ is attainable.

**Conclusion.** The columns do span $\mathbb{R}^4$.

### Exercise 1.4.41: Delete a redundant column while retaining the span

The pivot columns in the reduction of Exercise 1.4.39 are 1, 2, 3, and 5. These four original columns remain spanning after column 4 is deleted. The reduced form also displays the relation $a_4 = a_1+a_2+a_3$; row operations preserve a relation among columns because they multiply every column by the same invertible operation. Thus the deleted column adds no new span.

**Conclusion.** Delete column 4. The remaining four pivot columns span $\mathbb{R}^4$.

### Exercise 1.4.42: One deletion is possible but two are not

Exercise 1.4.40 has pivot columns 1, 2, 4, and 5. Delete column 3; the remaining four original pivot columns still span $\mathbb{R}^4$. Indeed the reduced matrix shows $a_3 = a_1 - a_2$, so this column was redundant. Deleting two columns from the original five would leave only three vectors. Exercise 1.4.32 proves that three vectors cannot span $\mathbb{R}^4$.

**Conclusion.** Column 3 can be deleted; no pair of columns can be deleted while preserving the span.

## 1.5 Solution sets of linear systems

A homogeneous system has right side zero: $Ax = 0$. It is always consistent, because $x = 0$ works. A nontrivial solution means any solution other than this entire zero vector; some individual coordinates may still be zero. A homogeneous system has a nontrivial solution exactly when there is a free variable.

To write a solution set parametrically, solve for pivot variables in terms of the free variables, then separate the resulting vector into its constant part and its parameter-dependent parts. If $p$ is one solution of $Ax = b$, all solutions are $p + v$, where $Av = 0$. Thus a consistent nonhomogeneous solution set is a translation of the homogeneous solution set.

### Practice problem 1.5.P1: Intersect two planes by solving their simultaneous equations

The two equations must hold at the same time. Starting with $x_1+4x_2-5x_3 = 0$ and $2x_1-x_2+8x_3 = 9$, subtract twice the first from the second to get $-9x_2 + 18x_3 = 9$. Thus $x_2 = -1 + 2x_3$ and the first equation gives $x_1 = 4 - 3x_3$. Write $x_3 = t$. Then

$$x = (4, -1, 0)^T + t(-3, 2, 1)^T, \qquad t \in \mathbb{R}.$$

Substitution gives $4 - 4 + (-3 + 8 - 5)t = 0$ in the first equation and $8 + 1 + (-6 - 2 + 8)t = 9$ in the second. The intersection is therefore the displayed line, not an empty set or an entire plane.

### Practice problem 1.5.P2: A translated plane

Solve $10x_1 - 3x_2 - 2x_3 = 7$ for its pivot variable: $x_1 = 7/10 + (3/10)x_2 + (1/5)x_3$. With $x_2 = s, x_3 = t$ this gives

$$x = (7/10, 0, 0)^T + s(3/10, 1, 0)^T + t(1/5, 0, 1)^T.$$

The two direction vectors are the same as for the homogeneous equation in Example 2. The constant vector translates that plane; it satisfies the new equation itself, since $10(7/10) = 7$.

### Exercise 1.5.1: Find a free variable with two eliminations

Add row 1 to row 2 and subtract twice row 1 from row 3. The coefficient matrix becomes

$$\begin{bmatrix} 2 & -5 & 8 \\ 0 & -12 & 9 \\ 0 & 12 & -9 \end{bmatrix}.$$

The last two rows are negatives. Their sum is zero, leaving only two pivot rows for three unknowns. Hence $x_3$ is free and a nontrivial solution exists. For an explicit check, $(-17, 6, 8)^T$ gives $-34-30+64 = 0$, $34 - 42 + 8 = 0$, and $-68 + 12 + 56 = 0$.

**Conclusion.** Yes, the system has nontrivial solutions.

### Exercise 1.5.2: Three pivots force the zero solution

The operations $R_2 \leftarrow R_2 + 2R_1$ and $R_3 \leftarrow R_3 - 2R_1$ give

$$\begin{bmatrix} 1 & -2 & 3 \\ 0 & -7 & 2 \\ 0 & 0 & 3 \end{bmatrix}.$$

All three columns have pivots. Back substitution gives $x_3 = 0$, then $x_2 = 0$, then $x_1 = 0$.

**Conclusion.** No: the only solution is the trivial solution.

### Exercise 1.5.3: Fewer equations guarantee a free variable here

A two-row coefficient matrix can have at most two pivots. With three unknowns, at least one variable must be free. Since the system is homogeneous, it is consistent, and the free variable gives a nontrivial solution. One such solution is $(-8, -4, 1)^T$: $24 - 16 - 8 = 0$ and $16 - 20 + 4 = 0$.

**Conclusion.** Yes; the pivot-count argument needs no row operations.

### Exercise 1.5.4: A second two-equation homogeneous system

There are three coefficient columns but only two rows. There cannot be a pivot in every column, so a free variable exists. Homogeneity ensures consistency, hence a nontrivial solution. For a numerical check, $(-2, 16, 29)^T$ gives $-10 - 48 + 58 = 0$ and $6 - 64 + 58 = 0$.

**Conclusion.** Yes, the system has nontrivial solutions.

### Exercise 1.5.5: Separate the common free parameter

The second equation is $-2$ times the first, so it adds no restriction. The last equation gives $x_2 = -x_3$. Divide the first equation by 2 to obtain $x_1 + x_2 + 2x_3 = 0$, hence $x_1 = -x_3$. Writing $x_3 = t$ gives

$$x = (-t, -t, t)^T = t(-1, -1, 1)^T, \qquad t \in \mathbb{R}.$$

Substitution into each original row gives zero.

**Conclusion.** The solution set is a line through the origin.

### Exercise 1.5.6: Three equal coordinates

The third equation gives $x_2 = x_1$. Substituting into the first produces $3x_1 - 3x_3 = 0$, so $x_1 = x_3$. The second equation then also holds, since $2x_3 + x_3 - 3x_3 = 0$. Thus, with $x_3 = t$,

$$x = t(1, 1, 1)^T, \qquad t \in \mathbb{R}.$$

**Conclusion.** This is another line through the origin.

### Exercise 1.5.7: Two free variables in four coordinates

Row equivalence preserves solutions to the homogeneous equation. The second displayed row gives $x_2 = 4x_3 - 5x_4$; substitute into the first to obtain $x_1 = -9x_3 + 8x_4$. Let $x_3 = s$ and $x_4 = t$.

$$x = s(-9, 4, 1, 0)^T + t(8, -5, 0, 1)^T, \qquad s, t \in \mathbb{R}.$$

These expressions satisfy both displayed rows, and every solution arises because its free coordinates may be chosen arbitrarily.

### Exercise 1.5.8: Solve the second pivot row first

The second row gives $x_2 = -2x_3 + 4x_4$. The first gives $x_1 = 3x_2 + 8x_3 - 5x_4 = 2x_3 + 7x_4$. Hence

$$x = s(2, -2, 1, 0)^T + t(7, 4, 0, 1)^T, \qquad s, t \in \mathbb{R}.$$

Here $s = x_3$ and $t = x_4$ are independent parameters; neither is silently set to zero.

### Exercise 1.5.9: The third coordinate is a pivot variable

The equations are $3x_1 - 6x_2 + 6x_3 = 0$ and $-2x_1 + 4x_2 - 2x_3 = 0$. Twice the first plus three times the second gives $6x_3 = 0$. Thus $x_3 = 0$ and $x_1 = 2x_2$.

$$x = t(2, 1, 0)^T, \qquad t \in \mathbb{R}.$$

The parameter is $x_2$, not $x_3$: the latter has been forced to zero.

### Exercise 1.5.10: A missing column gives a completely free coordinate

The equations are $-x_1 - 4x_2 - 4x_4 = 0$ and $2x_1 - 8x_2 + 8x_4 = 0$. Adding twice the first to the second gives $-16x_2 = 0$. Hence $x_2 = 0$ and $x_1 = -4x_4$. The variable $x_3$ never appears, so it is independently free.

$$x = s(0, 0, 1, 0)^T + t(-4, 0, 0, 1)^T, \qquad s, t \in \mathbb{R}.$$

### Exercise 1.5.11: Three free coordinates in a six-coordinate vector

The third nonzero row gives $x_5 = 4x_6$ and the second gives $x_3 = x_6$. Substitution into the first yields

$$x_1 = 4x_2 + 2x_3 - 3x_5 + 5x_6 = 4x_2 - 5x_6.$$

The free variables are $x_2 = s$, $x_4 = t$, and $x_6 = u$. Therefore

$$x = s(4, 1, 0, 0, 0, 0)^T + t(0, 0, 0, 1, 0, 0)^T + u(-5, 0, 1, 0, 4, 1)^T.$$

All three parameters range independently over $\mathbb{R}$. The zero fourth row imposes no additional equation.

### Exercise 1.5.12: A pivot in the last coefficient column is not inconsistency

The third row is $x_6 = 0$; this is a coefficient matrix for $Ax = 0$, not an augmented matrix with a nonzero right-side pivot. The second row gives $x_4 = -4x_5 + 6x_6 = -4x_5$. The first then gives $x_1 = 2x_2 - 3x_3 + 6x_4 - 5x_5 = 2x_2 - 3x_3 - 29x_5$. Thus

$$x = s(2,1,0,0,0,0)^T + t(-3,0,1,0,0,0)^T + u(-29,0,0,-4,1,0)^T, \qquad s,t,u \in \mathbb{R}.$$

The parameters are precisely the free coordinates $x_2, x_3, x_5$.

### Exercise 1.5.13: Read the point and direction of a line

Put $x_3 = t$ in the given coordinate description:

$$x = (5 + 4t, -2 - 7t, t)^T = (5,-2,0)^T + t(4,-7,1)^T.$$

Thus the line passes through $(5,-2,0)^T$ and has nonzero direction $(4,-7,1)^T$. Every real $t$ is allowed.

### Exercise 1.5.14: A line can lie in four-dimensional space

With $x_4 = t$, the given formulas assemble into

$$x = (5t, 3 - 2t, 2 + 5t, t)^T = (0,3,2,0)^T + t(5,-2,5,1)^T.$$

There is just one free parameter and one nonzero direction vector, so the set is a line in $\mathbb{R}^4$. A line does not require its ambient space to have only two or three dimensions.

### Exercise 1.5.15: Parallel planes from the same left side

Solve for $x_1$ and set $x_2 = s, x_3 = t$. The homogeneous equation gives

$$x = s(-5,1,0)^T + t(3,0,1)^T.$$

The equation with right side $-2$ gives

$$x = (-2,0,0)^T + s(-5,1,0)^T + t(3,0,1)^T.$$

The same two independent directions occur in both descriptions. The second plane is the first translated by $(-2,0,0)^T$. They are distinct and disjoint: the same point cannot make the left-hand expression equal both 0 and $-2$.

### Exercise 1.5.16: Translate the homogeneous plane by a particular solution

Using $x_2 = s, x_3 = t$, the homogeneous solutions are

$$x = s(2,1,0)^T + t(-3,0,1)^T.$$

For right side 4 the solutions become

$$x = (4,0,0)^T + s(2,1,0)^T + t(-3,0,1)^T.$$

Both are planes with the same two directions. One passes through zero, while the translated plane does not, since the left side at zero is 0, not 4.

### Exercise 1.5.17: Translate the line from Exercise 5

The last equation gives $x_2 = -4 - x_3$. Divide the first by 2: $x_1 + x_2 + 2x_3 = 4$, so $x_1 = 8 - x_3$. The second equation is again $-2$ times the first and adds nothing. With $x_3 = t$,

$$x = (8, -4, 0)^T + t(-1, -1, 1)^T, \qquad t \in \mathbb{R}.$$

The constant vector satisfies the original right side $(8, -16, 12)^T$. The direction agrees with Exercise 1.5.5, so the two solution lines are parallel; the current line is translated away from the origin.

### Exercise 1.5.18: Translate the line from Exercise 6

The third equation gives $x_2 = x_1 - 8$. The first becomes $3x_1 - 16 - 3x_3 = 5$, so $x_1 = 7 + x_3$ and $x_2 = -1 + x_3$. The second equation checks: $2(7 + x_3) + (-1 + x_3) - 3x_3 = 13$. Thus

$$x = (7, -1, 0)^T + t(1, 1, 1)^T, \qquad t \in \mathbb{R}.$$

This line has the same direction as the homogeneous line in Exercise 1.5.6 and is its translation by $(7, -1, 0)^T$.

### Exercise 1.5.19: A line through a point with a specified direction

A point on the required line is obtained by starting at $a$ and moving any real multiple of $b$. Thus

$$x = (-2, 0)^T + t(-5, 3)^T, \qquad t \in \mathbb{R}.$$

At $t = 0$ the point is $a$; differences between two points are scalar multiples of the nonzero vector $b$, so the line is parallel to $b$.

### Exercise 1.5.20: A line through a point with a specified direction

A point on the required line is obtained by starting at $a$ and moving any real multiple of $b$. Thus

$$x = (3, -2)^T + t(-7, 6)^T, \qquad t \in \mathbb{R}.$$

At $t = 0$ the point is $a$; differences between two points are scalar multiples of the nonzero vector $b$, so the line is parallel to $b$.

### Exercise 1.5.21: Use the difference of two points as direction

Subtract the starting point from the endpoint: $q - p = (1, 4)^T$. Hence the line is

$$x = (3, -3)^T + t(1, 4)^T, \qquad t \in \mathbb{R}.$$

The value $t = 0$ gives $p$ and $t = 1$ gives $q$; all displacements are parallel to $q - p$.

### Exercise 1.5.22: Use the difference of two points as direction

Subtract the starting point from the endpoint: $q - p = (3, -5)^T$. Hence the line is

$$x = (-3, 2)^T + t(3, -5)^T, \qquad t \in \mathbb{R}.$$

The value $t = 0$ gives $p$ and $t = 1$ gives $q$; all displacements are parallel to $q - p$.

### Exercise 1.5.23: Five statements about homogeneous and translated systems

(a) **True.** $x = 0$ always solves a homogeneous equation.

(b) **False.** $Ax = 0$ is an implicit condition. Parametric vector form is an explicit description in terms of free parameters.

(c) **False.** The trivial solution exists whether or not there is a free variable. A free variable is equivalent to existence of a *nontrivial* solution.

(d) **False.** $x = p + tv$ describes the line through $p$ parallel to $v$, not in general a line through $v$ parallel to $p$. For example $p = (1, 0)^T, v = (0, 1)^T$ give the vertical line $x_1 = 1$.

(e) **False as stated.** Such a representation requires $p$ to be a particular solution and $Ax = b$ to be consistent. Without those hypotheses it need not hold. With them it is Theorem 6.

### Exercise 1.5.24: Five further statements and their justifications

(a) **False.** A homogeneous system has the solution 0, so it cannot be inconsistent.

(b) **False.** Nontrivial means not *all* coordinates are zero. For example $(1, 0)^T$ is a nontrivial solution of $[0\ 1]x = 0$ even though its second entry is zero.

(c) **True.** The displacement from $v$ to $v + p$ is $(v + p) - v = p$, so it is in the direction of $p$ (with no movement when $p = 0$).

(d) **True.** If 0 solves $Ax = b$, then $b = A0 = 0$; that is exactly the homogeneous case.

(e) **True.** Consistency supplies a particular solution $p$; Theorem 6 states the entire solution set is $p + \{v : Av = 0\}$.

### Exercise 1.5.25: Prove the translation theorem in both directions

**What we must show.** Assume $Ap = b$. Relate every solution to the homogeneous system.

**(a) Add a homogeneous solution.**

If $Av_h = 0$ and $w = p + v_h$, additivity of multiplication gives $Aw = Ap + Av_h = b + 0 = b$. Thus every proposed vector really is a solution.

**(b) Recover the homogeneous part from any solution.**

If $Aw = b$, let $v_h = w - p$. Then $Av_h = A(w - p) = Aw - Ap = b - b = 0$, and $w = p + v_h$.

**Conclusion.** Both inclusions hold, so the two described solution sets are exactly equal, not merely one contained in the other.

### Exercise 1.5.26: The zero coefficient matrix imposes no restriction

For the $3 \times 3$ zero matrix, $Ax = (0,0,0)^T$ for every input $x \in \mathbb{R}^3$. Thus every vector satisfies the equation.

$$x = s(1,0,0)^T + t(0,1,0)^T + u(0,0,1)^T, \qquad s,t,u \in \mathbb{R}.$$

**Conclusion.** The solution set is all of $\mathbb{R}^3$.

### Exercise 1.5.27: Uniqueness is equivalent to a trivial homogeneous kernel

Because a solution exists, fix one and call it $p$. By Theorem 6, all solutions are $p + v_h$ with $Av_h = 0$. If the only homogeneous solution is 0, the only translated solution is $p + 0 = p$, so uniqueness holds. Conversely, if there were a nonzero homogeneous $v_h$, the vector $p + v_h \neq p$ would give a second solution. This contradicts uniqueness.

**Conclusion.** The equivalence follows in both directions. Consistency is needed in order to fix $p$.

### Exercise 1.5.28: Use the numbers of pivots, rows, and columns

There are 3 unknowns and 3 coefficient pivots. Consequently there are 0 free variables.

**(a) Nontrivial homogeneous solutions.**

Every variable is a pivot variable, so the only homogeneous solution is zero; there is no nontrivial solution.

**(b) Consistency for every right side.**

Every one of the 3 rows contains a pivot. Theorem 4 therefore guarantees consistency for every $b \in \mathbb{R}^3$.

### Exercise 1.5.29: Use the numbers of pivots, rows, and columns

There are 4 unknowns and 3 coefficient pivots. Consequently there are 1 free variables.

**(a) Nontrivial homogeneous solutions.**

Since there is at least one free variable, the homogeneous system has nontrivial solutions.

**(b) Consistency for every right side.**

There are fewer pivots than rows. Theorem 4 therefore says that some right sides have no solution.

### Exercise 1.5.30: Use the numbers of pivots, rows, and columns

There are 5 unknowns and 2 coefficient pivots. Consequently there are 3 free variables.

**(a) Nontrivial homogeneous solutions.**

Since there is at least one free variable, the homogeneous system has nontrivial solutions.

**(b) Consistency for every right side.**

Every one of the 2 rows contains a pivot. Theorem 4 therefore guarantees consistency for every $b \in \mathbb{R}^2$.

### Exercise 1.5.31: Use the numbers of pivots, rows, and columns

There are 2 unknowns and 2 coefficient pivots. Consequently there are 0 free variables.

**(a) Nontrivial homogeneous solutions.**

Every variable is a pivot variable, so the only homogeneous solution is zero; there is no nontrivial solution.

**(b) Consistency for every right side.**

There are fewer pivots than rows. Theorem 4 therefore says that some right sides have no solution.

### Exercise 1.5.32: A nonzero right side excludes the origin

If a solution set were a plane through the origin, it would contain $x = 0$. But $A0 = 0 \neq b$ when $b \neq 0$.

**Conclusion.** No. A nonzero right side makes the origin impossible as a solution, regardless of the dimension of the solution set.

### Exercise 1.5.33: Arrange for row sums to vanish

Take

$$A = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & -1 \end{bmatrix}.$$

This is nonzero, and multiplying by $(1, 1, 1)^T$ gives row sums $1 - 1 + 0 = 0$, $0 + 1 - 1 = 0$, and $1 + 0 - 1 = 0$.

**Conclusion.** The displayed matrix has the required nonzero null vector.

### Exercise 1.5.34: Make every row orthogonal to the prescribed vector

Take

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & -2 \end{bmatrix}.$$

The row products with $(2, -1, 1)^T$ are respectively $2 - 2 = 0$, $-1 + 1 = 0$, and $2 - 2 = 0$. Thus $A(2, -1, 1)^T = 0$, and $A$ is plainly not the zero matrix.

### Exercise 1.5.35: Cancel proportional columns by inspection

The second column is three times the first. Therefore $-3a_1 + a_2 = 0$. By the column definition this means $A(-3, 1)^T = 0$. The vector is nonzero, so it is nontrivial. No elimination is needed.

### Exercise 1.5.36: Cancel a different proportionality

The columns satisfy $2a_1 + 3a_2 = 0$, since $2(3, -6, 12)^T + 3(-2, 4, -8)^T = (0, 0, 0)^T$. Thus $(2, 3)^T$ is a nontrivial solution of $Ax = 0$.

### Exercise 1.5.37: An inconsistent right side is not a translated line

Choose

$$A = \begin{bmatrix} 1 & -4 \\ 0 & 0 \end{bmatrix}.$$

The homogeneous equation is $x_1 = 4x_2$, with solution line $x = t(4, 1)^T$, exactly the line through $(4, 1)^T$ and zero. Now choose $b = (0, 1)^T$. The second row of $Ax = b$ says $0 = 1$, so its solution set is empty, not a line parallel to the original one.

**Conclusion.** There is no contradiction: Theorem 6 assumes consistency, and this selected right side violates that assumption.

### Exercise 1.5.38: Scalar multiples stay in the homogeneous solution set

The scalar-multiplication rule gives

$$A(cw) = c(Aw) = c0 = 0.$$

This holds for every real $c$, including $c = 0$ and negative values.

**Conclusion.** Every multiple $cw$ is another homogeneous solution.

### Exercise 1.5.39: Sums and linear combinations stay in the solution set

Additivity first gives $A(v+w) = Av+Aw = 0+0 = 0$. For any scalars $c, d$, combine both multiplication rules:

$$A(cv + dw) = A(cv) + A(dw) = cAv + dAw = c0 + d0 = 0.$$

**Conclusion.** Every such linear combination is a homogeneous solution.

### Exercise 1.5.40: A missing right side rules out any unique solution

Suppose there were a right-side vector $y$ for which $Ax = y$ had a unique solution. Uniqueness would mean all three coefficient columns have pivots. A $3 \times 3$ matrix would then have a pivot in every row. Theorem 4 would make $Ax = b$ consistent for every $b \in \mathbb{R}^3$, contradicting the specified missing right side. Equivalently, the missing right side implies at most two pivots, so at least one variable is free. Every consistent system with this coefficient matrix consequently has infinitely many solutions.

**Conclusion.** No vector on the right can produce a unique solution.

## 1.6 Applications of linear systems

Three modeling ideas organize this section.

- In an exchange economy, column $j$ of the exchange table records where sector $j$'s output goes. Each column sums to 1. If $E$ is this table and $p$ lists the values of total outputs, equilibrium means $Ep = p$, or $(E - I)p = 0$.
- In a chemical reaction, conserve each type of atom separately. Assign positive unknown coefficients to all compounds; write reactant atom counts minus product atom counts equal to zero. Choose the smallest positive integer solution.
- In a flow network, write incoming flow equals outgoing flow at every node. After solving the linear equations, apply the required nonnegativity inequalities. A free flow parameter is not automatically allowed to take negative values.

### Practice problem 1.6.P1: Build the exchange table in the stated order

Use the order Agriculture, Mining, Manufacturing for both rows and columns. The three columns are respectively $(.65, .05, .30)^T$, $(.20, .10, .70)^T$, and $(.20, .30, .50)^T$:

$$E = \begin{bmatrix} .65 & .20 & .20 \\ .05 & .10 & .30 \\ .30 & .70 & .50 \end{bmatrix}.$$

For example Agriculture retains $1 - .05 - .30 = .65$, so that value belongs in the first entry of the first column. Each displayed column sums to 1; this checks that all output was allocated exactly once.

### Practice problem 1.6.P2: Use flow inequalities after solving the equations

Example 2 gives $x_1 = 600 - x_5$, $x_2 = 200 + x_5$, $x_3 = 400$, $x_4 = 500 - x_5$. Nonnegative $x_5$ and $x_4$ imply $0 \le x_5 \le 500$. The remaining two inequalities are then automatic. Consequently $100 \le x_1 \le 600$ and $200 \le x_2 \le 700$. Both endpoints occur, at $x_5 = 500$ and $x_5 = 0$, respectively. The two flows vary together: as $x_1$ decreases, $x_2$ increases.

### Exercise 1.6.1: Two-sector equilibrium is a ratio, not a unique price list

Use the order Goods, Services. The exchange matrix and equations are

$$E = \begin{bmatrix} .20 & .70 \\ .80 & .30 \end{bmatrix}, \qquad .20p_G + .70p_S = p_G, \quad .80p_G + .30p_S = p_S.$$

Both equations simplify to $8p_G = 7p_S$. Thus

$$p = t(7, 8)^T, \qquad t > 0.$$

For example output values 7 and 8 work: Goods spends $.20(7) + .70(8) = 7$, while Services spends $.80(7) + .30(8) = 8$.

**Conclusion.** Any common positive multiple of $(7, 8)^T$ is an equilibrium.

### Exercise 1.6.2: A currency change rescales every output value

In the Coal–Electric–Steel order of Example 1, the exact equations are

$$p_C = .4p_E + .6p_S, \qquad .9p_E = .6p_C + .2p_S.$$

Insert the first into the second: $.9p_E = .24p_E + .56p_S$, so $.66p_E = .56p_S$ and $p_E = (28/33)p_S$. Then $p_C = (31/33)p_S$. Thus an exact alternative is $(p_C, p_E, p_S) = (62, 56, 66)$. Checking the third row gives $.4(62) + .5(56) + .2(66) = 66$. The example rounds its decimal answer; these rational ratios avoid that rounding error. If one dollar is $k > 0$ yen, every component is multiplied by $k$. Because $E(kp) = kEp = kp$, the exchange proportions and equilibrium relations do not change, only the numerical currency units do.

### Exercise 1.6.3: A three-sector exchange economy

#### (a) Place each sector's sales in its column.

In the order Fuels and Power, Manufacturing, Services,

$$E = \begin{bmatrix} .10 & .10 & .20 \\ .80 & .10 & .40 \\ .10 & .80 & .40 \end{bmatrix}.$$

The retained amounts are $.10, .10, .40$, respectively. Each column sums to 1.

#### (b) Set the output value equal to the cost of purchased inputs.

Writing the values as $f, m, s$, the equations $Ep = p$ are

$$.10f + .10m + .20s = f, \quad .80f + .10m + .40s = m, \quad .10f + .80m + .40s = s.$$

Multiplying $(I - E)p = 0$ by 10 gives the augmented matrix

$$\left[\begin{array}{rrr|c} 9 & -1 & -2 & 0 \\ -8 & 9 & -4 & 0 \\ -1 & -8 & 6 & 0 \end{array}\right].$$

#### (c) Set the Services value to 100 and solve exactly.

The first equation gives $m = 9f - 2s$. Insert this in the second: $-8f + 9(9f - 2s) - 4s = 0$, hence $73f = 22s$. Thus $f = 22s/73$ and $m = 52s/73$. For $s = 100$,

$$p = (2200/73, 5200/73, 100)^T \approx (30.136986, 71.232877, 100)^T.$$

The third equation checks because $-22 - 8(52) + 6(73) = 0$ after factoring out $s/73$.

### Exercise 1.6.4: An exact four-sector equilibrium

#### (a) Construct the exchange matrix.

Use Mining, Lumber, Energy, Transportation in that order:

$$E = \begin{bmatrix} .30 & .15 & .20 & .20 \\ .10 & .15 & .15 & .10 \\ .60 & .50 & .45 & .50 \\ 0 & .20 & .20 & .20 \end{bmatrix}.$$

For instance Mining retains $1-.10-.60=.30$, and the Energy column retains $1-.20-.15-.20=.45$.

**(b) Solve $(I-E)p=0$.**

Multiplying by 20 avoids decimals. The coefficient matrix is

$$20(I-E) = \begin{bmatrix} 14 & -3 & -4 & -4 \\ -2 & 17 & -3 & -2 \\ -12 & -10 & 11 & -10 \\ 0 & -4 & -4 & 16 \end{bmatrix}.$$

Its reduced equations are

$$p_M = \frac{193}{141}p_T, \qquad p_L = \frac{118}{141}p_T, \qquad p_E = \frac{446}{141}p_T.$$

For clarity one can obtain these ratios without decimal arithmetic: the last equation gives $p_L + p_E = 4p_T$. Substituting $p_E = 4p_T - p_L$ in the second gives $20p_L = 2p_M + 14p_T$; the first gives $14p_M + p_L = 20p_T$. Solving these two equations gives the displayed ratios. Choose $p_T = 141$. Then $p = (193, 118, 446, 141)^T$ is positive. The rows of $20(I-E)p$ are $2702 - 354 - 1784 - 564 = 0$, $-386 + 2006 - 1338 - 282 = 0$, $-2316 - 1180 + 4906 - 1410 = 0$, and $-472 - 1784 + 2256 = 0$.

**Conclusion.** One exact equilibrium is $(193, 118, 446, 141)^T$; every positive common multiple also works.

### Exercise 1.6.5: Compare the equilibrium before and after a change in trade

**(a) Build the original table.**

Use Agriculture, Manufacturing, Services, Transportation:

$$E = \begin{bmatrix} .20 & .35 & .10 & .20 \\ .20 & .10 & .20 & .30 \\ .30 & .35 & .50 & .20 \\ .30 & .20 & .20 & .30 \end{bmatrix}.$$

Each column is the allocation of the sector at its top, including its retained share on the diagonal.

**(b) Normalize Transportation to 10.**

Writing $a, m, s, t$ for the four output values, multiply $(I-E)p=0$ by 20. The equations are

$$\begin{aligned} 16a - 7m - 2s - 4t &= 0, & -4a + 18m - 4s - 6t &= 0, \\ -6a - 7m + 10s - 4t &= 0, & -6a - 4m - 4s + 14t &= 0. \end{aligned}$$

The second and fourth give $2m + 2s = 7t - 3a$ and $-2a + 9m - 2s = 3t$, so $11m = 10t - a$. The first equation together with $2m + 2s = 7t - 3a$ gives $19a - 5m = 11t$. Substitute $11m = 10t - a$ to

obtain $209a - 5(10t - a) = 121t$, or $214a = 171t$. Therefore $a = (171/214)t$, $m = (179/214)t$, and $s = (627/428)t$. For $t = 10$,

$$p = (855/107, 895/107, 3135/214, 10)^T \approx (7.990654, 8.364486, 14.649533, 10)^T.$$

**(c) Change just the Agriculture column.**

The new shares from Agriculture are $.20, .10, .40, .30$; the other columns stay the same. Thus

$$\widetilde{E} = \begin{bmatrix} .20 & .35 & .10 & .20 \\ .10 & .10 & .20 & .30 \\ .40 & .35 & .50 & .20 \\ .30 & .20 & .20 & .30 \end{bmatrix}.$$

**(d) Solve and compare with the same normalization.**

The new coefficient matrix is

$$20(I - \widetilde{E}) = \begin{bmatrix} 16 & -7 & -2 & -4 \\ -2 & 18 & -4 & -6 \\ -8 & -7 & 10 & -4 \\ -6 & -4 & -4 & 14 \end{bmatrix}.$$

Its reduced equations give $a = (57/73)t$, $m = (56/73)t$, and $s = (114/73)t$. For example the last row again gives $2m + 2s = 7t - 3a$, and the second becomes $11m = 10t - 2a$; the first still gives $19a - 5m = 11t$. Substitution now yields $209a - 5(10t - 2a) = 121t$, hence $219a = 171t$ and $a = (57/73)t$, from which the other two ratios follow. At $t = 10$ the exact solution is

$$\widetilde{p} = (570/73, 560/73, 1140/73, 10)^T \approx (7.808219, 7.671233, 15.616438, 10)^T.$$

Direct multiplication verifies both $Ep = p$ and $\widetilde{E}\widetilde{p} = \widetilde{p}$. With Transportation fixed, Agriculture and Manufacturing decrease by about 0.182435 and 0.693253, while Services increases by about 0.966905. These are changes in the values of total sector outputs under the model's convention, not independently determined retail prices of individual products.

## Exercise 1.6.6: Balance aluminum oxide and carbon

Let the four coefficients, in the printed order, be $a, b, c, d$. Using atom order Al, O, C gives the vector equation

$$a(2, 3, 0)^T + b(0, 0, 1)^T = c(1, 0, 0)^T + d(0, 2, 1)^T.$$

Thus $2a = c$, $3a = 2d$, and $b = d$. Choose the smallest positive $a$ making all coefficients integral: $a = 2$ gives $d = 3, b = 3, c = 4$.

**Conclusion.** $2\mathrm{Al_2O_3} + 3\mathrm{C} \longrightarrow 4\mathrm{Al} + 3\mathrm{CO_2}$. The atom totals are Al: $4 = 4$, O: $6 = 6$, C: $3 = 3$.

## Exercise 1.6.7: Count every hydrogen in citric acid

Let the coefficients of $\mathrm{NaHCO_3}, \mathrm{H_3C_6H_5O_7}, \mathrm{Na_3C_6H_5O_7}, \mathrm{H_2O}, \mathrm{CO_2}$ be $a, b, c, d, e$. The citric acid molecule has $3 + 5 = 8$ hydrogen atoms. In atom order Na, H, C, O, the conservation equations are

$$a = 3c, \quad a + 8b = 5c + 2d, \quad a + 6b = 6c + e, \quad 3a + 7b = 7c + d + 2e.$$

This is the coordinate form of the vector equation obtained by listing those four atom counts for each compound. Set $c = 1$, so $a = 3$. The hydrogen equation gives $d = 4b - 1$; the carbon equation gives $e = 6b - 3$. The oxygen equation becomes $9 + 7b = 7 + (4b - 1) + 2(6b - 3) = 16b$, hence $b = 1$. Thus $d = e = 3$.

**Conclusion.** $3NaHCO_3 + H_3C_6H_5O_7 \longrightarrow Na_3C_6H_5O_7 + 3H_2O + 3CO_2$. Atom totals on both sides are Na: 3, H: 11, C: 9, O: 16.

### Exercise 1.6.8: Balance limestone neutralization

Use coefficients $a, b, c, d, e$ in the printed order, and atom order H, O, Ca, C. The vector equation gives

$$3a = 2c, \quad a + 3b = c + 2e, \quad b = d, \quad b = e.$$

Substitute $d = e = b$ in the oxygen equation to get $c = a + b$. Then $3a = 2(a + b)$ gives $a = 2b$. The smallest choice $b = 1$ therefore gives $(a, b, c, d, e) = (2, 1, 3, 1, 1)$.

**Conclusion.** $2H_3O + CaCO_3 \longrightarrow 3H_2O + Ca + CO_2$. The H, O, Ca, C totals are respectively $6, 5, 1, 1$ on each side.

### Exercise 1.6.9: Conserve boron, sulfur, hydrogen, and oxygen

With coefficients $a, b, c, d$, the atom-count vector equation yields

$$2a = c, \qquad 3a = d, \qquad 2b = 3c + 2d, \qquad b = 3c.$$

The first, second, and fourth give $c = 2a, d = 3a, b = 6a$. The hydrogen equation checks: $2(6a) = 3(2a) + 2(3a)$. Taking $a = 1$ gives the primitive integer coefficients.

**Conclusion.** $B_2S_3 + 6H_2O \longrightarrow 2H_3BO_3 + 3H_2S$. The atom totals B, S, H, O are $2, 3, 12, 6$.

### Exercise 1.6.10: Use exact fractions in a six-compound reaction

Let the coefficients of the six compounds, in printed order, be $a, b, c, d, e, f$. The vector equation in atom order Pb, N, Cr, Mn, O has coefficient matrix

$$\begin{bmatrix} 1 & 0 & -3 & 0 & 0 & 0 \\ 6 & 0 & 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & -2 & 0 & 0 \\ 0 & 2 & 0 & 0 & -1 & 0 \\ 0 & 8 & -4 & -3 & -2 & -1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \\ e \\ f \end{bmatrix} = 0.$$

The first four rows give $a = 3c$, $f = 6a = 18c$, $b = 2d$, and $e = 2b = 4d$. The last row says

$$8b = 4c + 3d + 2e + f \quad \Longrightarrow \quad 16d = 22c + 11d, \quad 5d = 22c.$$

To get positive integers choose $c = 5$, so $d = 22$, $a = 15, b = 44, e = 88, f = 90$. The entries have common divisor 1.

**Conclusion.** $15PbN_6 + 44CrMn_2O_8 \longrightarrow 5Pb_3O_4 + 22Cr_2O_3 + 88MnO_2 + 90NO$. The oxygen check is $44(8) = 5(4) + 22(3) + 88(2) + 90 = 352$. The other rows already check each remaining element.

### Exercise 1.6.11: Balance seven compounds without decimal rounding

Use coefficients $a, b, c, d, e, f, g$ for the seven printed compounds. Conserve Mn, S, As, Cr, H, O, respectively:

$$\begin{aligned} a &= d, & a + c &= 3f, & 2b &= e, \\ 10b &= f, & 2c &= d + 3e + 2g, & 35b + 4c &= 4d + 12f + g. \end{aligned}$$

These are the coordinates of a single atom-count vector equation. The first four give $d = a$, $e = 2b$, $f = 10b$, $c = 30b - a$. The hydrogen equation becomes $2g = 54b - 3a$. The oxygen equation becomes $g = 35b - 8a$. Equating the two formulas gives

$$54b - 3a = 70b - 16a, \qquad 13a = 16b.$$

Choose $b = 13, a = 16$. Then $c = 374, d = 16, e = 26, f = 130, g = 327$.

$$16\mathrm{MnS} + 13\mathrm{As_2Cr_{10}O_{35}} + 374\mathrm{H_2SO_4} \longrightarrow 16\mathrm{HMnO_4} + 26\mathrm{AsH_3} + 130\mathrm{CrS_3O_{12}} + 327\mathrm{H_2O}.$$

Check sulfur: $16 + 374 = 390 = 3(130)$; hydrogen: $2(374) = 748 = 16 + 3(26) + 2(327)$; oxygen: $35(13) + 4(374) = 1951 = 4(16) + 12(130) + 327$. Mn, As, and Cr match by the first equations.

**Conclusion.** The displayed positive integer coefficients balance every atom; their common divisor is 1.

### Exercise 1.6.12: Use junction equations before imposing nonnegative flow

Read the arrows at the three nodes. Node A receives $x_1 + x_4$ and sends $x_2$; node B receives $x_2$ and sends $x_3 + 100$; node C receives $x_3 + 80$ and sends $x_4$. Thus

$$x_1 + x_4 = x_2, \qquad x_2 = x_3 + 100, \qquad x_4 = x_3 + 80.$$

The last two inserted in the first give $x_1 = 20$. Set $x_3 = t$. The general algebraic flow is

$$(x_1, x_2, x_3, x_4) = (20, 100 + t, t, 80 + t).$$

Nonnegative flows require $t \geq 0$; that condition makes every other component nonnegative too. Hence $x_4 = 80 + t \geq 80$, and $t = 0$ attains the lower bound.

**Conclusion.** The smallest possible $x_4$ is 80.

### Exercise 1.6.13: A five-node network with two independent free flows

#### (a) Record all node balances.

Following the arrows in the supplied figure gives

$$\begin{aligned} A:&\quad 30 + x_2 = 80 + x_1, & E:&\quad 60 + x_1 = 20 + x_3, \\ B:&\quad x_3 + x_5 = x_2 + x_4, & C:&\quad 100 + x_6 = 40 + x_5, \\ D:&\quad 40 + x_4 = 90 + x_6. \end{aligned}$$

Let $s = x_1$ and $t = x_6$. The first, second, fourth, and fifth balances give

$$(x_1, x_2, x_3, x_4, x_5, x_6) = (s, s + 50, s + 40, t + 50, t + 60, t).$$

The balance at B checks identically: $(s + 40) + (t + 60) = (s + 50) + (t + 50)$.

**(b) Apply the arrow-direction restrictions.**

Nonnegative $x_1$ and $x_6$ require $s, t \geq 0$, and these inequalities make all other flows nonnegative. Their individual minima are therefore

$$\min x_2 = 50, \quad \min x_3 = 40, \quad \min x_4 = 50, \quad \min x_5 = 60.$$

They can all be attained at once by taking $s = t = 0$.

### Exercise 1.6.14: A road closure specializes a two-parameter traffic pattern

**(a) Write flow in equals flow out at each intersection.**

The four balances read

$$80 = x_1 + x_5, \quad 100 + x_1 + x_2 = x_4, \quad x_3 = x_2 + 90, \quad x_4 + x_5 = x_3 + 90.$$

The last is dependent on the first three; total external inflow $80 + 100$ equals total external outflow $90 + 90$. Put $s = x_2$ and $t = x_5$. Then

$$(x_1, x_2, x_3, x_4, x_5) = (80 - t, s, 90 + s, 180 + s - t, t).$$

For the indicated traffic directions require $s \geq 0$ and $0 \leq t \leq 80$; the remaining inequalities then follow.

**(b) Close the road carrying $x_5$.**

Setting $t = 0$ gives

$$(x_1, x_2, x_3, x_4, x_5) = (80, s, 90 + s, 180 + s, 0), \qquad s \geq 0.$$

This is the entire feasible family with the road closed, rather than one arbitrarily selected flow pattern.

**(c) Minimize $x_4$.**

The formula $x_4 = 180 + s$ with $s \geq 0$ gives the lower bound 180. It is attained when $s = 0$.

**Conclusion.** With $x_5 = 0$, the minimum flow in branch 4 is 180 cars/minute.

### Exercise 1.6.15: Circulating flow around a one-way roundabout

Read the successive balances around A, B, C, D, E, F:

$$x_6 + 60 = x_1, \quad x_1 = x_2 + 70, \quad x_2 + 100 = x_3,$$

$$x_3 = x_4 + 90, \quad x_4 + 80 = x_5, \quad x_5 = x_6 + 80.$$

Choose $t = x_6$. Working around the circle yields

$$(x_1, x_2, x_3, x_4, x_5, x_6) = (t + 60, t - 10, t + 90, t, t + 80, t).$$

The final equation checks because $(t + 80) - t = 80$. To respect every arrow all six flows must be nonnegative. The strongest restriction is $x_2 = t - 10 \geq 0$, namely $t \geq 10$; this also ensures each other component is nonnegative.

**Conclusion.** All feasible flows are the displayed family with $t \geq 10$, and the smallest possible $x_6$ is 10.

## 1.7 Linear independence

**The main question.** For vectors $v_1, \ldots, v_p$, write $c_1v_1+\cdots+c_pv_p = 0$. The list is *linearly independent* if its only solution is $c_1 = \cdots = c_p = 0$. It is dependent if a solution exists with at least one nonzero coefficient. Notice that the zero-coefficient solution always exists, so its existence proves nothing. Putting the vectors into the columns of $A$ turns the question into $Ac = 0$. A pivot in every column means no free variable, hence independence. A free variable supplies a nontrivial dependence relation.

### Practice problem 1.7.P1: Check every pair, not only the whole list

The four vectors are $u = (3, 2, -4)$, $v = (-6, 1, 7)$, $w = (0, -5, 2)$, and $z = (3, 7, -5)$. None is zero. For two nonzero vectors, dependence means one is a scalar multiple of the other. We can disprove this by looking just at their first two coordinates: the quantities $a_1b_2 - a_2b_1$ for $(u, v), (u, w), (u, z), (v, w), (v, z), (w, z)$ are respectively $15, -15, 15, 30, -45, 15$. None is zero. For example, if $v = tu$, then $3(1) - 2(-6)$ would have to be zero; it is 15.

**Conclusion.** All six pairs are linearly independent.

### Practice problem 1.7.P2: Pairwise independence is weaker than independence

No. Checking pairs rules out a relation using only two vectors, but it does not rule out a relation using three or four. For instance, $e_1, e_2, e_1 + e_2$ are pairwise independent yet $e_1 + e_2 - (e_1 + e_2) = 0$ is a nontrivial relation among all three. The same logical distinction applies to the four given vectors.

### Practice problem 1.7.P3: Why testing a selected vector can miss dependence

It is not a reliable test to ask only whether $w$ is in the span of the other three. A dependence relation may have coefficient zero on $w$. For these data, solving with the actual coordinates gives $z = 3u + v$. Thus $3u + v - z = 0$ already establishes dependence without involving $w$. Moreover $w$ is outside $\text{Span}\{u, v\}$: the equations $3a - 6b = 0$, $2a + b = -5$ force $b = -1$, $a = -2$, but then $-4a + 7b = 1 \neq 2$. So $w \notin \text{Span}\{u, v, z\}$ despite dependence of the whole list. Testing one chosen vector could therefore mislead us.

### Practice problem 1.7.P4: More vectors than coordinates

There are four vectors in $\mathbb{R}^3$, so their $3 \times 4$ column matrix cannot have a pivot in all four columns. At least one variable is free in the homogeneous system, and the list is dependent. The relation $3u + v - z = 0$ from the preceding problem gives a direct check.

### Exercise 1.7.1: Test the column equation

**What we must show.** Determine whether the given vectors are independent. Form $Ac = 0$; row operations preserve exactly its solutions.

$$A = \begin{bmatrix} 5 & 7 & 9 \\ 0 & 2 & 4 \\ 0 & -6 & -8 \end{bmatrix}.$$

Use $R_1 \leftarrow \frac{1}{5}R_1$. Then

$$\begin{bmatrix} 1 & \frac{7}{5} & \frac{9}{5} \\ 0 & 2 & 4 \\ 0 & -6 & -8 \end{bmatrix}.$$

Use $R_2 \leftarrow \frac{1}{2}R_2, \quad R_1 \leftarrow R_1 - (\frac{7}{5})R_2, \quad R_3 \leftarrow R_3 - (-6)R_2$. Then

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 4 \end{bmatrix}.$$

Use $R_3 \leftarrow \frac{1}{4}R_3, \quad R_1 \leftarrow R_1 - (-1)R_3, \quad R_2 \leftarrow R_2 - (2)R_3$. Then

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Every column has a pivot. The reduced equations force every coefficient in $c$ to be zero, so there is no nontrivial dependence relation.

**Conclusion.** The vectors are linearly independent.

### Exercise 1.7.2: Test the column equation

**What we must show.** Determine whether the given vectors are independent. Form $Ac = 0$; row operations preserve exactly its solutions.

$$A = \begin{bmatrix} 0 & 0 & -1 \\ 2 & 0 & 3 \\ 3 & -8 & 1 \end{bmatrix}.$$

Use $R_1 \leftrightarrow R_2, \quad R_1 \leftarrow \frac{1}{2}R_1, \quad R_3 \leftarrow R_3 - (3)R_1$. Then

$$\begin{bmatrix} 1 & 0 & \frac{3}{2} \\ 0 & 0 & -1 \\ 0 & -8 & -\frac{7}{2} \end{bmatrix}.$$

Use $R_2 \leftrightarrow R_3, \quad R_2 \leftarrow -\frac{1}{8}R_2$. Then

$$\begin{bmatrix} 1 & 0 & \frac{3}{2} \\ 0 & 1 & \frac{7}{16} \\ 0 & 0 & -1 \end{bmatrix}.$$

Use $R_3 \leftarrow -1R_3, \quad R_1 \leftarrow R_1 - (\frac{3}{2})R_3, \quad R_2 \leftarrow R_2 - (\frac{7}{16})R_3$. Then

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Every column has a pivot. The reduced equations force every coefficient in $c$ to be zero, so there is no nontrivial dependence relation.

**Conclusion.** The vectors are linearly independent.

### Exercise 1.7.3: Test the column equation

**What we must show.** Determine whether the given vectors are independent. Form $Ac = 0$; row operations preserve exactly its solutions.

$$A = \begin{bmatrix} 2 & -4 \\ -3 & 6 \end{bmatrix}.$$

Use $R_1 \leftarrow \frac{1}{2}R_1, \quad R_2 \leftarrow R_2 - (-3)R_1$. Then

$$\begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix}.$$

There is a free variable. For example, the nonzero coefficient vector $c = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ satisfies $Ac = 0$ (substitution checks every coordinate). This is a nontrivial relation among the columns.

**Conclusion.** The vectors are linearly dependent.

### Exercise 1.7.4: Test the column equation

**What we must show.** Determine whether the given vectors are independent. Form $Ac = 0$; row operations preserve exactly its solutions.

$$A = \begin{bmatrix} -1 & -3 \\ 3 & -9 \end{bmatrix}.$$

Use $R_1 \leftrightarrow R_2, \quad R_1 \leftarrow \frac{1}{3}R_1, \quad R_2 \leftarrow R_2 - (-1)R_1$. Then

$$\begin{bmatrix} 1 & -3 \\ 0 & -6 \end{bmatrix}.$$

Use $R_2 \leftarrow -\frac{1}{6}R_2, \quad R_1 \leftarrow R_1 - (-3)R_2$. Then

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Every column has a pivot. The reduced equations force every coefficient in $c$ to be zero, so there is no nontrivial dependence relation.

**Conclusion.** The vectors are linearly independent.

### Exercise 1.7.5: Test the column equation

**What we must show.** Determine whether the given vectors (the columns of the matrix) are independent. Form $Ac = 0$; row operations preserve exactly its solutions.

$$A = \begin{bmatrix} 0 & -3 & 9 \\ 2 & 1 & -7 \\ -1 & 4 & -5 \\ 1 & -4 & -2 \end{bmatrix}.$$

Use $R_1 \leftrightarrow R_4$, $R_2 \leftarrow R_2 - (2)R_1$, $R_3 \leftarrow R_3 - (-1)R_1$. Then

$$\begin{bmatrix} 1 & -4 & -2 \\ 0 & 9 & -3 \\ 0 & 0 & -7 \\ 0 & -3 & 9 \end{bmatrix}.$$

Use $R_2 \leftarrow \frac{1}{9}R_2$, $R_1 \leftarrow R_1 - (-4)R_2$, $R_4 \leftarrow R_4 - (-3)R_2$. Then

$$\begin{bmatrix} 1 & 0 & -\frac{10}{3} \\ 0 & 1 & -\frac{1}{3} \\ 0 & 0 & -7 \\ 0 & 0 & 8 \end{bmatrix}.$$

Use $R_3 \leftrightarrow R_4$, $R_3 \leftarrow \frac{1}{8}R_3$, $R_1 \leftarrow R_1 - (-\frac{10}{3})R_3$, $R_2 \leftarrow R_2 - (-\frac{1}{3})R_3$, $R_4 \leftarrow R_4 - (-7)R_3$. Then

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Every column has a pivot. The reduced equations force every coefficient in $c$ to be zero, so there is no nontrivial dependence relation.

**Conclusion.** The vectors are linearly independent.

### Exercise 1.7.6: Test the column equation

**What we must show.** Determine whether the given vectors (the columns of the matrix) are independent. Form $Ac = 0$; row operations preserve exactly its solutions.

$$A = \begin{bmatrix} -4 & -3 & 0 \\ 0 & -1 & 5 \\ 1 & 1 & -5 \\ 2 & 1 & -10 \end{bmatrix}.$$

Use $R_1 \leftrightarrow R_3$, $R_3 \leftarrow R_3 - (-4)R_1$, $R_4 \leftarrow R_4 - (2)R_1$. Then

$$\begin{bmatrix} 1 & 1 & -5 \\ 0 & -1 & 5 \\ 0 & 1 & -20 \\ 0 & -1 & 0 \end{bmatrix}.$$

Use $R_2 \leftrightarrow R_3$, $R_1 \leftarrow R_1 - (1)R_2$, $R_3 \leftarrow R_3 - (-1)R_2$, $R_4 \leftarrow R_4 - (-1)R_2$. Then

$$\begin{bmatrix} 1 & 0 & 15 \\ 0 & 1 & -20 \\ 0 & 0 & -15 \\ 0 & 0 & -20 \end{bmatrix}.$$

Use $R_3 \leftarrow -\frac{1}{15}R_3$, $R_1 \leftarrow R_1 - (15)R_3$, $R_2 \leftarrow R_2 - (-20)R_3$, $R_4 \leftarrow R_4 - (-20)R_3$. Then

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Every column has a pivot. The reduced equations force every coefficient in $c$ to be zero, so there is no nontrivial dependence relation.

**Conclusion.** The vectors are linearly independent.

### Exercise 1.7.7: Test the column equation

**What we must show.** Determine whether the given vectors (the columns of the matrix) are independent. Form $Ac = 0$; row operations preserve exactly its solutions.

$$A = \begin{bmatrix} 1 & 4 & -3 & 0 \\ -2 & -7 & 5 & 1 \\ -4 & -5 & 7 & 5 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - (-2)R_1$, $R_3 \leftarrow R_3 - (-4)R_1$. Then

$$\begin{bmatrix} 1 & 4 & -3 & 0 \\ 0 & 1 & -1 & 1 \\ 0 & 11 & -5 & 5 \end{bmatrix}.$$

Use $R_1 \leftarrow R_1 - (4)R_2$, $R_3 \leftarrow R_3 - (11)R_2$. Then

$$\begin{bmatrix} 1 & 0 & 1 & -4 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 6 & -6 \end{bmatrix}.$$

Use $R_3 \leftarrow \frac{1}{6}R_3$, $R_1 \leftarrow R_1 - (1)R_3$, $R_2 \leftarrow R_2 - (-1)R_3$. Then

$$\begin{bmatrix} 1 & 0 & 0 & -3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}.$$

There is a free variable. For example, the nonzero coefficient vector $c = \begin{bmatrix} 3 \\ 0 \\ 1 \\ 1 \end{bmatrix}$ satisfies $Ac = 0$ (substitution checks every coordinate). This is a nontrivial relation among the columns.

**Conclusion.** The vectors are linearly dependent.

### Exercise 1.7.8: Test the column equation

**What we must show.** Determine whether the given vectors (the columns of the matrix) are independent. Form $Ac = 0$; row operations preserve exactly its solutions.

$$A = \begin{bmatrix} 1 & -2 & 3 & 2 \\ -2 & 4 & -6 & 2 \\ 0 & 1 & -1 & 3 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - (-2)R_1$. Then

$$\begin{bmatrix} 1 & -2 & 3 & 2 \\ 0 & 0 & 0 & 6 \\ 0 & 1 & -1 & 3 \end{bmatrix}.$$

Use $R_2 \leftrightarrow R_3, \quad R_1 \leftarrow R_1 - (-2)R_2$. Then

$$\begin{bmatrix} 1 & 0 & 1 & 8 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & 6 \end{bmatrix}.$$

Use $R_3 \leftarrow \frac{1}{6}R_3, \quad R_1 \leftarrow R_1 - (8)R_3, \quad R_2 \leftarrow R_2 - (3)R_3$. Then

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

There is a free variable. For example, the nonzero coefficient vector $c = \begin{bmatrix} -1 \\ 1 \\ 1 \\ 0 \end{bmatrix}$ satisfies $Ac = 0$ (substitution checks every coordinate). This is a nontrivial relation among the columns.

**Conclusion.** The vectors are linearly dependent.

### Exercise 1.7.9: A span question can differ from a dependence question

#### (a) Reduce the span of the first two vectors.

Here $v_1 = (1, -3, 2)$ and $v_2 = (-3, 9, -6) = -3v_1$. Their span is the line of multiples of $v_1$. For $v_3 = (5, -7, h)$ to lie on that line, its first coordinate would force the multiplier to be 5, but its second would then have to be $-15$, not $-7$. No choice of $h$ fixes that contradiction.

#### (b) Test the three-vector list instead.

The relation $3v_1 + v_2 + 0v_3 = 0$ is nontrivial for every $h$.

**Conclusion.** (a) No real $h$. (b) Every real $h$. The first two vectors are already dependent, whether or not the third lies in their span.

### Exercise 1.7.10: The same distinction with another line

Here $v_1 = (1, -3, -5)$, $v_2 = (-3, 9, 15) = -3v_1$, and $v_3 = (2, -5, h)$.

**(a) Compare coordinates of a proposed multiple.**

If $v_3 = tv_1$, its first coordinate gives $t = 2$, but then the second coordinate would be $-6$, not $-5$. Thus $v_3$ is outside $\text{Span}\{v_1, v_2\}$ for every $h$.

**(b) Exhibit a relation that does not use $v_3$.**

Again $3v_1 + v_2 + 0v_3 = 0$, with coefficients not all zero.

**Conclusion.** (a) No $h$. (b) All $h \in \mathbb{R}$.

### Exercise 1.7.11: Locate the parameter value that removes a pivot

**Plan.** Keep $h$ symbolic, but eliminate using only the first two columns. Those pivots are nonzero constants, so these operations do not assume anything about $h$.

$$A = \begin{bmatrix} 2 & 4 & -2 \\ -2 & -6 & 2 \\ 4 & 7 & h \end{bmatrix}.$$

Use $R_1 \leftarrow \frac{1}{2}R_1$, $\quad R_2 \leftarrow R_2 - (-2)R_1$, $\quad R_3 \leftarrow R_3 - (4)R_1$. Then

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & -2 & 0 \\ 0 & -1 & h+4 \end{bmatrix}.$$

Use $R_2 \leftrightarrow R_3$, $\quad R_2 \leftarrow -1R_2$, $\quad R_1 \leftarrow R_1 - (2)R_2$, $\quad R_3 \leftarrow R_3 - (-2)R_2$. Then

$$\begin{bmatrix} 1 & 0 & 2h+7 \\ 0 & 1 & -h-4 \\ 0 & 0 & -2h-8 \end{bmatrix}.$$

The first two pivot columns are present for all $h$. The three columns are dependent precisely when the last row contributes no third pivot, that is, when $-2\,(h+4) = 0$.

**Conclusion.** The required value is $h = -4$. At other values all three coefficients are forced to zero, giving independence.

### Exercise 1.7.12: Locate the parameter value that removes a pivot

**Plan.** Keep $h$ symbolic, but eliminate using only the first two columns. Those pivots are nonzero constants, so these operations do not assume anything about $h$.

$$A = \begin{bmatrix} 3 & -6 & 9 \\ -6 & 4 & h \\ 1 & -3 & 3 \end{bmatrix}.$$

Use $R_1 \leftrightarrow R_3$, $R_2 \leftarrow R_2 - (-6)R_1$, $R_3 \leftarrow R_3 - (3)R_1$. Then

$$\begin{bmatrix} 1 & -3 & 3 \\ 0 & -14 & h+18 \\ 0 & 3 & 0 \end{bmatrix}.$$

Use $R_2 \leftrightarrow R_3$, $R_2 \leftarrow \frac{1}{3}R_2$, $R_1 \leftarrow R_1 - (-3)R_2$, $R_3 \leftarrow R_3 - (-14)R_2$. Then

$$\begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & h+18 \end{bmatrix}.$$

The first two pivot columns are present for all $h$. The three columns are dependent precisely when the last row contributes no third pivot, that is, when $h + 18 = 0$.

**Conclusion.** The required value is $h = -18$. At other values all three coefficients are forced to zero, giving independence.

### Exercise 1.7.13: Locate the parameter value that removes a pivot

**Plan.** Keep $h$ symbolic, but eliminate using only the first two columns. Those pivots are nonzero constants, so these operations do not assume anything about $h$.

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 5 & -9 & h \\ -3 & 6 & -9 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - (5)R_1$, $R_3 \leftarrow R_3 - (-3)R_1$. Then

$$\begin{bmatrix} 1 & -2 & 3 \\ 0 & 1 & h-15 \\ 0 & 0 & 0 \end{bmatrix}.$$

Use $R_1 \leftarrow R_1 - (-2)R_2$. Then

$$\begin{bmatrix} 1 & 0 & 2h-27 \\ 0 & 1 & h-15 \\ 0 & 0 & 0 \end{bmatrix}.$$

The first two pivot columns are present for all $h$. The three columns are dependent precisely when the last row contributes no third pivot, that is, when $0 = 0$.

This condition holds identically. The reduced equations are solved nontrivially by $c_3 = 1$, $c_2 = 15 - h$, and $c_1 = 27 - 2h$.

**Conclusion.** The list is dependent for every real $h$.

### Exercise 1.7.14: Locate the parameter value that removes a pivot

**Plan.** Keep $h$ symbolic, but eliminate using only the first two columns. Those pivots are nonzero constants, so these operations do not assume anything about $h$.

$$A = \begin{bmatrix} 1 & -3 & 2 \\ -2 & 7 & 1 \\ -4 & 6 & h \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - (-2)R_1, \quad R_3 \leftarrow R_3 - (-4)R_1$. Then

$$\begin{bmatrix} 1 & -3 & 2 \\ 0 & 1 & 5 \\ 0 & -6 & h+8 \end{bmatrix}.$$

Use $R_1 \leftarrow R_1 - (-3)R_2, \quad R_3 \leftarrow R_3 - (-6)R_2$. Then

$$\begin{bmatrix} 1 & 0 & 17 \\ 0 & 1 & 5 \\ 0 & 0 & h+38 \end{bmatrix}.$$

The first two pivot columns are present for all $h$. The three columns are dependent precisely when the last row contributes no third pivot, that is, when $h + 38 = 0$.

**Conclusion.** The required value is $h = -38$. At other values all three coefficients are forced to zero, giving independence.

### Exercise 1.7.15: Four vectors in a two-coordinate space

The list has four vectors in $\mathbb{R}^2$. Its $2 \times 4$ column matrix has at most two pivots, leaving at least two free variables in $Ac = 0$.

**Conclusion.** Dependent. More than two vectors in $\mathbb{R}^2$ cannot be independent.

### Exercise 1.7.16: Spot a scalar multiple

The second vector equals $-\frac{3}{2}$ times the first: $(-3, 6, -12) = -\frac{3}{2}(2, -4, 8)$. Thus $3v_1 + 2v_2 = 0$ is a relation with nonzero coefficients.

**Conclusion.** Dependent.

### Exercise 1.7.17: A zero vector is enough

The second vector is zero. The coefficients $(0, 1, 0)$ produce $0v_1 + 1v_2 + 0v_3 = 0$ without being all zero.

**Conclusion.** Dependent. The values of the other vectors do not matter.

### Exercise 1.7.18: Count the coordinates first

There are four vectors in $\mathbb{R}^2$, so a $2 \times 4$ column matrix cannot have four pivot columns. Its homogeneous system has a free variable.

**Conclusion.** Dependent. Row reduction is unnecessary.

### Exercise 1.7.19: Two vectors that are not multiples

For $u = (-8, 12, -4)$ and $v = (2, -3, -1)$, the first coordinate of $v = tu$ would force $t = -1/4$. The third coordinate would then be 1, whereas $v$ has third coordinate $-1$. Neither vector is zero, so the failure of scalar proportionality proves independence.

**Conclusion.** Independent.

### Exercise 1.7.20: Another zero-vector relation

The third vector is zero. The coefficient list $(0, 0, 1)$ therefore solves the column equation nontrivially.

**Conclusion.** Dependent.

### Exercise 1.7.21: Four statements about dependence

(a) **False.** Every homogeneous equation has the trivial solution. Independence requires it to be the *only* solution. The zero matrix has the trivial solution and many others.

(b) **False.** At least one vector is a combination of the others, not necessarily every vector. The list $e_1, 2e_1, e_2$ is dependent, but $e_2$ is not in the span of the first two vectors.

(c) **True.** Five columns in $\mathbb{R}^4$ give more variables than possible pivots, so the homogeneous system has a free variable.

(d) **True.** A nontrivial relation $ax + by + cz = 0$ must have $c \neq 0$; otherwise independence of $x, y$ would give $a = b = 0$. Solving gives $z = -(a/c)x - (b/c)y$.

### Exercise 1.7.22: Avoid converses that have not been proved

(a) **True.** If $w = au + bv$, then $au + bv - w = 0$ is a relation with coefficient $-1$ on $w$, hence a nontrivial relation.

(b) **False as written.** A plane need not pass through zero. The independent vectors $e_1, e_2, e_3$ all lie in the plane $x_1 + x_2 + x_3 = 1$. Vectors lying in a plane *through zero* are dependent when there are three of them, but the question does not impose that condition.

(c) **False.** In $\mathbb{R}^3$, the two vectors $e_1, 2e_1$ are dependent. Having few vectors makes independence possible, not automatic.

(d) **False.** The two vectors $e_1, 2e_1$ in $\mathbb{R}^3$ give a counterexample: dependence with only two vectors, fewer than three.

### Exercise 1.7.23: Every rank-deficient two-by-two echelon pattern

Use $p$ for a nonzero pivot and $*$ for an arbitrary entry. Dependent columns mean fewer than two pivots. The possible echelon patterns are

$$\begin{bmatrix} p & * \\ 0 & 0 \end{bmatrix}, \qquad \begin{bmatrix} 0 & p \\ 0 & 0 \end{bmatrix}, \qquad \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

These exhaust rank one and rank zero. A second pivot would force independence.

### Exercise 1.7.24: Three independent columns

Independence requires all three columns to have pivots. Since pivots move right as one moves down, their only possible positions are $(1,1),(2,2),(3,3)$. Thus the echelon pattern is

$$\begin{bmatrix} p & * & * \\ 0 & q & * \\ 0 & 0 & r \end{bmatrix}, \qquad p,q,r \neq 0.$$

The arbitrary entries are above pivots; entries below them are zero.

### Exercise 1.7.25: Do not silently assume the first column is nonzero

If $a_1 \neq 0$ and $a_2$ is not its multiple, the two columns are independent. But the hypothesis also permits $a_1 = 0$, in which case $a_2 \neq 0$ and there is only one pivot, in column two. Thus both patterns are possible:

$$\begin{bmatrix} p & * \\ 0 & q \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 0 & p \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \qquad p,q \neq 0.$$

For the second pattern, no nonzero vector is a multiple of the zero vector, so it really does satisfy the stated condition.

### Exercise 1.7.26: A third column outside a plane

Independence of $a_1, a_2$ supplies two pivots. If the third column failed to add a pivot, it would be a combination of the first two, contrary to the assumption. Hence all three columns are pivot columns:

$$\begin{bmatrix} p & * & * \\ 0 & q & * \\ 0 & 0 & r \\ 0 & 0 & 0 \end{bmatrix}, \qquad p,q,r \neq 0.$$

### Exercise 1.7.27: Count pivots for independence

A $6 \times 4$ matrix with independent columns must have **four** pivot columns: one for each variable in $Ax = 0$. Fewer would leave a free variable and hence a nonzero solution. Extra rows do not require extra pivots.

### Exercise 1.7.28: Count pivots for spanning

For the columns of a $4 \times 6$ matrix to span $\mathbb{R}^4$, $Ax = b$ must be consistent for every four-coordinate $b$. This requires a pivot in every row, so there must be **four** pivot columns. Six are impossible because there are only four rows.

### Exercise 1.7.29: Build dependent and independent examples

Take

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Then $A(-2,1)^T = 0$ is a nontrivial solution. On the other hand, $Bx = 0$ has first two equations $x_1 = 0, x_2 = 0$, so its only solution is zero. Both matrices have the required size $3 \times 2$.

### Exercise 1.7.30: State the exact pivot condition

#### (a) Fill the blank.

The number is $\boxed{n}$.

#### (b) Prove both directions.

There are $n$ columns. If all $n$ are pivot columns, $Ax = 0$ has no free variable, so its only solution is zero and the columns are independent. Conversely, if there are fewer than $n$ pivot columns, some variable is free. Set one free variable to 1 and the others to 0, and solve for the pivot variables. This gives a nonzero homogeneous solution, contradicting independence. Thus independence forces $n$ pivot columns.

### Exercise 1.7.31: Read a relation from the columns

The given relation is $a_3 = a_1 + a_2$. Move all terms to one side: $a_1 + a_2 - a_3 = 0$. Therefore

$$x = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

is a nonzero solution of $Ax = 0$. No row reduction is needed, because matrix multiplication is the weighted sum of the columns.

### Exercise 1.7.32: Read another relation without elimination

Here $a_1 - 3a_2 = a_3$, so $a_1 - 3a_2 - a_3 = 0$. Consequently $x = (1, -3, -1)^T$ is a nontrivial solution. The signs come from moving $a_3$ to the left, not from changing the columns themselves.

### Exercise 1.7.33: A relation already gives the answer

**True.** The assumption $v_3 = 2v_1 + v_2$ gives $2v_1 + v_2 - v_3 + 0v_4 = 0$. The coefficients are not all zero, so the four-vector list is dependent regardless of $v_4$.

### Exercise 1.7.34: The overlooked zero-vector exception

**False.** In $\mathbb{R}^4$, take $v_1 = 0$ and $v_2 = e_1$. The nonzero $v_2$ is not a scalar multiple of $v_1$, but $1v_1 + 0v_2 = 0$ is a nontrivial dependence relation. For two nonzero vectors the proposed test would be valid; the statement did not require both vectors to be nonzero.

### Exercise 1.7.35: A zero column yields a relation

**True.** If $v_3 = 0$, the coefficient list $(0, 0, 1, 0, 0)$ gives a zero weighted sum and is not itself zero. Thus the list is dependent.

### Exercise 1.7.36: The first two vectors might already be dependent

**False.** Let $v_1 = e_1$, $v_2 = 2e_1$, and $v_3 = e_2$ in $\mathbb{R}^3$. The third vector is outside the span of the first two, but $2v_1 - v_2 + 0v_3 = 0$ proves dependence. Adding a new direction cannot remove a relation that was already present.

### Exercise 1.7.37: Dependence survives enlarging the list

**True.** A nontrivial relation among $v_1, v_2, v_3$ remains a nontrivial relation among $v_1, v_2, v_3, v_4$ after assigning coefficient zero to $v_4$. Therefore the larger list is dependent.

### Exercise 1.7.38: Independence survives taking a sublist

**True.** Suppose $c_1v_1 + c_2v_2 + c_3v_3 = 0$. Then $c_1v_1 + c_2v_2 + c_3v_3 + 0v_4 = 0$. Independence of the four-vector list forces $c_1 = c_2 = c_3 = 0$. This is precisely the independence test for the three-vector sublist.

### Exercise 1.7.39: Use the special target zero

**What we must show.** Prove column independence from the assumption of at most one solution for every $b$. Apply the assumption to $b = 0$. The zero vector is already one solution of $Ax = 0$, so "at most one" says there is no other solution. Thus every relation among the columns has all coefficients zero, which is the definition of independence. Existence for other $b$ is not needed.

### Exercise 1.7.40: No free variable means no second solution

There are $n$ pivot columns and only $n$ columns altogether, so there are no free variables. For a given $b$, reduction of $[A \mid b]$ may produce a contradiction, giving no solution; if it does not, every variable is a pivot variable with one determined value. Thus there can never be infinitely many solutions or two distinct solutions. Equivalently, if $Ax = Ay = b$, then $A(x - y) = 0$; the $n$ pivots force $x - y = 0$, and hence $x = y$.

### Exercise 1.7.41: Choose a maximal independent collection of original columns

**What we must show.** Retain as many original columns as possible while making $Bx = 0$ have only the zero solution. The source matrix is

$$A = \begin{bmatrix} 3 & -4 & 10 & 7 & -4 \\ -5 & -3 & -7 & -11 & 15 \\ 4 & 3 & 5 & 2 & 1 \\ 8 & -7 & 23 & 4 & 15 \end{bmatrix}.$$

Exact row reduction gives

$$\operatorname{rref}(A) = \begin{bmatrix} 1 & 0 & 2 & 0 & 2 \\ 0 & 1 & -1 & 0 & -1 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Its pivot columns are 1, 2, 4. Take these columns from the *original* matrix, not from its reduced matrix:

$$B = \begin{bmatrix} 3 & -4 & 7 \\ -5 & -3 & -11 \\ 4 & 3 & 2 \\ 8 & -7 & 4 \end{bmatrix}.$$

Reducing $B$ gives

$$\operatorname{rref}(B) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Thus $Bx = 0$ forces every one of its 3 variables to be zero. No larger independent subcollection is possible, because the reduced $A$ has only 3 pivot positions; every nonpivot column is a combination of the pivot columns. The next related exercise exhibits such a combination explicitly.

### Exercise 1.7.42: Choose a maximal independent collection of original columns

**What we must show.** Retain as many original columns as possible while making $Bx = 0$ have only the zero solution. The source matrix is

$$A = \begin{bmatrix} 12 & 10 & -6 & 8 & 4 & -14 \\ -7 & -6 & 4 & -5 & -7 & 9 \\ 9 & 9 & -9 & 9 & 9 & -18 \\ -4 & -3 & -1 & 0 & -8 & 1 \\ 8 & 7 & -5 & 6 & 1 & -11 \end{bmatrix}.$$

Exact row reduction gives

$$\operatorname{rref}(A) = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 & 0 & -2 \\ 0 & 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Its pivot columns are 1, 2, 3, 5. Take these columns from the *original* matrix, not from its reduced matrix:

$$B = \begin{bmatrix} 12 & 10 & -6 & 4 \\ -7 & -6 & 4 & -7 \\ 9 & 9 & -9 & 9 \\ -4 & -3 & -1 & -8 \\ 8 & 7 & -5 & 1 \end{bmatrix}.$$

Reducing $B$ gives

$$\operatorname{rref}(B) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Thus $Bx = 0$ forces every one of its 4 variables to be zero. No larger independent subcollection is possible, because the reduced $A$ has only 4 pivot positions; every nonpivot column is a combination of the pivot columns. The next related exercise exhibits such a combination explicitly.

### Exercise 1.7.43: Recover an omitted column

For the choice $B = [a_1\ a_2\ a_4]$ in Exercise 41, choose $v = a_3 = (10, -7, 5, 23)^T$. The reduced column relation from that exercise is $a_3 = 2a_1 - a_2$. Directly,

$$2\begin{bmatrix} 3 \\ -5 \\ 4 \\ 8 \end{bmatrix} - \begin{bmatrix} -4 \\ -3 \\ 3 \\ -7 \end{bmatrix} = \begin{bmatrix} 10 \\ -7 \\ 5 \\ 23 \end{bmatrix}.$$

So $Bc = v$ is solved by $c = (2, -1, 0)^T$.

**Conclusion.** $v$ is in the span of the columns of $B$.

### Exercise 1.7.44: Why maximality forces every omitted column into the span

For Exercise 42, $B = [a_1\ a_2\ a_3\ a_5]$. Choose the omitted column $v = a_4$. Its coordinates in the retained columns are $(1, -1, -1, 0)^T$, because

$$a_1 - a_2 - a_3 = \begin{bmatrix} 12 \\ -7 \\ 9 \\ -4 \\ 8 \end{bmatrix} - \begin{bmatrix} 10 \\ -6 \\ 9 \\ -3 \\ 7 \end{bmatrix} - \begin{bmatrix} -6 \\ 4 \\ -9 \\ -1 \\ -5 \end{bmatrix} = \begin{bmatrix} 8 \\ -5 \\ 9 \\ 0 \\ 6 \end{bmatrix} = a_4.$$

Thus $v \in \operatorname{Col} B$. In general, if an omitted column were outside this span, adjoining it to the independent columns of $B$ would produce a larger independent list: a relation with nonzero coefficient on the new column would express it in the old span, and a relation with zero coefficient would contradict independence of the old list. That contradicts the requirement that $B$ use as many columns as possible.

**Conclusion.** Every omitted column lies in the span of a maximal independent subcollection, not only the particular column checked here.

## 1.8 Introduction to linear transformations

**A matrix as a rule.** An $m \times n$ matrix acts on an $n$-coordinate input and produces an $m$-coordinate output. The *domain* is $\mathbb{R}^n$, the *codomain* is $\mathbb{R}^m$, and the *range* is the subset of outputs actually attained. A linear transformation satisfies

$$T(cu + dv) = cT(u) + dT(v)$$

for all inputs $u, v$ and scalars $c, d$. In particular $T(0) = 0$. To find inputs producing $b$, solve $Ax = b$. To find inputs sent to zero, solve $Ax = 0$. These are different questions: existence concerns whether $b$ is attained; uniqueness concerns whether it is attained by only one input.

### Practice problem 1.8.P1: Read matrix dimensions from input and output

Five input coordinates require five columns, so that each column receives one coefficient. Two output coordinates require two rows, one row computation per output entry. Thus $A$ is $2 \times 5$.

### Practice problem 1.8.P2: Interpret the sign change

Multiplication by $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ takes $(x_1, x_2)$ to $(x_1, -x_2)$. Horizontal position is unchanged and vertical position is reversed, so this is reflection across the $x_1$-axis. Points on that axis are fixed.

### Practice problem 1.8.P3: Map a segment starting at zero

A point of the segment has the form $x = tu$ with $0 \le t \le 1$. Linearity gives $T(x) = tT(u)$, which has exactly the form of a point on the segment from 0 to $T(u)$ with the same parameter. Conversely each such output comes from $tu$. Thus the image is the whole segment, possibly collapsed to one point if $T(u) = 0$.

### Exercise 1.8.1: Multiply by a scalar matrix

Here $A = 2I_2$. Hence

$$T(1, -3)^T = (2, -6)^T, \qquad T(a, b)^T = (2a, 2b)^T.$$

Each output entry is twice the corresponding input entry. Geometrically this doubles lengths from the origin without changing direction.

### Exercise 1.8.2: A uniform contraction

Here $A = \frac{1}{3} I_3$, so multiplying each input coordinate by $1/3$ gives

$$T(3, 6, -9)^T = (1, 2, -3)^T, \qquad T(a, b, c)^T = (a/3, b/3, c/3)^T.$$

The factor is $1/3$, not 3: it is the value on each diagonal entry.

### Exercise 1.8.3: Solve for an input with the specified image

**What we must show.** Solve $Ax = b$ and decide whether the input is unique. Put the target vector in the last column and apply the same row operations to the whole augmented matrix.

$$[A \mid b] = \begin{bmatrix} 1 & 0 & -3 & -2 \\ -3 & 1 & 6 & 3 \\ 2 & -2 & -1 & -1 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - (-3)R_1, \quad R_3 \leftarrow R_3 - (2)R_1$. Then

$$\begin{bmatrix} 1 & 0 & -3 & -2 \\ 0 & 1 & -3 & -3 \\ 0 & -2 & 5 & 3 \end{bmatrix}.$$

Use $R_3 \leftarrow R_3 - (-2)R_2$. Then

$$\begin{bmatrix} 1 & 0 & -3 & -2 \\ 0 & 1 & -3 & -3 \\ 0 & 0 & -1 & -3 \end{bmatrix}.$$

Use $R_3 \leftarrow -1R_3, \quad R_1 \leftarrow R_1 - (-3)R_3, \quad R_2 \leftarrow R_2 - (-3)R_3$. Then

$$\begin{bmatrix} 1 & 0 & 0 & 7 \\ 0 & 1 & 0 & 6 \\ 0 & 0 & 1 & 3 \end{bmatrix}.$$

The reduced equations give $x = (7, 6, 3)^T$. Every one of the three variable columns has a pivot, so the solution is unique.

As an independent check, multiplying the displayed particular input by the original $A$ gives $\begin{bmatrix} -2 \\ 3 \\ -1 \end{bmatrix}$, exactly the required target.

### Exercise 1.8.4: Solve for an input with the specified image

**What we must show.** Solve $Ax = b$ and decide whether the input is unique. Put the target vector in the last column and apply the same row operations to the whole augmented matrix.

$$[A \mid b] = \begin{bmatrix} 1 & -2 & 3 & -6 \\ 0 & 1 & -3 & -4 \\ 2 & -5 & 6 & -5 \end{bmatrix}.$$

Use $R_3 \leftarrow R_3 - (2)R_1$. Then

$$\begin{bmatrix} 1 & -2 & 3 & -6 \\ 0 & 1 & -3 & -4 \\ 0 & -1 & 0 & 7 \end{bmatrix}.$$

Use $R_1 \leftarrow R_1 - (-2)R_2, \quad R_3 \leftarrow R_3 - (-1)R_2$. Then

$$\begin{bmatrix} 1 & 0 & -3 & -14 \\ 0 & 1 & -3 & -4 \\ 0 & 0 & -3 & 3 \end{bmatrix}.$$

Use $R_3 \leftarrow -\frac{1}{3}R_3, \quad R_1 \leftarrow R_1 - (-3)R_3, \quad R_2 \leftarrow R_2 - (-3)R_3$. Then

$$\begin{bmatrix} 1 & 0 & 0 & -17 \\ 0 & 1 & 0 & -7 \\ 0 & 0 & 1 & -1 \end{bmatrix}.$$

The reduced equations give $x = (-17, -7, -1)^T$. Every variable is a pivot variable, so this input is unique.

As an independent check, multiplying the displayed particular input by the original $A$ gives $\begin{bmatrix} -6 \\ -4 \\ -5 \end{bmatrix}$, exactly the required target.

### Exercise 1.8.5: Solve for an input with the specified image

**What we must show.** Solve $Ax = b$ and decide whether the input is unique. Put the target vector in the last column and apply the same row operations to the whole augmented matrix.

$$[A \mid b] = \begin{bmatrix} 1 & -5 & -7 & -2 \\ -3 & 7 & 5 & -2 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - (-3)R_1$. Then

$$\begin{bmatrix} 1 & -5 & -7 & -2 \\ 0 & -8 & -16 & -8 \end{bmatrix}.$$

Use $R_2 \leftarrow -\frac{1}{8}R_2, \quad R_1 \leftarrow R_1 - (-5)R_2$. Then

$$\begin{bmatrix} 1 & 0 & 3 & 3 \\ 0 & 1 & 2 & 1 \end{bmatrix}.$$

The reduced equations give $x = (3 - 3t, 1 - 2t, t)^T$, $t \in \mathbb{R}$. Here $x_3 = t$ is free. For example $t = 0$ gives $(3, 1, 0)^T$. Varying $t$ gives different inputs with the same output, so uniqueness fails.

As an independent check, multiplying the displayed particular input by the original $A$ gives $\begin{bmatrix} -2 \\ -2 \end{bmatrix}$, exactly the required target.

### Exercise 1.8.6: Solve for an input with the specified image

**What we must show.** Solve $Ax = b$ and decide whether the input is unique. Put the target vector in the last column and apply the same row operations to the whole augmented matrix.

$$[A \mid b] = \begin{bmatrix} 1 & -3 & 2 & 1 \\ 3 & -8 & 8 & 6 \\ 0 & 1 & 2 & 3 \\ 1 & 0 & 8 & 10 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - (3)R_1, \quad R_4 \leftarrow R_4 - (1)R_1$. Then

$$\begin{bmatrix} 1 & -3 & 2 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 1 & 2 & 3 \\ 0 & 3 & 6 & 9 \end{bmatrix}.$$

Use $R_1 \leftarrow R_1 - (-3)R_2, \quad R_3 \leftarrow R_3 - (1)R_2, \quad R_4 \leftarrow R_4 - (3)R_2$. Then

$$\begin{bmatrix} 1 & 0 & 8 & 10 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The reduced equations give $x = (10 - 8t, 3 - 2t, t)^T$, $t \in \mathbb{R}$. Here $x_3 = t$ is free. Taking $t = 0$ gives $(10, 3, 0)^T$, but other values give additional inputs. The zero rows impose no extra restrictions.

As an independent check, multiplying the displayed particular input by the original $A$ gives $\begin{bmatrix} 1 \\ 6 \\ 3 \\ 10 \end{bmatrix}$, exactly the required target.

### Exercise 1.8.7: The domain and codomain of a six-by-five matrix

There are five columns, so $Ax$ requires $x \in \mathbb{R}^5$; there are six rows, so $Ax \in \mathbb{R}^6$. Therefore $\boxed{a = 5, b = 6}$. The range can be smaller than $\mathbb{R}^6$; the question asks for the codomain.

### Exercise 1.8.8: Construct the required shape

A map $\mathbb{R}^5 \to \mathbb{R}^7$ implemented by $Ax$ needs $\boxed{\text{7 rows and 5 columns}}$. Five entries are consumed and seven entries are produced. Reversing these numbers would define a different map.

### Exercise 1.8.9: Find every input sent to zero

**What we must show.** Solve $Ax = 0$. The right-hand side stays zero under row operations, so we can reduce the coefficient matrix alone.

$$A = \begin{bmatrix} 1 & -3 & 5 & -5 \\ 0 & 1 & -3 & 5 \\ 2 & -4 & 4 & -4 \end{bmatrix}.$$

Use $R_3 \leftarrow R_3 - (2)R_1$. Then

$$\begin{bmatrix} 1 & -3 & 5 & -5 \\ 0 & 1 & -3 & 5 \\ 0 & 2 & -6 & 6 \end{bmatrix}.$$

Use $R_1 \leftarrow R_1 - (-3)R_2, \quad R_3 \leftarrow R_3 - (2)R_2$. Then

$$\begin{bmatrix} 1 & 0 & -4 & 10 \\ 0 & 1 & -3 & 5 \\ 0 & 0 & 0 & -4 \end{bmatrix}.$$

Use $R_3 \leftarrow -\frac{1}{4}R_3, \quad R_1 \leftarrow R_1 - (10)R_3, \quad R_2 \leftarrow R_2 - (5)R_3$. Then

$$\begin{bmatrix} 1 & 0 & -4 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The equations are $x_1 - 4x_3 = 0$, $x_2 - 3x_3 = 0$, $x_4 = 0$. Let $x_3 = t$. Then

$$x = t \begin{bmatrix} 4 \\ 3 \\ 1 \\ 0 \end{bmatrix}, \qquad t \in \mathbb{R}.$$

Each displayed generator is sent to zero by the original matrix. Conversely, every solution must obey the reduced equations, so no additional inputs are missing.

### Exercise 1.8.10: Find every input sent to zero

**What we must show.** Solve $Ax = 0$. The right-hand side stays zero under row operations, so we can reduce the coefficient matrix alone.

$$A = \begin{bmatrix} 3 & 2 & 10 & -6 \\ 1 & 0 & 2 & -4 \\ 0 & 1 & 2 & 3 \\ 1 & 4 & 10 & 8 \end{bmatrix}.$$

Use $R_1 \leftrightarrow R_2, \quad R_2 \leftarrow R_2 - (3)R_1, \quad R_4 \leftarrow R_4 - (1)R_1$. Then

$$\begin{bmatrix} 1 & 0 & 2 & -4 \\ 0 & 2 & 4 & 6 \\ 0 & 1 & 2 & 3 \\ 0 & 4 & 8 & 12 \end{bmatrix}.$$

Use $R_2 \leftrightarrow R_3, \quad R_3 \leftarrow R_3 - (2)R_2, \quad R_4 \leftarrow R_4 - (4)R_2$. Then

$$\begin{bmatrix} 1 & 0 & 2 & -4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The equations are $x_1 + 2x_3 - 4x_4 = 0$ and $x_2 + 2x_3 + 3x_4 = 0$. Set $x_3 = s, x_4 = t$. Then

$$x = s \begin{bmatrix} -2 \\ -2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 4 \\ -3 \\ 0 \\ 1 \end{bmatrix}, \quad s, t \in \mathbb{R}.$$

Each displayed generator is sent to zero by the original matrix. Conversely, every solution must obey the reduced equations, so no additional inputs are missing.

### Exercise 1.8.11: Test membership in the range

The target belongs to the range exactly when $Ax = b$ is consistent. Reducing the augmented matrix gives

$$[A \mid b] \sim \begin{bmatrix} 1 & 0 & -4 & 0 & 2 \\ 0 & 1 & -3 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}.$$

The equations give $x_1 = 2 + 4t$, $x_2 = 1 + 3t$, $x_3 = t$, $x_4 = 0$. In particular $x = (2, 1, 0, 0)^T$ produces $b = (-1, 1, 0)^T$. Thus the target is in the range. A free variable does not prevent existence; it only prevents uniqueness.

### Exercise 1.8.12: Test membership in the range

The target belongs to the range exactly when $Ax = b$ is consistent. Reducing the augmented matrix gives

$$[A \mid b] \sim \begin{bmatrix} 1 & 0 & 2 & -4 & 0 \\ 0 & 1 & 2 & 3 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

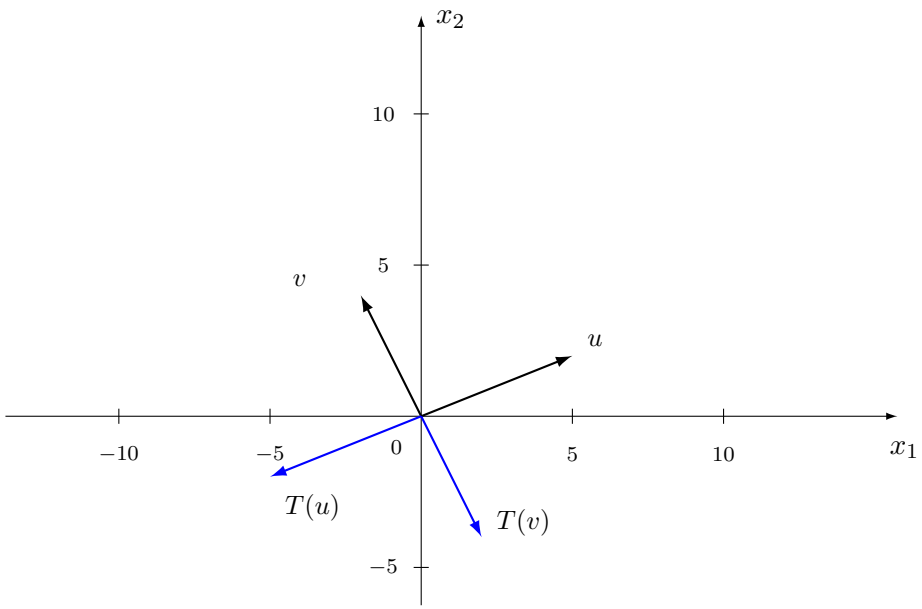
The third row says $0 = 1$, which is impossible. There is no input with image $(-1, 3, -1, 4)^T$, so the target is not in the range. The fact that it lies in the codomain $\mathbb{R}^4$ is not enough.

### Exercise 1.8.13: Calculate and sketch the transformed vectors

Multiply each row of $A$ by the input coordinates:

$$T(u) = \begin{bmatrix} -5 \\ -2 \end{bmatrix}, \qquad T(v) = \begin{bmatrix} 2 \\ -4 \end{bmatrix}.$$

For a general point, $T(x_1, x_2) = (-x_1, -x_2)$. This is reflection through the origin (equivalently a rotation through 180°). The black arrows show the original vectors; the blue arrows show their images.

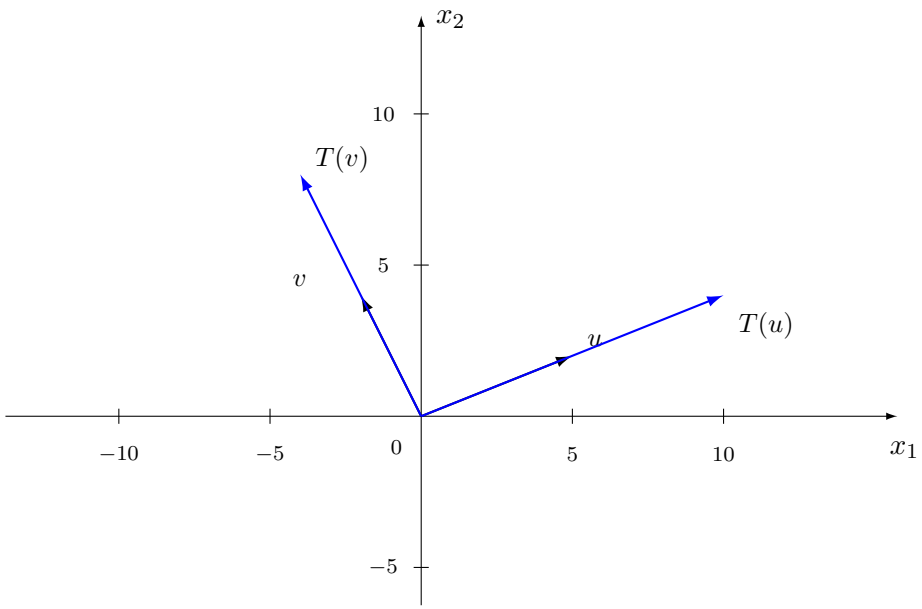



**Exercise 1.8.14: Calculate and sketch the transformed vectors**

Multiply each row of $A$ by the input coordinates:

$$T(u) = \begin{bmatrix} 10 \\ 4 \end{bmatrix}, \qquad T(v) = \begin{bmatrix} -4 \\ 8 \end{bmatrix}.$$

For a general point, $T(x_1, x_2) = (2x_1, 2x_2)$. This is expansion by a factor of 2 about the origin. The black arrows show the original vectors; the blue arrows show their images.

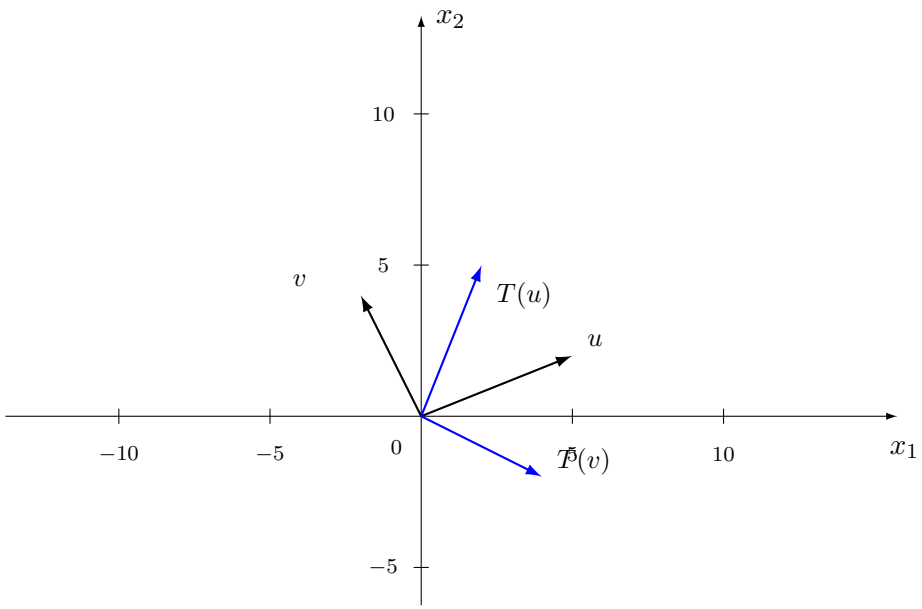



**Exercise 1.8.15: Calculate and sketch the transformed vectors**

Multiply each row of $A$ by the input coordinates:

$$T(u) = \begin{bmatrix} 2 \\ 5 \end{bmatrix}, \qquad T(v) = \begin{bmatrix} 4 \\ -2 \end{bmatrix}.$$

For a general point, $T(x_1, x_2) = (x_2, x_1)$. This is reflection through the line $x_2 = x_1$. The black arrows show the original vectors; the blue arrows show their images.

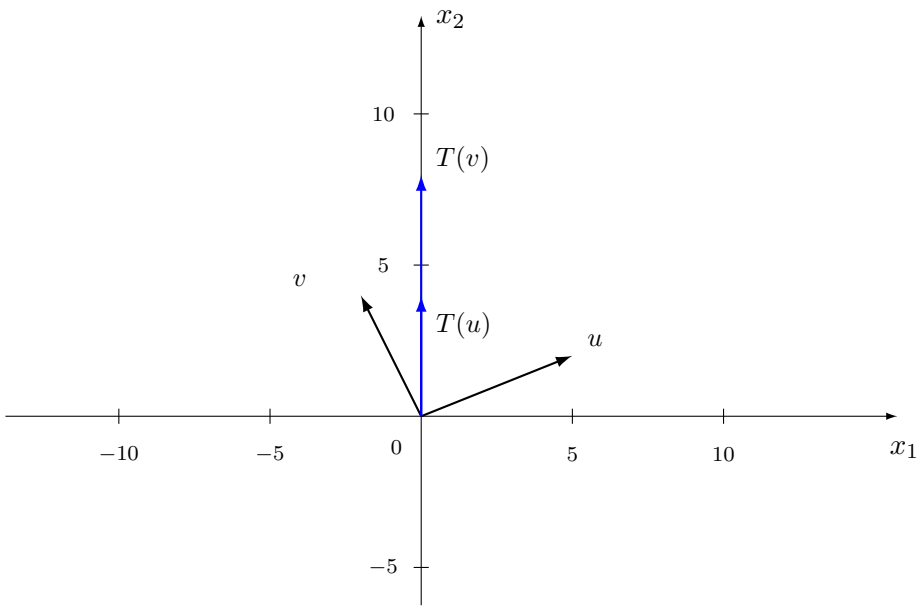



### Exercise 1.8.16: Calculate and sketch the transformed vectors

Multiply each row of $A$ by the input coordinates:

$$T(u) = \begin{bmatrix} 0 \\ 4 \end{bmatrix}, \qquad T(v) = \begin{bmatrix} 0 \\ 8 \end{bmatrix}.$$

For a general point, $T(x_1, x_2) = (0, 2x_2)$. This is projection onto the $x_2$-axis followed by multiplication of the remaining coordinate by 2. The black arrows show the original vectors; the blue arrows show their images.




### Exercise 1.8.17: Apply linearity before looking for a matrix

The given images are $T(u) = (4, 1)^T$ and $T(v) = (-1, 3)^T$. Homogeneity and then additivity give

$$T(2u) = 2T(u) = (8, 2)^T, \qquad T(3v) = 3T(v) = (-3, 9)^T,$$

$$T(2u + 3v) = 2T(u) + 3T(v) = (8, 2)^T + (-3, 9)^T = (5, 11)^T.$$

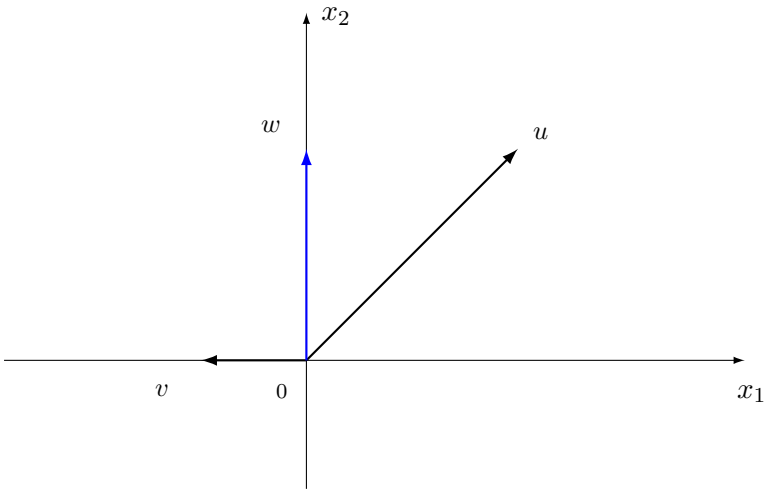
We do not need to reconstruct $T$ on every input to answer these questions.

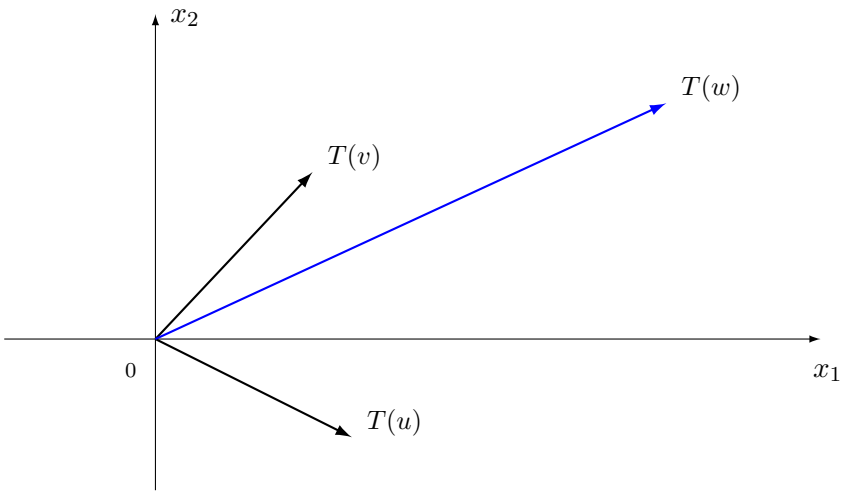
### Exercise 1.8.18: Transfer a graphical linear combination

In the input diagram, $w = u+2v$: adding two copies of the left-pointing vector $v$ cancels the horizontal component of $u$ without changing its vertical component. By linearity,

$$T(w) = T(u) + 2T(v).$$

In the output diagram, begin at the tip of $T(u)$, append two copies of $T(v)$ head to tail, and draw the arrow from the origin to the final point. The blue arrow below gives the result. These are schematic coordinate drawings of the relation, not new numerical measurements assigned to the textbook's unlabeled vectors.




**Exercise 1.8.19: Use the standard coordinate vectors**

Write $(5,-3)^T = 5e_1 - 3e_2$. Then

$$T(5,-3)^T = 5(2,5)^T - 3(-1,6)^T = (13,7)^T.$$

More generally, $x = x_1e_1 + x_2e_2$ gives

$$T(x) = x_1(2,5)^T + x_2(-1,6)^T = (2x_1 - x_2, 5x_1 + 6x_2)^T.$$

The images of the two coordinate vectors determine every image.

### Exercise 1.8.20: Make the two image vectors into columns

The rule is $T(x) = x_1(-3,5)^T + x_2(7,-2)^T$. Matrix multiplication forms exactly this column combination, so

$$A = \begin{bmatrix} -3 & 7 \\ 5 & -2 \end{bmatrix}, \qquad T(x) = Ax.$$

Putting the image vectors into rows instead would give the wrong formula.

### Exercise 1.8.21: Functions, codomains, ranges, and linearity

(a) **True.** A transformation is a function; a linear transformation is one satisfying additional addition and scaling identities.

(b) **False.** Five columns require inputs in $\mathbb{R}^5$. Three rows determine the codomain $\mathbb{R}^3$, not the domain.

(c) **False.** The range is $\operatorname{Col} A$, which need not equal $\mathbb{R}^m$. For the zero matrix the only output is zero.

(d) **True in the setting of this chapter, $\mathbb{R}^n \to \mathbb{R}^m$.** For any such linear $T$, write $x = \sum_j x_j e_j$; then $T(x) = \sum_j x_j T(e_j) = [T(e_1) \ \cdots \ T(e_n)]x$. Thus it has a matrix implementation. A transformation on other vector spaces requires a choice of coordinates before this literal formula applies.

(e) **True.** Additivity and homogeneity imply the displayed combination rule. Conversely, setting both scalars to 1 gives additivity, and setting the second scalar to 0 gives homogeneity. Those are exactly the two defining properties.

### Exercise 1.8.22: Separate an existence question from a uniqueness question

(a) **True.** The products $Ax$ are precisely all weighted combinations of the columns, by the definition of matrix multiplication.

(b) **True.** $A(cu + dv) = cAu + dAv$, by distributivity and scaling of matrix products.

(c) **False.** Membership of $c$ in the range asks whether *some* input gives $c$: it is an existence question. A uniqueness question asks whether two different inputs can give that same output.

(d) **True.** These are the defining properties of a linear map.

(e) **True.** $T(0) = T(0+0) = T(0) + T(0)$; subtracting $T(0)$ from both sides gives $T(0) = 0$.

### Exercise 1.8.23: A straight-line graph need not define a linear map

**(a) Put $b = 0$.**

For $f(x) = mx$ and any real $c, d, u, v$, $f(cu + dv) = m(cu + dv) = c(mu) + d(mv) = cf(u) + df(v)$. Hence it is linear.

**(b) Inspect zero when $b \neq 0$.**

Then $f(0) = b \neq 0$, violating the necessary condition that a linear transformation sends zero to zero.

**(c) Explain the terminology.**

In elementary algebra and calculus, "linear function" often means a function whose graph is a straight line, allowing an intercept. Linear algebra uses the stricter operation-preserving meaning. A nonzero intercept makes the function affine rather than linear in this stricter sense.

### Exercise 1.8.24: An affine displacement

For $T(x) = Ax + b$, the zero input gives $T(0) = b$. If $b \neq 0$, this cannot equal zero, whereas every linear transformation sends zero to zero. Thus the affine map is not linear. The matrix part is linear; it is the added fixed displacement that violates the condition.

### Exercise 1.8.25: The image of an entire line

Write the input line as $x = p + tv$ for all $t \in \mathbb{R}$, with $v \neq 0$. Then $T(x) = T(p) + tT(v)$. If $T(v) \neq 0$, these outputs form the line through $T(p)$ in direction $T(v)$. Every point of that line comes from the same parameter $t$, so the map is onto that line. If $T(v) = 0$, every input on the line maps to the single point $T(p)$.

**Conclusion.** The image is a line or a single point.

### Exercise 1.8.26: Affine combinations describe lines and segments

**(a) Rewrite a line's usual parametrization.**

For distinct $p, q$, the line is $p + t(q - p)$. Distributing gives $(1 - t)p + tq$, for all real $t$. Conversely these formulas are equal for every $t$, so they describe the same set. If $p = q$, the formula instead describes the degenerate line consisting of that one point.

**(b) Restrict the parameter to the segment.**

For $0 \leq t \leq 1$, linearity gives $T((1 - t)p + tq) = (1 - t)T(p) + tT(q)$. As $t$ runs over the whole interval $[0, 1]$, the outputs cover the segment between $T(p)$ and $T(q)$. If these endpoints coincide, that segment is a single point.

### Exercise 1.8.27: A plane can collapse in two ways

Every input in $P$ is $su+tv$ with $s, t \in \mathbb{R}$. Its image is $sT(u)+tT(v)$, and conversely every combination of those two image vectors comes from such an input. Thus

$$T(P) = \text{Span}\{T(u), T(v)\}.$$

If the image vectors are independent, their span is a plane through zero. If they are dependent but at least one is nonzero, their span is a line through zero. If both are zero, the span is just $\{0\}$.

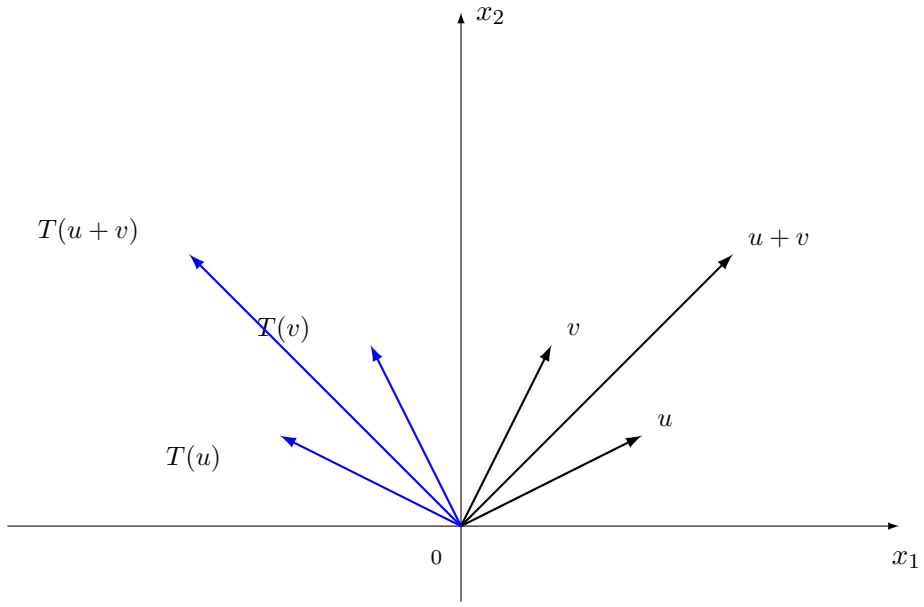
**Conclusion.** The image is a plane precisely when $T(u), T(v)$ are independent.

### Exercise 1.8.28: A parallelogram maps into the corresponding parallelogram

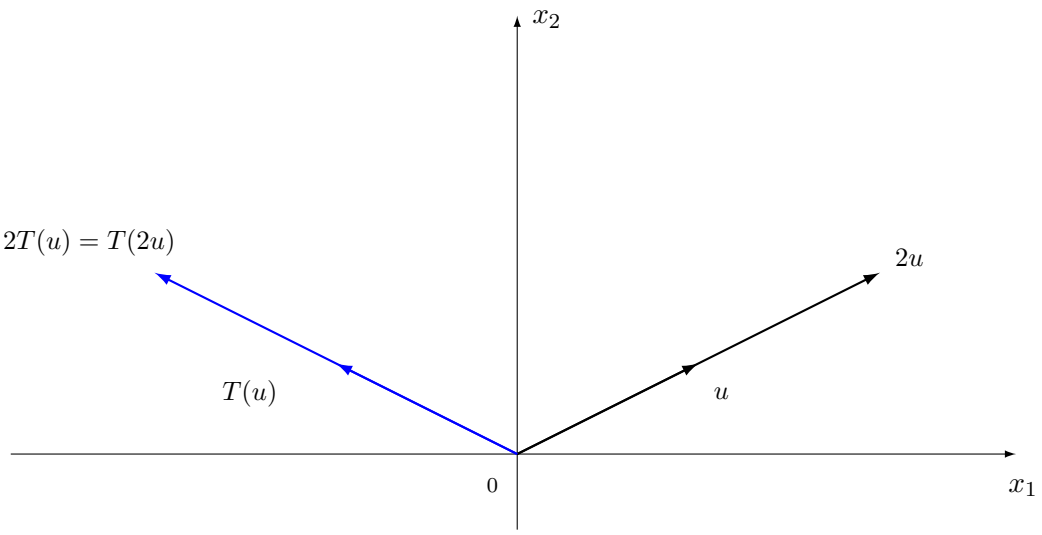
An input in the given parallelogram has the form $au+bv$ with $0 \le a, b \le 1$. Its image is $aT(u)+bT(v)$ with the same two coefficients in the same ranges. This is precisely a point of the parallelogram generated by $T(u), T(v)$. In fact the image is the whole of that set, since every allowed pair $a, b$ also gives an input. The output parallelogram may degenerate to a segment or a point.

### Exercise 1.8.29: Two sketches for reflection across the vertical axis

Here $T(x_1, x_2) = (-x_1, x_2)$. For addition, choose $u = (2, 1)$, $v = (1, 2)$. Then $u + v = (3, 3)$, while $T(u) = (-2, 1)$, $T(v) = (-1, 2)$, and $T(u) + T(v) = (-3, 3) = T(u + v)$. The paired arrows below have matching vertical coordinates and opposite horizontal coordinates.




For scaling, take $u = (2, 1)$ and $c = 2$. Then $T(2u) = (-4, 2) = 2T(u)$. Both the original ray and its reflected ray double their lengths; their directions are reflected across the vertical axis.




### Exercise 1.8.30: A linear map vanishing on a spanning list is zero

Take any $x \in \mathbb{R}^n$. Since $v_1, \ldots, v_p$ span the domain, there are scalars $c_j$ with $x = \sum_{j=1}^{p} c_j v_j$. Repeated use of linearity gives

$$T(x) = \sum_{j=1}^{p} c_j T(v_j) = \sum_{j=1}^{p} c_j 0 = 0.$$

The choice of $x$ was arbitrary, so $T$ sends every input to zero. The spanning assumption is essential: vanishing on just one line would not generally determine a map on the whole domain.

### Exercise 1.8.31: A dependence relation survives applying a linear map

Dependence gives scalars $c_1, c_2, c_3$, not all zero, such that $c_1 v_1 + c_2 v_2 + c_3 v_3 = 0$. Apply $T$ and use $T(0) = 0$:

$$c_1 T(v_1) + c_2 T(v_2) + c_3 T(v_3) = T(c_1 v_1 + c_2 v_2 + c_3 v_3) = 0.$$

The same coefficients are still not all zero. Thus they form a nontrivial relation among the three image vectors, proving dependence.

### Exercise 1.8.32: An absolute value violates homogeneity

For $T(x_1, x_2) = (x_1 - 2|x_2|, x_1 - 4x_2)$, choose $u = (0, 1)$. Then $T(u) = (-2, -4)$, so $-T(u) = (2, 4)$. But $T(-u) = T(0, -1) = (-2, 4)$, which differs in its first coordinate. Hence $T(-u) \neq -T(u)$, violating homogeneity for scalar $-1$.

**Conclusion.** The transformation is not linear.

### Exercise 1.8.33: A constant term violates preservation of zero

The displayed rule gives $T(0,0) = (0,-3,0) \neq (0,0,0)$. Since every linear transformation must map the zero input to the zero output, this single counterexample disproves linearity. No complicated comparison of general inputs is needed.

### Exercise 1.8.34: Reflection through a coordinate plane is linear

Let $u, v \in \mathbb{R}^3$ and $c, d \in \mathbb{R}$. For the stated rule,

$$\begin{aligned} T(cu+dv) &= (cu_1+dv_1, cu_2+dv_2, -cu_3-dv_3) \\ &= c(u_1,u_2,-u_3) + d(v_1,v_2,-v_3) \\ &= cT(u) + dT(v). \end{aligned}$$

This is the combined linearity condition for arbitrary inputs and scalars. Equivalently, $T$ is multiplication by $\operatorname{diag}(1,1,-1)$.

### Exercise 1.8.35: A coordinate projection is linear

For $T(x_1,x_2,x_3) = (x_1,0,x_3)$,

$$\begin{aligned} T(cu+dv) &= (cu_1+dv_1, 0, cu_3+dv_3) \\ &= c(u_1,0,u_3) + d(v_1,0,v_3) = cT(u) + dT(v). \end{aligned}$$

The middle zero is preserved under addition and scaling. Thus the map is linear; its matrix is $\operatorname{diag}(1,0,1)$.

### Exercise 1.8.36: Dependence created in the output forces a nonzero zero-input

Because $T(u), T(v)$ are dependent, some $a, b$, not both zero, satisfy $aT(u)+bT(v) = 0$. Set $x = au+bv$. Linearity gives $T(x) = 0$. Moreover $x \neq 0$: if $au+bv = 0$, independence of $u, v$ would force $a = b = 0$, contrary to their choice.

**Conclusion.** $x$ is a nontrivial solution of $T(x) = 0$.

### Exercise 1.8.37: Compute all inputs in the kernel

The supplied matrix is

$$A = \begin{bmatrix} 2 & 3 & 5 & -5 \\ -7 & 7 & 0 & 0 \\ -3 & 4 & 1 & 3 \\ -9 & 3 & -6 & -4 \end{bmatrix}.$$

Exact row reduction yields

$$\operatorname{rref}(A) = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Choose the sole free variable to be $t$. Reading the pivot equations gives

$$x = t \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \qquad t \in \mathbb{R}.$$

Direct multiplication of the original $A$ by this generator gives zero in every row. Conversely the pivot equations determine all remaining coordinates from $t$, so the displayed family is the complete solution set.

### Exercise 1.8.38: Compute all inputs in the kernel

The supplied matrix is

$$A = \begin{bmatrix} 3 & 4 & -7 & 0 \\ 5 & -8 & 7 & 4 \\ 6 & -8 & 6 & 4 \\ 9 & -7 & -2 & 0 \end{bmatrix}.$$

Exact row reduction yields

$$\operatorname{rref}(A) = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Choose the sole free variable to be $t$. Reading the pivot equations gives

$$x = t \begin{bmatrix} -1 \\ -1 \\ -1 \\ 1 \end{bmatrix}, \qquad t \in \mathbb{R}.$$

Direct multiplication of the original $A$ by this generator gives zero in every row. Conversely the pivot equations determine all remaining coordinates from $t$, so the displayed family is the complete solution set.

### Exercise 1.8.39: Find an input for a specified output

Use the matrix in Exercise 37 and append the stated target. The exact reduced matrix is

$$[A \mid b] \sim \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

No row says $0 = 1$, so the equations are consistent. A particular input is $x = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \end{bmatrix}$. Substitution gives

$$Ax = \begin{bmatrix} 8 \\ 7 \\ 5 \\ -3 \end{bmatrix} = b.$$

All inputs with that same image are

$$x = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \quad t \in \mathbb{R}.$$

The homogeneous part changes the input without changing its image, because its product with $A$ is zero.

### Exercise 1.8.40: Find an input for a specified output

Use the matrix in Exercise 38 and append the stated target. The exact reduced matrix is

$$[A \mid b] \sim \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 2 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

No row says $0 = 1$, so the equations are consistent. A particular input is $x = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \end{bmatrix}$. Substitution gives

$$Ax = \begin{bmatrix} 4 \\ -4 \\ -4 \\ -7 \end{bmatrix} = b.$$

All inputs with that same image are

$$x = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ -1 \\ -1 \\ 1 \end{bmatrix}, \quad t \in \mathbb{R}.$$

The homogeneous part changes the input without changing its image, because its product with $A$ is zero.

## 1.9 The matrix of a linear transformation

For a linear map $T : \mathbb{R}^n \to \mathbb{R}^m$, its standard matrix is

$$A = [T(e_1) \ \cdots \ T(e_n)].$$

The image of $e_j$ is column $j$, not row $j$. This works because $x = \sum_j x_j e_j$ and linearity gives $T(x) = \sum_j x_j T(e_j) = Ax$. The map is one-to-one exactly when every column has a pivot, so $Ax = 0$ has only the trivial solution. It is onto $\mathbb{R}^m$ exactly when every row has a pivot. One-to-one means at most one input for any output; onto means at least one input for every output in the codomain. For a composition, the transformation applied first corresponds to the matrix farther to the right in the product.

### Practice problem 1.9.P1: A shear followed by reflection

The shear fixes $e_1$ and takes $e_2$ to $e_2 - \frac{1}{2}e_1$. Reflection across the $x_2$-axis negates the first coordinate and leaves the second unchanged. Thus the final images are $T(e_1) = (-1, 0)^T$ and $T(e_2) = (1/2, 1)^T$. Therefore

$$A = \begin{bmatrix} -1 & 1/2 \\ 0 & 1 \end{bmatrix}.$$

We follow the image through both operations before making it a column.

### Exercise 1.9.1: Put two four-coordinate images into columns

The domain has dimension two and the output has four coordinates. Use the two specified images in their stated order:

$$A = [T(e_1) \ T(e_2)] = \begin{bmatrix} 3 & -5 \\ 1 & 2 \\ 3 & 0 \\ 1 & 0 \end{bmatrix}.$$

Indeed its first column is $T(e_1)$ and its second is $T(e_2)$, which is enough to determine the linear map on every input.

### Exercise 1.9.2: Three input coordinates and two output coordinates

There must be three columns and two rows. The standard matrix is

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 4 & 9 & -8 \end{bmatrix}.$$

For example, the second column is the given image $(-2, 9)^T$. Multiplying $A$ by any $(x_1, x_2, x_3)^T$ forms the required weighted combination of the three images.

### Exercise 1.9.3: A vertical shear

The given images are $T(e_1) = e_1 - 3e_2 = (1, -3)^T$ and $T(e_2) = e_2 = (0, 1)^T$. Hence

$$A = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}, \qquad T(x_1, x_2) = (x_1, x_2 - 3x_1).$$

The horizontal coordinate stays fixed while the vertical coordinate changes in proportion to it.

### Exercise 1.9.4: A horizontal shear

Here $T(e_1) = e_1$ and $T(e_2) = e_2 + 2e_1 = (2, 1)^T$. Thus

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \qquad T(x_1, x_2) = (x_1 + 2x_2, x_2).$$

This time the vertical coordinate stays fixed.

### Exercise 1.9.5: A counterclockwise quarter turn

Rotation through $\pi/2$ takes $e_1$ to $e_2$ and $e_2$ to $-e_1$. Therefore

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

The resulting rule $T(x_1, x_2) = (-x_2, x_1)$ agrees with both coordinate-vector images.

### Exercise 1.9.6: A clockwise three-quarter turn

An angle of $-3\pi/2$ differs from $\pi/2$ by a full turn. Consequently the final coordinate-vector images are again $e_1 \mapsto e_2$ and $e_2 \mapsto -e_1$, giving

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

The direction of travel is different, but the final transformation is the same.

### Exercise 1.9.7: Rotate, then reflect

Rotation through $-3\pi/4$ has matrix

$$R = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix}.$$

Reflection across the horizontal axis negates the second output coordinate, so it changes the signs in the second row of $R$. Thus

$$A = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}.$$

In particular the first column is $(-1/\sqrt{2}, 1/\sqrt{2})^T$, agreeing with the hint. The order is important: reflect the rotated image.

### Exercise 1.9.8: Shear, then reflect across a sloping line

The shear is $S(x_1, x_2) = (x_1+2x_2, x_2)$. Reflection across $x_2 = -x_1$ takes $(a, b)$ to $(-b, -a)$. Composing these two explicit rules gives

$$T(x_1, x_2) = (-x_2, -x_1 - 2x_2), \qquad A = \begin{bmatrix} 0 & -1 \\ -1 & -2 \end{bmatrix}.$$

The columns can also be checked by applying the two rules to $e_1, e_2$.

### Exercise 1.9.9: Reflect, then rotate clockwise

The first step takes $(x_1, x_2)$ to $(x_1, -x_2)$. A clockwise quarter turn takes $(a, b)$ to $(b, -a)$. Hence the final image is $(-x_2, -x_1)$, and

$$A = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}.$$

Applying the two steps in the opposite order would produce a different map.

### Exercise 1.9.10: Two reflections make a rotation

The horizontal reflection gives $(x_1, -x_2)$; reflection across $x_2 = x_1$ exchanges its coordinates. Thus

$$T(x_1, x_2) = (-x_2, x_1), \qquad A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

The result is a counterclockwise quarter turn.

### Exercise 1.9.11: Reflect across the two coordinate axes

After horizontal reflection, $(x_1, x_2)$ becomes $(x_1, -x_2)$. Vertical reflection then gives $(-x_1, -x_2)$. The standard matrix is $-I_2$, whose columns are $-e_1, -e_2$. A rotation by $\pi$ radians has the same two images, so it is the same linear transformation.

**Conclusion.** A half turn, through $\pi$ radians (equivalently $-\pi$).

### Exercise 1.9.12: Identify the angle from the coordinate-vector images

Exercise 10 gives $e_1 \mapsto e_2$, $e_2 \mapsto -e_1$. These are precisely the images under counterclockwise rotation by $\pi/2$. Since a linear map is determined by its values on the standard coordinate vectors, the two transformations agree on every vector.

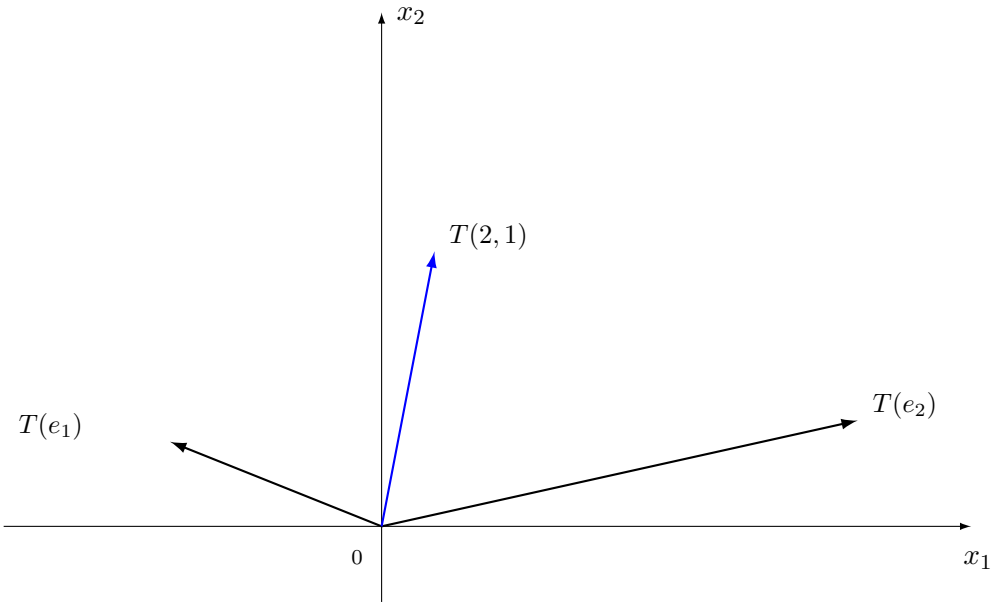
**Conclusion.** The rotation angle is $\pi/2$ modulo $2\pi$.

### Exercise 1.9.13: Draw a weighted sum of the two column images

The vector $(2, 1)^T$ equals $2e_1 + e_2$, so linearity gives

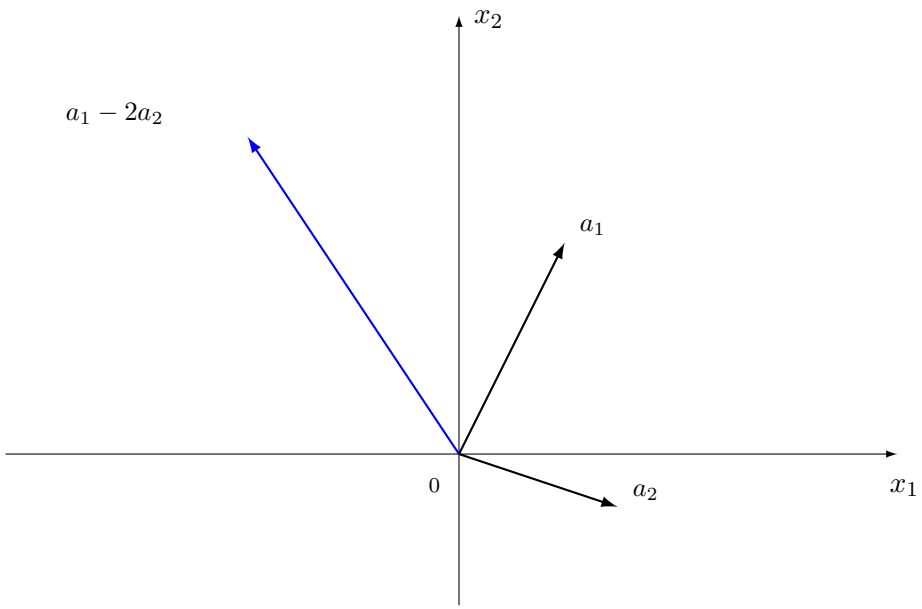
$$T(2, 1)^T = 2T(e_1) + T(e_2).$$

Take two copies of the up-left arrow $T(e_1)$, followed head to tail by one copy of the up-right arrow $T(e_2)$. The final arrow from zero ends high above the origin and slightly to its right. The following schematic preserves the source directions; the exact statement is the linear-combination formula, since the source gives no numerical coordinates for these image vectors.




### Exercise 1.9.14: Subtract twice the second column

The product $A(1,-2)^T$ is $a_1 - 2a_2$. Reverse the up-right/down-right source arrow $a_2$ and double its length, then attach it at the tip of $a_1$. The resultant points up and to the left. This construction uses the original columns as vectors, not their lengths alone. A schematic is shown below.




### Exercise 1.9.15: Read each coefficient into the correct row and column

Each output coordinate becomes one row. In that row, use the coefficients of $x_1, x_2, \ldots$ in order, inserting 0 for a missing variable. This gives

$$A = \begin{bmatrix} 2 & -4 & 0 \\ 1 & 0 & -1 \\ 0 & -1 & 3 \end{bmatrix}.$$

Indeed direct row-by-column multiplication yields

$$Ax = \begin{bmatrix} 2x_1 - 4x_2 \\ x_1 - x_3 \\ -x_2 + 3x_3 \end{bmatrix}.$$

The identity is required for *all* inputs, so these coefficients are forced; one numerical input would not be enough to determine all the entries.

### Exercise 1.9.16: Read each coefficient into the correct row and column

Each output coordinate becomes one row. In that row, use the coefficients of $x_1, x_2, \ldots$ in order, inserting 0 for a missing variable. This gives

$$A = \begin{bmatrix} 3 & -2 \\ 1 & 4 \\ 0 & 1 \end{bmatrix}.$$

Indeed direct row-by-column multiplication yields

$$Ax = \begin{bmatrix} 3x_1 - 2x_2 \\ x_1 + 4x_2 \\ x_2 \end{bmatrix}.$$

The identity is required for *all* inputs, so these coefficients are forced; one numerical input would not be enough to determine all the entries.

### Exercise 1.9.17: Read each coefficient into the correct row and column

Each output coordinate becomes one row. In that row, use the coefficients of $x_1, x_2, \ldots$ in order, inserting 0 for a missing variable. This gives

$$A = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & -1 \end{bmatrix}.$$

Indeed direct row-by-column multiplication yields

$$Ax = \begin{bmatrix} x_1 + 2x_2 \\ 0 \\ 2x_2 + x_4 \\ x_2 - x_4 \end{bmatrix}.$$

This is exactly the stated rule $T(x)$, so $T$ is a matrix transformation and therefore linear. For any $u, v, c, d$, $A(cu + dv) = cAu + dAv$ verifies the definition.

### Exercise 1.9.18: Read each coefficient into the correct row and column

Each output coordinate becomes one row. In that row, use the coefficients of $x_1, x_2, \ldots$ in order, inserting 0 for a missing variable. This gives

$$A = \begin{bmatrix} 1 & 4 \\ 0 & 0 \\ 1 & -3 \\ 1 & 0 \end{bmatrix}.$$

Indeed direct row-by-column multiplication yields

$$Ax = \begin{bmatrix} x_1 + 4x_2 \\ 0 \\ x_1 - 3x_2 \\ x_1 \end{bmatrix}.$$

This is exactly the stated rule $T(x)$, so $T$ is a matrix transformation and therefore linear. For any $u, v, c, d$, $A(cu + dv) = cAu + dAv$ verifies the definition.

### Exercise 1.9.19: Read each coefficient into the correct row and column

Each output coordinate becomes one row. In that row, use the coefficients of $x_1, x_2, \ldots$ in order, inserting 0 for a missing variable. This gives

$$A = \begin{bmatrix} 1 & -5 & 4 \\ 0 & 1 & -6 \end{bmatrix}.$$

Indeed direct row-by-column multiplication yields

$$Ax = \begin{bmatrix} x_1 - 5x_2 + 4x_3 \\ x_2 - 6x_3 \end{bmatrix}.$$

This is exactly the stated rule $T(x)$, so $T$ is a matrix transformation and therefore linear. For any $u, v, c, d$, $A(cu + dv) = cAu + dAv$ verifies the definition.

### Exercise 1.9.20: Read each coefficient into the correct row and column

Each output coordinate becomes one row. In that row, use the coefficients of $x_1, x_2, \ldots$ in order, inserting 0 for a missing variable. This gives

$$A = \begin{bmatrix} 3 & 0 & 4 & -2 \end{bmatrix}.$$

Indeed direct row-by-column multiplication yields

$$Ax = \begin{bmatrix} 3x_1 + 4x_3 - 2x_4 \end{bmatrix}.$$

This is exactly the stated rule $T(x)$, so $T$ is a matrix transformation and therefore linear. For any $u, v, c, d$, $A(cu + dv) = cAu + dAv$ verifies the definition.

### Exercise 1.9.21: Find the input producing two prescribed coordinates

The equations are $x_1 + x_2 = 3$ and $4x_1 + 5x_2 = 8$. Subtract four times the first from the second: $x_2 = -4$. Substitute into the first: $x_1 = 3 - (-4) = 7$. A check gives $T(7, -4) = (7 - 4, 28 - 20) = (3, 8)$.

**Conclusion.** $x = (7, -4)^T$. The elimination determines both variables uniquely.

### Exercise 1.9.22: Check all three output coordinates

From $2x_1 - x_2 = 0$, obtain $x_2 = 2x_1$. The second equation $-3x_1 + x_2 = -1$ then becomes $-x_1 = -1$, so $x_1 = 1$ and $x_2 = 2$. The third equation must still be checked: $2(1) - 3(2) = -4$, as required.

**Conclusion.** $x = (1, 2)^T$. The third equation is consistent with the values forced by the first two.

### Exercise 1.9.23: What coordinate-vector images determine

(a) **True.** Every $x \in \mathbb{R}^n$ is $\sum_j x_j e_j$, so linearity determines $T(x)$ from all the images $T(e_j)$.

(b) **True.** The rotation matrix $\begin{bmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{bmatrix}$ implements the stated rotation, and multiplication by a matrix is linear.

(c) **False.** A composition of linear maps is always linear: $T(S(cu+dv)) = T(cS(u)+dS(v)) = cT(S(u)) + dT(S(v))$.

(d) **False.** Every map with that codomain sends each input somewhere in $\mathbb{R}^m$, whether or not it is onto. Onto means every *output* in $\mathbb{R}^m$ is reached by some input. The zero map to $\mathbb{R}$ is a counterexample to the claimed implication.

(e) **False.** The $3 \times 2$ matrix with columns $e_1, e_2$ in $\mathbb{R}^3$ has independent columns, so its map is one-to-one. Having more output coordinates than input coordinates does not obstruct injectivity.

### Exercise 1.9.24: Distinguish being a function from being one-to-one

(a) **False.** A $4 \times 3$ matrix has at most three pivots, so it cannot have a pivot in all four rows. Its columns cannot span $\mathbb{R}^4$.

(b) **True.** The matrix with columns $T(e_j)$ implements any linear map between the stated coordinate spaces.

(c) **True.** This is exactly the column construction of the standard matrix.

(d) **False.** Having one output for each input merely says $T$ is a function. One-to-one requires different inputs to give different outputs. The zero map has a unique output at every input but is not one-to-one on a nontrivial domain.

(e) **False.** A horizontal shear has matrix $\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$, with nonzero shear amount $k$. The displayed diagonal matrices instead change signs of coordinate axes.

### Exercise 1.9.25: An unused input coordinate and a zero output coordinate

For the map in Exercise 17, the third column is zero, so $T(e_3) = 0 = T(0)$ with $e_3 \neq 0$. Thus it is **not one-to-one**. The second output coordinate is always zero, so the vector $(0,1,0,0)^T$ is never attained. Therefore it is also **not onto** $\mathbb{R}^4$. These two failures concern different sides of the same matrix: a zero column creates nonuniqueness, and a zero row here prevents some outputs.

### Exercise 1.9.26: A wide matrix can be onto but not one-to-one

The matrix in Exercise 2 is $\begin{bmatrix} 1 & -2 & 3 \\ 4 & 9 & -8 \end{bmatrix}$. Replacing $R_2$ by $R_2 - 4R_1$ gives $\begin{bmatrix} 1 & -2 & 3 \\ 0 & 17 & -20 \end{bmatrix}$. There is a pivot in both rows, so the map is **onto** $\mathbb{R}^2$. But three columns and only two pivots leave a free variable, so it is **not one-to-one**. For example $(-11, 20, 17)^T$ is a nonzero input sent to zero, as substitution verifies.

### Exercise 1.9.27: Two pivot rows with one free variable

For Exercise 19, the matrix $\begin{bmatrix} 1 & -5 & 4 \\ 0 & 1 & -6 \end{bmatrix}$ is already in echelon form. Its two nonzero rows each have a pivot, making the map **onto** $\mathbb{R}^2$. The homogeneous equations yield $x_2 = 6x_3$ and $x_1 = 26x_3$, so $(26, 6, 1)^T$ is a nonzero input sent to zero. Hence the map is **not one-to-one**.

### Exercise 1.9.28: Use the geometry of two column directions

In Exercise 14 the vectors $a_1$ and $a_2$ are nonzero and not parallel: the first points up-right and the second down-right. They are therefore independent, giving two pivot columns. In a $2 \times 2$ matrix those are also two pivot rows. Thus the map is **one-to-one and onto** $\mathbb{R}^2$. Equivalently, the two independent column directions span the whole plane, and each combination has unique coefficients.

### Exercise 1.9.29: A tall one-to-one matrix

For $T : \mathbb{R}^3 \to \mathbb{R}^4$ to be one-to-one, its $4 \times 3$ matrix must have a pivot in every column. The sole echelon pattern is

$$\begin{bmatrix} p & * & * \\ 0 & q & * \\ 0 & 0 & r \\ 0 & 0 & 0 \end{bmatrix}, \qquad p, q, r \neq 0.$$

The last row must be zero because the three available pivots occupy the first three nonzero rows.

### Exercise 1.9.30: Every full-row-rank three-by-four echelon pattern

Onto $\mathbb{R}^3$ requires three pivots, one in each row. They can occupy any three of the four columns, subject to moving right as one moves down. Thus the possible patterns are

$$\begin{bmatrix} p & * & * & * \\ 0 & q & * & * \\ 0 & 0 & r & * \end{bmatrix}, \qquad \begin{bmatrix} p & * & * & * \\ 0 & q & * & * \\ 0 & 0 & 0 & r \end{bmatrix},$$

$$\begin{bmatrix} p & * & * & * \\ 0 & 0 & q & * \\ 0 & 0 & 0 & r \end{bmatrix}, \qquad \begin{bmatrix} 0 & p & * & * \\ 0 & 0 & q & * \\ 0 & 0 & 0 & r \end{bmatrix}, \qquad p, q, r \neq 0.$$

Here $*$ denotes any entry. There are no additional patterns, because there are exactly four ways to choose three pivot columns out of four.

### Exercise 1.9.31: Count columns for injectivity

The blank is $\boxed{n}$. One-to-one is equivalent to $Ax = 0$ having only zero as a solution. With $n$ variables, this happens exactly when none is free, that is, when all $n$ columns are pivot columns. Conversely any missing column pivot leaves a free variable and thus a nonzero input with the same image as zero.

### Exercise 1.9.32: Count rows for surjectivity

The blank is $\boxed{m}$. Theorem 12 says onto is equivalent to columns spanning $\mathbb{R}^m$. Theorem 4 in Section 1.4 says this is equivalent to a pivot in every row. There are $m$ rows, so there must be $m$ pivots and hence $m$ pivot columns. A row without a pivot would allow an inconsistent target after row reduction.

### Exercise 1.9.33: The standard matrix is unique

Suppose $T(x) = Ax = Bx$ for every $x \in \mathbb{R}^n$. Substitute $x = e_j$. The vector $Ae_j$ is column $j$ of $A$ and $Be_j$ is column $j$ of $B$. Both equal $T(e_j)$, so these columns agree for every $j = 1, \ldots, n$. Matrices with identical corresponding columns are identical, proving $A = B$. Equality on just a single nonbasis input would not imply equality of all columns.

### Exercise 1.9.34: A composition preserves both operations

Take $u, v \in \mathbb{R}^p$ and scalars $c, d$. Linearity of $S$ first gives $S(cu + dv) = cS(u) + dS(v)$, where the two image vectors lie in $\mathbb{R}^n$, the domain of $T$. Applying linearity of $T$ then gives

$$T(S(cu + dv)) = T(cS(u) + dS(v)) = cT(S(u)) + dT(S(v)).$$

This is the combined linearity criterion for $T \circ S$ on $\mathbb{R}^p$, with outputs in $\mathbb{R}^m$. The domain matching is what makes the composition defined.

### Exercise 1.9.35: Compare input and output dimensions

If $T$ is onto, its $m \times n$ matrix has $m$ pivot rows, so it needs at least $m$ columns: $\boxed{m \leq n}$. If it is one-to-one, the matrix has $n$ pivot columns and therefore needs at least $n$ rows: $\boxed{n \leq m}$. If both properties hold, both inequalities force $m = n$. These are necessary dimension conditions, not sufficient ones; even a square zero matrix has neither property in positive dimension.

### Exercise 1.9.36: Onto asks for the existence of a preimage

Saying $T$ is onto means: for *every* $b$ in the codomain, there *exists* an input $x$ with $T(x) = b$. Thus it asks whether each target can be reached at all. It does not require that such an input be unique. A wide full-row-rank matrix can be onto while each target has infinitely many preimages.

### Exercise 1.9.37: Use exact reduction to test the requested property

The source matrix and its exact reduced echelon form are

$$A = \begin{bmatrix} -5 & 6 & -5 & -6 \\ 8 & 3 & -3 & 8 \\ 2 & 9 & 5 & -12 \\ -3 & 2 & 7 & -12 \end{bmatrix},$$

$$\operatorname{rref}(A) = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There are only three pivots in four columns. Taking the free variable $x_4 = 1$ gives the nonzero solution $x = \begin{bmatrix} -1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$ of $Ax = 0$. Direct multiplication verifies this relation. Hence $T$ is not one-to-one.

### Exercise 1.9.38: Use exact reduction to test the requested property

The source matrix and its exact reduced echelon form are

$$A = \begin{bmatrix} 7 & 5 & 9 & -9 \\ 5 & 6 & 4 & -4 \\ 4 & 8 & 0 & 7 \\ -6 & -6 & 6 & 5 \end{bmatrix},$$

$$\operatorname{rref}(A) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Every one of the four columns has a pivot, so $Ax = 0$ forces $x = 0$. Thus $T$ is one-to-one. Since there is also a pivot in every row, this particular map is onto as well, although injectivity is the requested property.

### Exercise 1.9.39: Use exact reduction to test the requested property

The source matrix and its exact reduced echelon form are

$$A = \begin{bmatrix} 4 & -7 & 3 & 7 & 5 \\ 6 & -8 & 5 & 12 & -8 \\ -7 & 10 & -8 & -9 & 14 \\ 3 & -5 & 4 & 2 & -6 \\ -5 & 6 & -6 & -7 & 3 \end{bmatrix},$$

$$\operatorname{rref}(A) = \begin{bmatrix} 1 & 0 & 0 & 5 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

There are four pivots and five rows. The zero row in the reduced coefficient matrix means there is not a pivot in every row, so the columns cannot span $\mathbb{R}^5$. Therefore $T$ does not map $\mathbb{R}^5$ onto $\mathbb{R}^5$.

### Exercise 1.9.40: Use exact reduction to test the requested property

The source matrix and its exact reduced echelon form are

$$A = \begin{bmatrix} 9 & 43 & 5 & 6 & -1 \\ 14 & 15 & -7 & -5 & 4 \\ -8 & -6 & 12 & -5 & -9 \\ -5 & -6 & -4 & 9 & 8 \\ 13 & 14 & 15 & 3 & 11 \end{bmatrix},$$

$$\operatorname{rref}(A) = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

There is a pivot in every one of the five rows. Appending any right-hand side cannot create an inconsistent zero-coefficient row, so $Ax = b$ is solvable for every $b \in \mathbb{R}^5$. Thus $T$ maps $\mathbb{R}^5$ onto $\mathbb{R}^5$; it is also one-to-one because every column has a pivot.

## 1.10 Linear models in business, science, and engineering

A model begins by defining its variables and their units. Matrix columns then describe what one unit of each input contributes, and rows record one output requirement at a time. A mathematically consistent nutritional system is not physically feasible if it requires negative amounts of food. In a circuit, signs depend on the chosen loop directions and battery polarities. In a migration model, column $j$ records where people starting in region $j$ go, so its entries sum to 1 when nobody leaves the system. The labels [M] in the text permit matrix software; exact reduced matrices and substitution checks are supplied below rather than treating a software answer as a substitute for interpreting the model.

### Practice problem 1.10.P1: Express the diet example as a matrix equation

Let $x_1, x_2, x_3$ be the amounts of nonfat milk, soy flour, and whey, respectively, measured in 100-gram units. Put the nutrient contributions per unit into columns and the targets into the right-hand side:

$$A = \begin{bmatrix} 36 & 51 & 13 \\ 52 & 34 & 74 \\ 0 & 7 & 1.1 \end{bmatrix}, \qquad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \qquad b = \begin{bmatrix} 33 \\ 45 \\ 3 \end{bmatrix}, \qquad Ax = b.$$

The rows specify protein, carbohydrate, and fat in grams. For example, the first row says $36x_1 + 51x_2 + 13x_3 = 33$, so it adds the protein from all three ingredients. The actual food masses in grams are $100x_j$.

### Exercise 1.10.1: A two-cereal mixture with four requirements

**(a) Define the serving counts and the vector equation.**

Let $x$ be servings of Cheerios and $y$ servings of Natural Cereal. With rows ordered as calories, protein, carbohydrate, and fat,

$$x \begin{bmatrix} 110 \\ 4 \\ 20 \\ 2 \end{bmatrix} + y \begin{bmatrix} 130 \\ 3 \\ 18 \\ 5 \end{bmatrix} = \begin{bmatrix} 295 \\ 9 \\ 48 \\ 8 \end{bmatrix}.$$

**(b) Solve and check all the requirements.**

The equivalent matrix equation is

$$\begin{bmatrix} 110 & 130 \\ 4 & 3 \\ 20 & 18 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 295 \\ 9 \\ 48 \\ 8 \end{bmatrix}.$$

Use the protein equation $4x + 3y = 9$ and fat equation $2x + 5y = 8$. Subtract twice the latter from the former: $-7y = -7$, hence $y = 1$. Then $2x + 5 = 8$ gives $x = 3/2$. Check the other rows: $110(3/2) + 130 = 295$ and $20(3/2) + 18 = 48$. Both serving counts are nonnegative, so the solution is physically meaningful.

**Conclusion.** Mix 1.5 servings of Cheerios with 1 serving of Natural Cereal.

### Exercise 1.10.2: A specified mixture and a new nutritional target

**(a) Build the contribution matrix.**

Order the rows as calories, protein, fiber, and fat, and the columns as Shredded Wheat and Crispix. Then

$$B = \begin{bmatrix} 160 & 110 \\ 5 & 2 \\ 6 & 0.1 \\ 1 & 0.4 \end{bmatrix}, \quad u = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \quad Bu = \begin{bmatrix} 700 \\ 19 \\ 18.2 \\ 3.8 \end{bmatrix}.$$

For instance the fiber amount is $6(3) + 0.1(2) = 18.2$ grams.

**(b) Solve for the requested different mixture.**

The desired equation is $B(x, y)^T = (130, 3.20, 2.46, 0.64)^T$. Its reduced augmented matrix is

$$\begin{bmatrix} 1 & 0 & 0.4 \\ 0 & 1 & 0.6 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Thus $x = 0.4$ and $y = 0.6$. To check all rows, the totals are $64 + 66 = 130$ calories, $2 + 1.2 = 3.20$ protein, $2.4 + 0.06 = 2.46$ fiber, and $0.4 + 0.24 = 0.64$ fat. These nonnegative proportions also add to one serving.

**Conclusion.** The target is attainable with 0.4 serving of Shredded Wheat and 0.6 serving of Crispix.

### Exercise 1.10.3: Compare two versions of a lunch

Let $x, y, z$ be servings of the pasta dish, broccoli, and chicken. Use calories, protein, and fiber as the three rows. Solving each system requires all three targets, not only the calories.

**(a) Use the original Mac and Cheese.**

$$\begin{bmatrix} 270 & 51 & 70 \\ 10 & 5.4 & 15 \\ 2 & 5.2 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 400 \\ 30 \\ 10 \end{bmatrix}.$$

Exact row reduction, treating the decimals as the stated rational numbers, gives

$$[A \mid b] \sim \begin{bmatrix} 1 & 0 & 0 & 8205/8323 \\ 0 & 1 & 0 & 12850/8323 \\ 0 & 0 & 1 & 6550/8323 \end{bmatrix}.$$

Hence $(x, y, z) \approx (0.98582, 1.54391, 0.78698)$. Every variable is determined by a pivot, so the solution is unique. Substituting the exact fractions into the original three rows gives $(400, 30, 10)^T$. All three fractions are positive, so the mixture is feasible.

**(b) Replace the pasta column by the whole-wheat-shells data.**

Only that ingredient's column changes:

$$\begin{bmatrix} 260 & 51 & 70 \\ 9 & 5.4 & 15 \\ 5 & 5.2 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 400 \\ 30 \\ 10 \end{bmatrix}.$$

Reduction gives $(x, y, z) = (5470, 4400, 5180)/5023$, approximately $(1.08899, 0.87597, 1.03126)$. Substitution again gives exactly 400 calories, 30 grams protein, and 10 grams fiber before rounding. The broccoli portion drops from about 1.54 to 0.88 serving, in line with the stated preference.

**Conclusion.** To two decimal places the mixtures are $(0.99, 1.54, 0.79)$ and $(1.09, 0.88, 1.03)$ servings in the stated ingredient order. Use the exact fractions for exact nutritional totals.

### Exercise 1.10.4: A consistent diet equation can be physically impossible

#### (a) Specify the four variables and four nutrient rows.

Let $x_1, x_2, x_3, x_4$ be 100-gram units of nonfat milk, soy flour, whey, and isolated soy protein. In the row order protein, carbohydrate, fat, calcium, the equation is

$$\begin{bmatrix} 36 & 51 & 13 & 80 \\ 52 & 34 & 74 & 0 \\ 0 & 7 & 1.1 & 3.4 \\ 1.26 & 0.19 & 0.8 & 0.18 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 33 \\ 45 \\ 3 \\ 0.8 \end{bmatrix}.$$

Each coefficient is grams of that nutrient per 100 grams of ingredient.

#### (b) Solve and interpret the signs.

Exact reduction yields the unique solution

$$x = \begin{bmatrix} 2264687/3531104 \\ 1921341/3531104 \\ -326887/3531104 \\ -1468529/7062208 \end{bmatrix} \approx \begin{bmatrix} 0.64135 \\ 0.54412 \\ -0.09257 \\ -0.20794 \end{bmatrix}.$$

Multiplication of the exact vector by the original matrix reproduces all four targets, so the algebraic equations are consistent. But the whey and isolated-protein amounts are negative. One cannot subtract food already absent from the mixture, so this unique algebraic solution is not a permissible recipe. Because it is unique, there is no different nonnegative solution to these same equations.

**Conclusion.** These four ingredients cannot meet all four targets with nonnegative amounts under the supplied nutrient data.

**Circuit convention for Exercises 5–8.** Use the counterclockwise loop directions drawn in the textbook. For each loop, add all resistances on its boundary to get the diagonal coefficient. A shared resistor carries the difference of the adjacent loop currents, so its resistance enters the neighboring column with a minus sign. On the right, crossing a battery from its short plate to its long plate contributes a positive voltage; the reverse crossing contributes a negative voltage. These rules explain every sign in $Ri = v$. A negative current means the actual direction is opposite to the assumed counterclockwise direction.

### Exercise 1.10.5: Derive the loop equations and solve them

The four diagonal resistance sums are $1+1+4+5=11$, $3+5+1+1=10$, $4+1+2+2=9$, and $1+2+3+4=10$. The successive shared resistances are $5, 1, 2$. Battery crossings give $20+30=50$, $-30-10=-40$, $10+20=30$, and $-20-10=-30$ volts.

Thus, in amperes, the matrix equation is

$$\begin{bmatrix} 11 & -5 & 0 & 0 \\ -5 & 10 & -1 & 0 \\ 0 & -1 & 9 & -2 \\ 0 & 0 & -2 & 10 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix} = \begin{bmatrix} 50 \\ -40 \\ 30 \\ -30 \end{bmatrix}.$$

Exact elimination of this augmented system gives a pivot in every variable column and hence the unique solution

$$i = \begin{bmatrix} \frac{265}{72} \\ -\frac{137}{72} \\ \frac{185}{72} \\ -\frac{179}{72} \end{bmatrix} \approx \begin{bmatrix} 3.6806 \\ -1.9028 \\ 2.5694 \\ -2.4861 \end{bmatrix}.$$

The displayed exact fractions satisfy $Ri = v$ in every row. For example, the first row computes 50, the first specified net voltage. All remaining rows were checked by the same substitution. Interpret a minus sign as a direction reversal rather than a negative physical magnitude.

### Exercise 1.10.6: Derive the loop equations and solve them

The diagonal sums are $3+2+1=6$, $1+4+4=9$, $4+1+2=7$, and $2+2+3=7$. Adjacent loops share resistances $1, 4, 2$. Every battery on the left is crossed from short plate to long plate in the chosen direction, giving voltages $30, 20, 40, 10$.

Thus, in amperes, the matrix equation is

$$\begin{bmatrix} 6 & -1 & 0 & 0 \\ -1 & 9 & -4 & 0 \\ 0 & -4 & 7 & -2 \\ 0 & 0 & -2 & 7 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix} = \begin{bmatrix} 30 \\ 20 \\ 40 \\ 10 \end{bmatrix}.$$

Exact elimination of this augmented system gives a pivot in every variable column and hence the unique solution

$$i = \begin{bmatrix} \frac{3630}{571} \\ \frac{4650}{571} \\ \frac{6700}{571} \\ \frac{2730}{571} \end{bmatrix} \approx \begin{bmatrix} 6.3573 \\ 8.1436 \\ 11.734 \\ 4.7811 \end{bmatrix}.$$

The displayed exact fractions satisfy $Ri = v$ in every row. For example, the first row computes 30, the first specified net voltage. All remaining rows were checked by the same substitution. Interpret a minus sign as a direction reversal rather than a negative physical magnitude.

### Exercise 1.10.7: Derive the loop equations and solve them

Keep the order $I_1$ upper left, $I_2$ lower left, $I_3$ lower right, $I_4$ upper right. The diagonal sums are $1+7+4=12$, $7+6+2=15$, $6+5+3=14$, and $4+5+4=13$. Shared resistances are 7 between loops 1 and 2, 6 between 2 and 3, 5 between 3 and 4, and 4 between 4 and 1. The source voltages are $40, 30, 20, -10$; the last is negative because loop 4 crosses that battery against its positive direction.

Thus, in amperes, the matrix equation is

$$\begin{bmatrix} 12 & -7 & 0 & -4 \\ -7 & 15 & -6 & 0 \\ 0 & -6 & 14 & -5 \\ -4 & 0 & -5 & 13 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix} = \begin{bmatrix} 40 \\ 30 \\ 20 \\ -10 \end{bmatrix}.$$

Exact elimination of this augmented system gives a pivot in every variable column and hence the unique solution

$$i = \begin{bmatrix} \frac{119910}{10487} \\ \frac{110640}{10487} \\ \frac{84270}{10487} \\ \frac{61240}{10487} \end{bmatrix} \approx \begin{bmatrix} 11.434 \\ 10.55 \\ 8.0357 \\ 5.8396 \end{bmatrix}.$$

The displayed exact fractions satisfy $Ri = v$ in every row. For example, the first row computes 40, the first specified net voltage. All remaining rows were checked by the same substitution. Interpret a minus sign as a direction reversal rather than a negative physical magnitude.

### Exercise 1.10.8: Derive the loop equations and solve them

The loop order is the four outer regions $1, 2, 3, 4$, then the central diamond 5. Their diagonal sums are $3+1+4+1=9$, $1+1+3+2=7$, $2+2+3+3=10$, $1+1+2+3=7$, and $4+3+3+2=12$. The central loop shares resistances $4, 3, 3, 2$ with the four outer loops. Additional shared stems are 1 between 1 and 4, 1 between 1 and 2, 2 between 2 and 3, and 3 between 3 and 4. Its own battery contribution is zero; the outer contributions are $50, -30, 20, -40$.

Thus, in amperes, the matrix equation is

$$\begin{bmatrix} 9 & -1 & 0 & -1 & -4 \\ -1 & 7 & -2 & 0 & -3 \\ 0 & -2 & 10 & -3 & -3 \\ -1 & 0 & -3 & 7 & -2 \\ -4 & -3 & -3 & -2 & 12 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \\ I_5 \end{bmatrix} = \begin{bmatrix} 50 \\ -30 \\ 20 \\ -40 \\ 0 \end{bmatrix}.$$

Exact elimination of this augmented system gives a pivot in every variable column and hence the unique solution

$$i = \begin{bmatrix} \frac{3019}{755} \\ -\frac{3309}{755} \\ -\frac{683}{755} \\ -\frac{4382}{755} \\ -\frac{722}{755} \end{bmatrix} \approx \begin{bmatrix} 3.9987 \\ -4.3828 \\ -0.90464 \\ -5.804 \\ -0.95629 \end{bmatrix}.$$

The displayed exact fractions satisfy $Ri = v$ in every row. For example, the first row computes 50, the first specified net voltage. All remaining rows were checked by the same substitution. Interpret a minus sign as a direction reversal rather than a negative physical magnitude.

**Migration convention for Exercises 9–13.** Put the current populations into a column vector. In each migration matrix, column $j$ tells what fractions of starting population $j$ arrive in the various destinations. Thus $x_{k+1} = Mx_k$ and two years later means $x_2 = M(Mx_0)$, not $Mx_0$.

### Exercise 1.10.9: A seven-percent outward migration rate

Let $x_k = (c_k, s_k)^T$ denote city and suburban populations $k$ years after 2010. The matrix and initial population are

$$M = \begin{bmatrix} 0.93 & 0.05 \\ 0.07 & 0.95 \end{bmatrix}, \qquad x_0 = \begin{bmatrix} 800000 \\ 500000 \end{bmatrix}, \qquad x_{k+1} = Mx_k.$$

For example $c_{k+1} = 0.93c_k + 0.05s_k$: city residents who stay plus suburban residents who move into the city. First multiplication gives $x_1 = (769000, 531000)^T$. Multiplying once more gives

$$x_2 = \begin{bmatrix} 741720 \\ 558280 \end{bmatrix}.$$

The total remains 1300000, as a check on the column sums and arithmetic.

**Conclusion.** In 2012 the model predicts 741720 city residents and 558280 suburban residents.

### Exercise 1.10.10: A six-percent outward migration rate

With city first and suburbs second,

$$M = \begin{bmatrix} 0.94 & 0.04 \\ 0.06 & 0.96 \end{bmatrix}, \quad x_0 = \begin{bmatrix} 10000000 \\ 800000 \end{bmatrix}, \quad x_{k+1} = Mx_k.$$

Here the first column says 94% of city residents stay and 6% move to the suburbs. The second says 4% move in and 96% stay in the suburbs. Successive multiplications give

$$x_1 = \begin{bmatrix} 9432000 \\ 1368000 \end{bmatrix}, \qquad x_2 = \begin{bmatrix} 8920800 \\ 1879200 \end{bmatrix}.$$

Both totals equal the initial 10800000.

**Conclusion.** The 2012 prediction is 8920800 in the city and 1879200 in the suburbs, subject to the model's exclusions.

### Exercise 1.10.11: Estimate migration rates before iterating

#### (a) Divide each flow by its population of origin.

The fraction leaving California is $516100/31524000 \approx 0.0163716533$, hence 0.01637 to five decimal places. The fraction moving into California from elsewhere is $381262/228680000 \approx 0.0016672293$, hence 0.00167. Put California first. The rounded migration model is

$$M = \begin{bmatrix} 0.98363 & 0.00167 \\ 0.01637 & 0.99833 \end{bmatrix}, \qquad x_0 = \begin{bmatrix} 31524000 \\ 228680000 \end{bmatrix}.$$

The diagonal entries are the complements of the corresponding outward rates, not independently rounded unrelated quantities.

**(b) Apply six annual steps to reach 2000.**

Compute $x_{k+1} = Mx_k$ six times, retaining full arithmetic with the specified five-place coefficients. The result is

$$x_6 \approx \begin{bmatrix} 30754526.4814 \\ 229449473.5186 \end{bmatrix}.$$

To the nearest person, these are 30754526 in California and 229449474 elsewhere. Their sum is the unchanged total 260204000. The fractional entries before rounding are population estimates, not literal fractional people. Using unrounded migration rates would yield slightly different estimates, so it is important to state which rates were used.

**Conclusion.** The model predicts about 30.755 million in California and 229.449 million elsewhere, with births, deaths, and international migration excluded as required.

### Exercise 1.10.12: Follow a fleet through two daily transitions

Use the order airport, east, west. The columns correspond to the rental origin and rows to the return location, so

$$M = \begin{bmatrix} 0.97 & 0.05 & 0.10 \\ 0 & 0.90 & 0.05 \\ 0.03 & 0.05 & 0.85 \end{bmatrix}, \qquad x_0 = \begin{bmatrix} 295 \\ 55 \\ 150 \end{bmatrix}.$$

Monday to Tuesday gives $x_1 = Mx_0 = (303.9, 57, 139.1)^T$. Tuesday to Wednesday gives

$$x_2 = Mx_1 = \begin{bmatrix} 311.543 \\ 58.255 \\ 130.202 \end{bmatrix}.$$

For example the first entry is $0.97(303.9) + 0.05(57) + 0.10(139.1)$. The three entries sum to 500 before rounding, matching the fleet size.

**Conclusion.** Approximately 312 cars at the airport, 58 at the east location, and 130 at the west location on Wednesday.

### Exercise 1.10.13: Twenty iterations and their limiting pattern

The source model uses $M = \begin{bmatrix} 0.95 & 0.03 \\ 0.05 & 0.97 \end{bmatrix}$. The total population is 1000000 in both parts. Therefore, if $c_k$ is the city population, the suburb population is $1000000 - c_k$, and the first row of the recurrence becomes

$$c_{k+1} = 0.95c_k + 0.03(1000000 - c_k) = 0.92c_k + 30000.$$

Subtract 375000 from both sides to obtain $c_{k+1} - 375000 = 0.92(c_k - 375000)$. Repeated substitution gives

$$c_k = 375000 + (c_0 - 375000)(0.92)^k, \qquad s_k = 1000000 - c_k.$$

This formula both checks the matrix iterations and explains their pattern.

**(a) Start from** $(600000, 400000)^T$**.**

The city component is $375000 + 225000(0.92)^k$.

**(b) Start from** $(350000, 650000)^T$**.**

The city component is $375000 - 25000(0.92)^k$. The following table lists all twenty requested vectors, with entries rounded to the nearest person only for display.

| | Part (a) | | Part (b) | |
|---|---|---|---|---|
| $k$ | City | Suburbs | City | Suburbs |
| 1 | 582000 | 418000 | 352000 | 648000 |
| 2 | 565440 | 434560 | 353840 | 646160 |
| 3 | 550205 | 449795 | 355533 | 644467 |
| 4 | 536188 | 463812 | 357090 | 642910 |
| 5 | 523293 | 476707 | 358523 | 641477 |
| 6 | 511430 | 488570 | 359841 | 640159 |
| 7 | 500515 | 499485 | 361054 | 638946 |
| 8 | 490474 | 509526 | 362170 | 637830 |
| 9 | 481236 | 518764 | 363196 | 636804 |
| 10 | 472737 | 527263 | 364140 | 635860 |
| 11 | 464918 | 535082 | 365009 | 634991 |
| 12 | 457725 | 542275 | 365808 | 634192 |
| 13 | 451107 | 548893 | 366544 | 633456 |
| 14 | 445018 | 554982 | 367220 | 632780 |
| 15 | 439417 | 560583 | 367843 | 632157 |
| 16 | 434264 | 565736 | 368415 | 631585 |
| 17 | 429522 | 570478 | 368942 | 631058 |
| 18 | 425161 | 574839 | 369427 | 630573 |
| 19 | 421148 | 578852 | 369872 | 630128 |
| 20 | 417456 | 582544 | 370283 | 629717 |

Since $0 < 0.92 < 1$, its powers tend to zero. In (a) the city population decreases toward 375000, passing below the suburban population between years 7 and 8 (first below at the year-8 update). In (b) it increases toward the same 375000. The suburb populations approach 625000 from the opposite sides. The common limiting proportions come from the migration rule, not from the initial distribution.

**Conclusion.** Both initial distributions tend to $(375000, 625000)^T$, but they approach it from different sides.

## Exercise 1.10.14: Temperature models obey superposition

**(a) Write one averaging equation for each interior point.**

Order the points as 1 upper left, 2 upper right, 3 lower right, 4 lower left, as in the diagrams. Moving

the two interior-neighbor temperatures to the left gives the common coefficient matrix

$$K = \begin{bmatrix} 4 & -1 & 0 & -1 \\ -1 & 4 & -1 & 0 \\ 0 & -1 & 4 & -1 \\ -1 & 0 & -1 & 4 \end{bmatrix}.$$

For plate A, each point touches boundary temperatures 20 and 0, so $KT_A = (20, 20, 20, 20)^T$. Direct substitution gives $T_A = (10, 10, 10, 10)^T$. For plate B, the two relevant boundary values sum to $10, 40, 50, 20$ in point order. Thus $KT_B = (10, 40, 50, 20)^T$, and solving gives

$$T_B = (10, 17.5, 20, 12.5)^T.$$

For example its first equation is $4(10) - 17.5 - 12.5 = 10$, and its third is $4(20) - 17.5 - 12.5 = 50$. Their sum is

$$T_A + T_B = (20, 27.5, 30, 22.5)^T,$$

exactly the temperature list cited from Section 1.1. The boundary data there are the sums of these two plates' boundary data, so the same additivity appears at the interior points.

**(b) Triple all of plate A's boundary temperatures.**

The new right-hand side is $(60, 60, 60, 60)^T$, three times the old one. Hence $T = (30, 30, 30, 30)^T = 3T_A$. This is also checked directly: each row is $4(30) - 30 - 30 = 60$.

**(c) Formulate and justify the general rule.**

The map from the eight boundary values to the four interior values is linear: adding boundary data adds interior solutions, and scaling boundary data scales the interior solution. To justify uniqueness, if two solutions existed, their difference $z$ would satisfy $Kz = 0$. Choose a component whose absolute value is largest, say $|z_i| = L$. Its equation gives $4L \le 2L$, since there are two neighboring components, each of absolute value at most $L$. Thus $L = 0$ and $z = 0$. There cannot be two distinct solutions. Existence for every right-hand side $b$ can be verified directly by substitution in the explicit solution formula

$$T = \frac{1}{24} \begin{bmatrix} 7 & 2 & 1 & 2 \\ 2 & 7 & 2 & 1 \\ 1 & 2 & 7 & 2 \\ 2 & 1 & 2 & 7 \end{bmatrix} b.$$

Each $b_i$ is a sum of two boundary values, so both stages of the construction use only fixed linear combinations.

**Conclusion.** The interior-temperature map is linear. Its additivity explains part (a), and its homogeneity explains part (b).

## 1.S Chapter 1 supplementary exercises

These exercises combine consistency, parametrization, independence, and transformations. Keep existence and uniqueness separate: a missing variable pivot creates freedom only in a *consistent* system. When a problem describes physical quantities such as floors, solve the real linear equations first, then impose the required integer and nonnegative restrictions.

### Exercise 1.S.1: Twenty-six statements and their logical pitfalls

(a) **False.** Reduced echelon form is unique; ordinary echelon form is not. The one-by-one matrices [1] and [2] are row equivalent and both in echelon form, but are different.

(b) **False.** For $n = 1$, the equation $0x = 0$ has infinitely many solutions, more than one. A square system can have zero, one, or infinitely many solutions.

(c) **True.** If $Ax = Ay = b$ and $x \neq y$, then $A(x + t(y - x)) = b$ for every real $t$. Different $t$ give distinct solutions because $y - x \neq 0$.

(d) **False.** No free variables guarantees at most one solution, not consistency. The equations $x = 0$ and $x = 1$ have no free variable and no solution.

(e) **True.** Elementary row operations are reversible, so each operation preserves the solution set in both directions.

(f) **True.** If $Ax = Ay = b$ with $x \neq y$, then $A(x - y) = 0$ with $x - y \neq 0$. Thus the homogeneous system has both zero and a distinct nonzero solution.

(g) **False.** Every $A$ has a consistent system for $b = 0$. For the zero $2 \times 2$ matrix the columns span only $\{0\}$, not $\mathbb{R}^2$.

(h) **False.** Every matrix has a reduced echelon form, including inconsistent augmented matrices. The single equation $0x = 1$ is already a counterexample.

(i) **True.** Each row-equivalence class has one reduced echelon form. Reducing either member therefore gives the same matrix.

(j) **False.** $Ax = 0$ always has the trivial solution, including when free variables exist. Take $A = [0]$.

(k) **True.** Consistency for every $b \in \mathbb{R}^m$ means a pivot in every one of the $m$ rows. There are exactly $m$ pivots and thus $m$ pivot columns.

(l) **False.** Row pivots imply existence for every target, not uniqueness. The matrix [1 0] has a pivot in every row, but $x_1 = b$, with $x_2$ arbitrary, gives infinitely many solutions.

(m) **True.** In a square $n \times n$ matrix, $n$ pivots occupy every column and every row. Reduced pivots are ones with zeros elsewhere in their columns, so the reduced matrix is $I_n$.

(n) **True.** Both matrices reduce to $I_3$. Reduce $A$ to $I_3$, then reverse the operations reducing $B$ to $I_3$ to reach $B$.

(o) **True.** The first system supplies a nonzero $z$ with $Az = 0$. If $Ay = c$, then $A(y + tz) = c$ for every real $t$, giving infinitely many distinct solutions of the second system.

(p) **True.** Row-equivalent matrices have the same number of pivots. If $A$ spans $\mathbb{R}^m$, its $m$ pivots fill all rows; the same is true of $B$, so its columns also span $\mathbb{R}^m$. This does not assert equality of their actual column vectors.

(q) **False.** The list $e_1, e_2, e_1 + e_2$ in $\mathbb{R}^3$ has no proportional pair but has the nontrivial relation $e_1 + e_2 - (e_1 + e_2) = 0$.

(r) **True.** Three vectors in $\mathbb{R}^2$ cannot be independent: a $2 \times 3$ matrix has at most two pivots. Thus the assumed independence excludes that ambient space.

(s) **False.** Four columns supply at most four pivots, not the five row pivots required to span $\mathbb{R}^5$.

(t) **True.** $-u = (-1)u + 0v$, a permitted linear combination.

(u) **False.** Let $u = e_1$, $v = 2e_1$, $w = e_2$ in $\mathbb{R}^2$. All are nonzero, but the first two span only the horizontal axis, which does not contain $w$.

(v) **False.** Take $u = e_1$, $v = e_2$, and $w = e_2$. Then $w = 0u + v$, but $u$ is outside Span$\{v, w\}$.

(w) **False.** The hypothesis does not exclude $v_1 = 0$. Take $v_1 = 0$, $v_2 = e_1$, $v_3 = e_2$ in $\mathbb{R}^5$. The second is not a multiple of the first, and the third is outside their span, but the zero vector makes the list dependent.

(x) **True.** A linear transformation is a function with additional operation-preserving properties.

(y) **True.** A $6 \times 5$ matrix cannot have six pivots, so its columns cannot span $\mathbb{R}^6$.

(z) **False.** $m$ pivot columns mean onto $\mathbb{R}^m$. For one-to-one we need $n$ pivots. Again [1 0] has $m = 1$ pivot but sends every $(0, t)^T$ to zero.

### Exercise 1.S.2: Every possible solution set of one equation

If $a \neq 0$, division gives the unique solution $x = b/a$. If $a = 0$ but $b \neq 0$, the equation says $0 = b$ and has no solution. If $a = b = 0$, the equation is true for every real $x$, so its solution set is all of $\mathbb{R}$. These three cases exhaust the values of $a, b$.

### Exercise 1.S.3: Three configurations of planes

(a) Use $x = 0$, $y = 0$, $x + y = 0$. The first two force $x = y = 0$, while $z$ is arbitrary; the third adds no restriction. The intersection is the $z$-axis, a single line.

(b) Use $x = 0$, $y = 0$, $z = 0$. Their only common point is the origin.

(c) Use $x = 0$, $x = 1$, $y = 0$. No point can satisfy the first two simultaneously, so there is no common intersection.

Every listed equation has at least one nonzero coefficient and therefore really does describe a plane, rather than the whole space or an empty set.

### Exercise 1.S.4: Why the square size matters

Three pivot columns in a $3 \times 3$ coefficient matrix also mean three pivot rows. Thus appending any right-hand side gives a consistent system, because no zero-coefficient row remains to create a contradiction. There are no free variables, so the solution is unique. Both properties are needed: column pivots alone in a taller matrix would not ensure existence.

### Exercise 1.S.5: Classify two parameter-dependent systems

#### (a) Eliminate the first variable.

Subtract four times $x_1 + 3x_2 = k$ from the second equation. The result is $(h - 12)x_2 = 8 - 4k$. If $h \neq 12$, this determines $x_2$ uniquely, then the first equation determines $x_1$. If $h = 12$, the result is $0 = 8 - 4k$, consistent exactly when $k = 2$.

**Conclusion.** (i) No solutions: $h = 12, k \neq 2$. (ii) One solution: $h \neq 12$ with any $k$. (iii) Infinitely many: $h = 12, k = 2$.

#### (b) Add three times the first equation to the second.

The eliminated equation is $(k + 3h)x_2 = 1$. If $k + 3h \neq 0$, then $x_2 = 1/(k + 3h)$ and $x_1 = (hx_2 - 1)/2$ are uniquely determined. If $k + 3h = 0$, we obtain the contradiction $0 = 1$.

**Conclusion.** (i) No solutions: $k = -3h$. (ii) One solution: $k \neq -3h$. (iii) No parameter values give infinitely many solutions.

### Exercise 1.S.6: Restate one consistency question three ways

Let $a_1 = (4, 8)^T$, $a_2 = (-2, -3)^T$, $a_3 = (7, 10)^T$, and $b = (-5, -3)^T$.

#### (a) Use a linear combination.

Ask whether $b = x_1a_1 + x_2a_2 + x_3a_3$ for some real weights. Eliminating twice the first scalar equation from the second gives $x_2 - 4x_3 = 7$. Let $x_3 = t$; then $x_2 = 7 + 4t$ and $4x_1 = 9 + t$. Thus the full solution family is

$$x = (9/4 + t/4,\ 7 + 4t,\ t)^T, \qquad t \in \mathbb{R}.$$

In particular $(9/4, 7, 0)^T$ works, proving consistency.

#### (b) Use matrix columns.

For $A = \begin{bmatrix} 4 & -2 & 7 \\ 8 & -3 & 10 \end{bmatrix}$, ask whether $b$ lies in the span of the columns of $A$. Part (a) answers yes.

#### (c) Use a transformation.

Define $T : \mathbb{R}^3 \to \mathbb{R}^2$ by $T(x) = Ax$. Ask whether $b$ is in the range of $T$, or equivalently whether any input has image $b$. The displayed family supplies all such inputs.

### Exercise 1.S.7: Restate consistency for every right-hand side

Set $v_1 = (2, -5, 7)^T$, $v_2 = (-4, 1, -5)^T$, and $v_3 = (-2, 1, -3)^T$, and put $A = [v_1\ v_2\ v_3]$.

**(a) Ask whether the span is all of $\mathbb{R}^3$.**

Row three of $A$ is row one minus row two. Hence every vector $b$ in the column span satisfies $b_3 = b_1 - b_2$. The first two rows have two independent leading columns: elimination of $x_1$ from them leaves a nonzero coefficient $-9$ on $x_2$ after multiplying the first row by $5/2$ and adding it to row two. Thus the first two equations can be solved for any $b_1, b_2$, but the third requires exactly $b_3 = b_1 - b_2$. Consequently $\mathrm{Span}\{v_1, v_2, v_3\}$ is this plane, not $\mathbb{R}^3$.

**(b) State the column version.**

The question asks whether the columns of $A$ span $\mathbb{R}^3$. They do not; for example $(0, 0, 1)^T$ is outside their span.

**(c) State the transformation version.**

For $T : \mathbb{R}^3 \to \mathbb{R}^3$, $T(x) = Ax$, the question is whether $T$ is onto. It is not, because its range is the proper plane above.

**Conclusion.** The system is consistent precisely for $b_3 = b_1 - b_2$, so it is not consistent for all triples of right-hand sides.

## Exercise 1.S.8: Echelon patterns for spanning columns

Use $p, q, r$ for nonzero pivots and $*$ for arbitrary entries.

**(a) Two rows and three columns.**

Spanning $\mathbb{R}^2$ requires two row pivots, which can occur in columns $(1, 2)$, $(1, 3)$, or $(2, 3)$:

$$\begin{bmatrix} p & * & * \\ 0 & q & * \end{bmatrix}, \quad \begin{bmatrix} p & * & * \\ 0 & 0 & q \end{bmatrix}, \quad \begin{bmatrix} 0 & p & * \\ 0 & 0 & q \end{bmatrix}.$$

**(b) Three rows and three columns.**

All three columns must also be pivot columns, leaving the sole pattern $\begin{bmatrix} p & * & * \\ 0 & q & * \\ 0 & 0 & r \end{bmatrix}$.

## Exercise 1.S.9: Decompose a vector along two different lines

A vector on $y = 2x$ is $(s, 2s)$; one on $y = x/2$ is $(t, t/2)$. Their sum equals $(5, 6)$ when $s + t = 5$ and $2s + t/2 = 6$. Substituting $t = 5 - s$ into the latter and multiplying by two gives $4s + 5 - s = 12$, so $s = 7/3$ and $t = 8/3$. Therefore

$$(5, 6) = (7/3, 14/3) + (8/3, 4/3).$$

Each summand satisfies its respective line equation, and their coordinate sums check the requested result.

### Exercise 1.S.10: Infer existence and uniqueness from the diagram

The two source arrows $a_1, a_2$ are nonzero and not parallel, so they are independent. Their two column directions span $\mathbb{R}^2$. Thus any $b$ shown in that plane is attained, and its two coefficients are unique. Negative coefficients are allowed; being outside the wedge of positive combinations does not prevent a solution of $Ax = b$.

**Conclusion.** Yes, there is exactly one solution.

### Exercise 1.S.11: A nonechelon matrix with a line of zero-images

Choose

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

Both rows lead in column one, so it is not in echelon form. The second equation in $Ax = 0$ gives $x_1 = 0$, then the first gives $x_2 = 0$, while $x_3$ is free. Thus the solution set is $\{(0, 0, t)^T : t \in \mathbb{R}\}$, the required line in $\mathbb{R}^3$.

### Exercise 1.S.12: A nonechelon matrix with a plane of zero-images

Choose $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$. The second row again leads under the first, so the matrix is not echelon. Its second equation is twice the first, leaving only $x_1 + 2x_2 + 3x_3 = 0$. Its solutions are

$$x = s(-2, 1, 0)^T + t(-3, 0, 1)^T, \qquad s, t \in \mathbb{R}.$$

The two generators are independent, so this is a plane, not merely a line.

### Exercise 1.S.13: A prescribed null vector determines the missing entries

The pivot information forces reduced echelon form $\begin{bmatrix} 1 & 0 & u \\ 0 & 1 & v \\ 0 & 0 & 0 \end{bmatrix}$. Applying it to $(3, -2, 1)^T$ gives $(3 + u, -2 + v, 0)^T$. For this to be zero, $u = -3$ and $v = 2$. Hence the required form is

$$\begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

Row operations preserve the null space, so the prescribed vector must satisfy these reduced equations even if the original $A$ was different.

### Exercise 1.S.14: Parameter values for two-vector independence

Put the two vectors into columns and replace $R_2$ by $R_2 - aR_1$:

$$\begin{bmatrix} 1 & a \\ a & a + 2 \end{bmatrix} \sim \begin{bmatrix} 1 & a \\ 0 & a + 2 - a^2 \end{bmatrix}.$$

There are two pivots precisely when $a+2-a^2 \neq 0$. Since $a+2-a^2 = -(a-2)(a+1)$, independence holds for $\boxed{a \neq 2, -1}$. At either excluded value there is one pivot, so a free coefficient supplies dependence.

### Exercise 1.S.15: Parameters in triangular-looking vector lists

**(a) Three vectors in $\mathbb{R}^3$.**

The column matrix is $\begin{bmatrix} a & b & d \\ 0 & c & e \\ 0 & 0 & f \end{bmatrix}$. Independence requires and is guaranteed by $a, c, f \neq 0$; the other entries $b, d, e$ may be any real numbers. For sufficiency, read a homogeneous relation from bottom to top: $fz = 0$ gives $z = 0$, then $cy = 0$ gives $y = 0$, then $ax = 0$ gives $x = 0$. For necessity, if $a = 0$ the first vector is zero; if $c = 0$, the first two vectors are dependent; if $f = 0$, all three vectors lie in the two-dimensional coordinate plane with third coordinate zero.

**(b) Three vectors in $\mathbb{R}^4$.**

There are *no restrictions* on $a, b, c, d, e, f$. A relation among the columns, read from the last coordinate upward, first forces the third coefficient to zero because that coordinate is 1 only in column three. The third coordinate then forces the second coefficient to zero, and the second coordinate forces the first to zero. Hence the list is independent for every choice of the six numbers.

### Exercise 1.S.16: Use the preceding-column characterization

The supplied matrix has nonzero diagonal entries $1, 5, 8, 10$ and zeros above the diagonal. Column one is nonzero. For column $j > 1$, suppose it were a combination of the preceding columns. Its first $j-1$ coordinates are zero. Reading those coordinates from top to bottom forces every coefficient of the preceding columns to be zero, because each relevant diagonal entry is nonzero. The proposed combination would then be zero, contradicting the nonzero $j$th coordinate of column $j$. Thus no later column is a combination of preceding ones. The characterization in Theorem 7 of Section 1.7 now proves that all four columns are independent.

### Exercise 1.S.17: Adjoin a vector outside the old span

Suppose $c_1v_1 + c_2v_2 + c_3v_3 + c_4v_4 = 0$. If $c_4 \neq 0$, solving for $v_4$ would put it in $\text{Span}\{v_1, v_2, v_3\}$, contrary to the hypothesis. So $c_4 = 0$. Independence of the first three then forces $c_1 = c_2 = c_3 = 0$. All coefficients vanish, proving independence of the four-vector list.

### Exercise 1.S.18: Add one basis vector to another

A relation $av_1 + b(v_1 + v_2) = 0$ can be collected as $(a+b)v_1 + bv_2 = 0$. Independence of $v_1, v_2$ forces $b = 0$ and $a + b = 0$, hence $a = 0$. Thus the new pair $v_1, v_1 + v_2$ is also independent. The argument proves the result for all coefficients, not just a selected numerical example.

### Exercise 1.S.19: Three collinear points give dependent position vectors

Because the points are distinct, $v_2 - v_1 \neq 0$ gives a direction for their common line. Some real $t$ satisfies $v_3 = v_1 + t(v_2 - v_1) = (1-t)v_1 + tv_2$. Therefore

$$(1-t)v_1 + tv_2 - v_3 = 0.$$

The coefficient of $v_3$ is $-1$, so this relation is nontrivial. It proves dependence even if the line does not pass through the origin: the line's position vectors still lie in the span of two of its points.

### Exercise 1.S.20: Reverse an input using homogeneity

Since $-u = (-1)u$, linearity gives $T(-u) = (-1)T(u) = -v$. This uses the scaling rule with the scalar $-1$ and does not require $T$ to be invertible.

### Exercise 1.S.21: A coordinate-plane reflection

The images of $e_1, e_2, e_3$ are $e_1, -e_2, e_3$. Put them into columns to get

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

It leaves the first and third coordinates fixed and reverses the second, exactly the specified reflection through $x_2 = 0$.

### Exercise 1.S.22: Surjectivity forces injectivity for this square size

Onto $\mathbb{R}^3$ means a pivot in every row of the $3 \times 3$ matrix. These three pivots also occupy every column, so there is no free variable in $Ax = 0$. Its only solution is zero, giving one-to-one. The equality of the input and output dimensions is crucial here.

### Exercise 1.S.23: A Givens rotation in two coordinates

Multiplying the displayed matrix by $(4, 3)^T$ gives $(4a - 3b, 4b + 3a)^T$. The target requires $4a - 3b = 5$ and $3a + 4b = 0$. From the second equation, $b = -3a/4$; substituting in the first gives $25a/4 = 5$. Thus $\boxed{a = 4/5, b = -3/5}$. The condition $a^2 + b^2 = 16/25 + 9/25 = 1$ is satisfied, and the image is $(5, 0)^T$, so both direction and length check.

### Exercise 1.S.24: Zero one coordinate without changing the middle one

The first and third coordinates give $2a - 4b = 2\sqrt{5}$ and $2b + 4a = 0$. The latter says $b = -2a$, so $10a = 2\sqrt{5}$. Therefore

$$a = 1/\sqrt{5}, \qquad b = -2/\sqrt{5}.$$

The middle coordinate remains 3 automatically, and $a^2 + b^2 = 1/5 + 4/5 = 1$. Substituting in the first and third coordinates gives $2\sqrt{5}$ and 0, as required.

### Exercise 1.S.25: A building design needs whole, nonnegative floor counts

**(a) Interpret a scalar multiple of the recipe column.**

The vector $x_1(3,7,8)^T$ gives the numbers of three-, two-, and one-bedroom apartments on the $x_1$ floors using plan A.

**(b) Add the contributions from all three plans.**

The total apartment-count vector is

$$x_1\begin{bmatrix}3\\7\\8\end{bmatrix}+x_2\begin{bmatrix}4\\4\\8\end{bmatrix}+x_3\begin{bmatrix}5\\3\\9\end{bmatrix}.$$

**(c) Solve the target equations, then impose the physical restrictions.**

Appending $(66,74,136)^T$ and reducing gives

$$\left[\begin{array}{ccc|c}3&4&5&66\\7&4&3&74\\8&8&9&136\end{array}\right]\sim\left[\begin{array}{ccc|c}1&0&-1/2&2\\0&1&13/8&15\\0&0&0&0\end{array}\right].$$

Thus the real solutions are $x_1=2+t/2$, $x_2=15-13t/8$, $x_3=t$. Floors must be nonnegative integers. Integer $t$ makes $x_2$ an integer only when $t$ is a multiple of 8, since 13 and 8 have no common factor. Nonnegativity requires $0\le t\le 120/13<10$. Hence the only allowed values are $t=0,8$, giving $(x_1,x_2,x_3)=(2,15,0)$ or $(6,2,8)$. Substitution yields the requested apartment totals in both cases.

**Conclusion.** Yes. There are exactly two designs with whole nonnegative floor counts, although the unrestricted real system has infinitely many solutions.